A computable prime-corrected Riemann–von Mangoldt clock: an exact defect identity, the prime-information threshold, and the location of the zeros of the Riemann zeta function in mean square

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Abstract

Let γ_k be the ordinates of the nontrivial zeros of the Riemann zeta function, counted with multiplicity, and let τ_k be the midpoint Riemann–von Mangoldt clock (RvM clock): the solution of $N_0(\tau_k) = k - \frac{1}{2}$, where N_0 is the main term of the Riemann–von Mangoldt formula, given in closed form by Lambert's function. The clock is close to the ordinates, $\tau_k - \gamma_k$ being bounded; in units of zeros, however, the ordinal displacement $s_k = k - \frac{1}{2} - N_0(\gamma_k)$ has mean square $\frac{1}{2\pi^2}\log\log T$ (Selberg), which is unbounded. We prove, unconditionally: (A) an exact identity that turns sums of $\mathcal{G}(u_k + \frac{1}{2}) - \mathcal{G}(u_k - \frac{1}{2})$, $u_k = k - \frac{1}{2} - F(\gamma_k)$, over consecutive ordinates into dF-integrals of $g(N - F)$, where $\mathcal{G}' = g$, for every continuous g and every C^1 model F of the counting function N; (B) for the clock equation corrected by the Euler phase S_X of the primes $p \le X$, the mean square of the ordinal defect at the ordinates is $\frac{1}{2\pi^2}\log\frac{\log T}{\log X} + O(1 + \sqrt{\log\frac{\log T}{\log X}})$ uniformly for $2 \le X \le T^{1/2}$, so that it is bounded if and only if $\log X \asymp \log T$; (C) an exact finite-height accounting of the gain of the one-step clock $\tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$, $\mathcal{L}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$; (D) with $X = T^{1/\sqrt{\log\log T}}$ the one-step clock, calculated from τ_k and the primes alone, has mean-square error $(\log\log\log T + O(\sqrt{\log\log\log T}))/\log^2 T$ in the ordinates, against $(2 + o(1))\log\log T/\log^2 T$ for the clock, and the Newton-type step iterated $\log^* T + O(1)$ times, with cutoffs depending only on T through absolute constants, locates the ordinates with a mean-square error of $O(1)$ mean spacings; sorting the final points gives a nondecreasing sequence with no larger mean-square error; the ordinal defect has exponential tails and the errors of the corrected clocks have all moments bounded in units of $\sqrt{H_m}$; (D′) with a fixed cutoff $X = T^c$, the best L^4 approximation of $N_0 + S_X$ on $[\frac{T}{2}, 3T]$ by piecewise linear functions with slopes between $\frac{\log T}{4\pi}$ and $\frac{\log T}{\pi}$ is unique and is obtained without zeros, its level points $\widehat{x}_k$ are increasing with gaps between $\pi/\log T$ and $4\pi/\log T$, and $\langle|\widehat{x}_k - \gamma_k|^4\rangle \ll (1 + \log\frac{1}{c})^2/\log^4 T$ uniformly for $2 \le X \le \sqrt{T/2}$, without Newton iteration; (D″) for the one-step clock with a fixed cutoff exponent, $\langle(\tau_k^{T^c} - \gamma_k)^2\rangle \ll (1 + \log\log\log T)/\log^2 T$ uniformly for $(\log\log T)^{-1/2} \le c \le \frac{1}{2}$; (E) the largest displacement satisfies $\limsup_k |\tau_k - \gamma_k| = 2\pi \limsup_t |S(t)|/\log t$, and Backlund's method with the convex hull of the exponent pairs of Bourgain and Heath-Brown and of their images under van der Corput's process gives $\limsup_k |\tau_k - \gamma_k| \le 0.6072889$ (0.6060224 with the exponent pairs of the preprints of Trudgian–Yang and Tao–Trudgian–Yang), against $2\pi \cdot 0.10076 = 0.6331\ldots$ from the explicit bound of Bellotti and Wong; this bound for the lim sup is asymptotic and is not an effective finite-height estimate; (F) an ordinate of multiplicity one carries a simple zero on the critical line, so for every model F the number of simple zeros on the line in $(T, 2T]$ is at least $\#\{T < \gamma_k \le 2T\} - \sum_k (2u_k)^{2j}$, and with (A) their proportion is at least $\frac{4}{3} - 4Q_2 - o(1)$,

where Q_2 is the dF_X-mean square of $N - F_X$, $F_X = N_0 + S_X$; (G) for a fixed cutoff, the zeta function whose zeros are $\frac{1}{2} \pm i\kappa_k$, κ_k the roots of the corrected equation in its monotone range, is the zeta function of the clock times the finite Euler product $\prod_{p \le X}(1 - p^{-s})^{-1}$, up to a factor holomorphic and without zeros in $\Re s > \frac{1}{2}$. No corrected point evaluates the Riemann zeta function or any of its zeros, and the constant of (E) is obtained without evaluating the zeta function. Our approach is similar to that of LeClair, who solves numerically the clock equation corrected by the first N primes and compares the solutions with known zeros, and the zeros of Gonek's functions ζ_X, built from finite Euler products, solve a version of the same equation with smoothed weights; here the corrected points are explicit or defined by a convex program, and their errors are proved.

Keywords: Riemann zeta function, zeros of the Riemann zeta function, Riemann–von Mangoldt formula, argument of the Riemann zeta function, truncated Euler product, Newton-type iteration
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1. Introduction

1.1. The Riemann–von Mangoldt clock and the ordinates

Let $\zeta(s)$ be the Riemann zeta function, let $N(T)$ be the number of its nontrivial zeros $\rho = \beta + i\gamma$ with $0 < \gamma \le T$, counted with multiplicity, and list their ordinates as $\gamma_1 \le \gamma_2 \le \cdots$, with multiplicity. No assumption is made on the real parts: γ_k is the ordinate of a zero whether or not it lies on the critical line. Throughout, t is a real variable, T is the height of a window $(T, 2T]$, and k, l are indices of ordinates. For T not an ordinate the Riemann–von Mangoldt formula reads [37, §§9.3–9.4]

$$N(T) = N_0(T) + \frac{1}{\pi}\arg\zeta\Big(\frac{1}{2} + iT\Big) + O\Big(\frac{1}{T}\Big), \qquad N_0(T) = \frac{T}{2\pi}\log\frac{T}{2\pi e} + \frac{7}{8}, \tag{1.1}$$

where $\arg\zeta(\frac{1}{2}+iT)$ has its classical determination [37, §9.3]: the value obtained by continuous variation along $2 \to 2 + iT \to \frac{1}{2} + iT$, starting from $\arg\zeta(2) = 0$. With this determination $S(T) = \frac{1}{\pi}\arg\zeta(\frac{1}{2} + iT)$ is the remainder of the counting function, and every result quoted below on S (its moments, its normal law, Selberg's approximation, the bound of Bellotti and Wong) refers to it. The *midpoint Riemann–von Mangoldt clock* (RvM clock) is the sequence of solutions $\tau_k > 2\pi$ of the main-term equation,

$$N_0(\tau_k) = k - \frac{1}{2}, \qquad \tau_k = \frac{2\pi\,(k - \frac{11}{8})}{W_0\Big((k - \frac{11}{8})/e\Big)} \qquad (k \ge 1). \tag{1.2}$$

Here W is the Lambert function [7], the inverse of $w \mapsto we^w$: for real $y \geq -1/e$ the equation $we^w = y$ has a real solution, unique if $y \geq 0$ and two if $-1/e < y < 0$. The principal branch W_0 is the solution with $W_0(y) \geq -1$; it is increasing on $[-1/e, \infty)$, $W_0(0) = 0$, and $W_0(y) = \log y - \log\log y + o(1)$ as $y \to \infty$; the other real branch, $W_{-1} \leq -1$, is defined only on $[-1/e, 0)$. Substituting $w = W_0((k - 11/8)/e)$ in $we^w = (k - 11/8)/e$ proves the inversion. For $k = 1$ the argument is negative and W_{-1} gives a value below 2π, excluded by the defining condition $\tau_k > 2\pi$. The clock is explicit: if g_n is the n-th Gram point, $\vartheta(g_n) = n\pi$ with ϑ the Riemann–Siegel theta function (Section 2.2), then $\vartheta(\tau_k) = \pi(k - \frac{3}{2}) + O(1/\tau_k)$, so τ_k lies between the consecutive Gram points g_{k-2} and g_{k-1} for k large. The explicit, fully determined Lambert values are an essential part of this paper: its aim is to improve and characterize the ordinates γ_k starting precisely from them. The prime information added to the clock is the Euler phase of the primes $p \leq X$,

$$S_X(t) = -\frac{1}{\pi} \sum_{p \leq X} \arg\bigl(1 - p^{-1/2-it}\bigr), \qquad F_X(t) = N_0(t) + S_X(t), \tag{1.3}$$

which turns the clock equation into the *corrected clock equation* $F_X(t) = k - \frac{1}{2}$. The relation between the clock and the ordinates is measured in units of zeros by the ordinal displacement and by its version for the corrected equation, the *ordinal defect*,

$$s_k = k - \frac{1}{2} - N_0(\gamma_k), \qquad u_k(X) = s_k - S_X(\gamma_k) = k - \frac{1}{2} - F_X(\gamma_k). \tag{1.4}$$

$u_k(X)$ is the residual of the corrected clock equation at the true ordinate. If $|u_k(X)| < \frac{1}{2}$, then $k - 1 < F_X(\gamma_k) < k$: the half-integer level of F_X nearest to $F_X(\gamma_k)$ is $k - \frac{1}{2}$, and the assignment $\hat{k} = \lfloor F_X(\gamma_k) \rfloor + 1$ gives γ_k its index k. This is not the assignment by rounding, $\lfloor F_X(\gamma_k) + \frac{1}{2} \rfloor$, which equals k exactly when $-1 < u_k(X) \leq 0$ (Proposition 5.5 concerns that rule). For a multiple ordinate, k is one of the indices $N(\gamma_k^-) + 1, \ldots, N(\gamma_k^+)$, and in general not $N(\gamma_k)$.

1.2. The approximation of the zeros by the clock and the primes

The approximation studied here starts from the explicit numbers τ_k and corrects them with the Euler phase of finitely many primes and prime powers, by one or several Newton-type steps for the corrected equation $N_0(t) + S_X(t) = k - \frac{1}{2}$ from $t = \tau_k$, with cutoffs that depend only on the height, or, with a fixed cutoff, by solving the same equation for a regularized version of $N_0 + S_X$, its best L^4 approximation by increasing piecewise linear functions with controlled slopes (Theorem D′). Every corrected point is calculated from k, N_0 and the primes: the construction evaluates neither the Riemann zeta function nor any of its zeros. The ordinates γ_k enter only in the statements, to measure the error. Theorems A and C are exact identities; Theorems B, D, D′ and D″ are asymptotic statements in mean square (in fourth mean for D′) over the zeros with $T < \gamma_k \leq 2T$, and for D′ also over the window $T \leq \tau_k < 2T$; Theorem E concerns the largest displacement $\limsup_k |\tau_k - \gamma_k|$. Theorem F is a consequence of Theorem A for the simple zeros on the critical line, and Theorem G identifies the zeta function of the roots of the corrected equation for a fixed cutoff.

1.3. Background

The exact statements are in Section 3.

- The displacement $\delta_k = \tau_k - \gamma_k$ is bounded, and $\limsup_k |\delta_k| \le 2\pi \cdot 0.10076 = 0.6331\ldots$, from the explicit bound of Bellotti and Wong for $N(T) - \frac{T}{2\pi}\log\frac{T}{2\pi e}$ [3], which improves those of Trudgian [39] and of Hasanalizade, Shen and Wong [15]. For large T only, Trudgian proved $|S(T)| \le 0.1013\log T$ with Huxley's exponent [38, §5 of the preprint], which gives $\limsup_k |\delta_k| \le 0.6365$.

- The ordinal displacement s_k differs by less than 1 from minus the integer displacement of γ_k with respect to the Gram intervals, whose moments are those of a normal variable with variance $\frac{1}{2\pi^2}\log\log T$, by Selberg's formulae as proved by Korolev [24, 21]. Hence $\langle s_k^2\rangle \sim \frac{1}{2\pi^2}\log\log T$ and $\langle \delta_k^2\rangle \sim 2\log\log T/\log^2 T$ over $T < \gamma_k \le 2T$.

- In t-mean, $S(t) = \frac{1}{\pi}\arg\zeta(\frac{1}{2}+it)$ is asymptotically normal (Selberg [32], Tsang [41]), and Selberg's approximation of $S(t)$ by prime sums up to a small power of T holds in $L^{2k}(T,2T)$ [32, 33], in the form proved by Fujii [11, §3], uniformly in k (Lemma 3.6), together with Fujii's moments of $S(t+h)-S(t)$ [12, 37]. These concern t-averages, not the ordinates.

- França and LeClair [10, (20), p. 11] write the condition $\vartheta(t_n) + \lim_{\delta\to 0^+}\arg\zeta(\frac{1}{2}+\delta+it_n) = (n-\frac{3}{2})\pi$, that is $N_0(t_n) + S^*(t_n) = n - \frac{1}{2}$ up to the terms $O(1/t)$ of Stirling's series, the limit being $\pi S^*(t_n)$ at a simple zero, as an equation satisfied by the zeros on the critical line (at a simple ordinate it is the identity $N_0(\gamma_k) + S^*(\gamma_k) + \varpi(\gamma_k)/\pi = k - \frac{1}{2}$ of Section 2.4); under the assumption that this limit exists for every t they conclude that the equation has a unique solution for every n [10, p. 11]; its first approximation, without the argument, is the clock equation $N_0(t) = n - \frac{1}{2}$, for which they use Lambert's function [10, (62)–(63), p. 24], as in (1.2). Their numerical solutions evaluate $\arg\zeta$ [10, §7.1, p. 28]. LeClair [26, §VI, (49), p. 18] replaces $\arg\zeta$ by the Euler phase of the first N primes; the resulting equation contains no ζ, and, up to the term $1/(48t)$ of Stirling's series, it is the clock equation corrected by the primes, $N_0(t) + S_X(t) = n - \frac{1}{2}$ with $X = p_N$. It is solved there by a root finder, adding one prime at a time from the Lambert value. Its accuracy is estimated numerically, by comparison with zeros computed by Odlyzko near $n = 10^{21}$ and $n = 10^{22}$ [26, Table I, p. 19], and is argued from the claim $S(t) = O(1)$ [26, Prop. 4, p. 12], made under the assumption of [26, Thm. 3]; that claim is incompatible with the normal law of $S(t)$ [32, 41], by which S is unbounded. No error bound for the solutions of either equation is proved in [10, 26]. Our approach is similar: we also start from the RvM clock and correct it with the Euler phase of the primes. The differences are the following. (i) The points of Theorem D are explicit expressions in k, τ_k and the primes, and those of Theorem D′ are defined by a convex program in N_0 and S_X; the points of [26] are numerical roots. (ii) Their errors are proved: one Newton-type step gives mean-square error $(\log\log\log T + O(\sqrt{\log\log\log T}))/\log^2 T$ in the ordinates (Theorem D), $\log^* T + O(1)$ steps give $O(1)$ mean spacings, and so does the regularization of Theorem D′ with a fixed cutoff exponent, in fourth mean. (iii) Theorem B measures the corrected equation at the true ordinates and shows how many primes are needed: with a fixed set of primes the mean-square defect grows like $\frac{1}{2\pi^2}\log\log T$, and it is bounded if

and only if $\log p_N \asymp \log T$. (iv) Theorem E gives the exact lim sup of the displacement in terms of S, and a bound for it; Theorems F and G relate the defect to the simple zeros on the line and the roots of the corrected equation to the finite Euler product. The range covered here is $p_N \le T^{1/2}$ for Theorems B and D and $p_N \le \sqrt{T/2}$ for Theorem D′ (Table 2); the larger numbers of primes allowed in [26, p. 19], up to $[t^2]$, are not covered.

- Gonek's functions $\zeta_X(s) = P_X(s) + \chi(s)\overline{P_X(s)}$, built from the finite Euler product P_X, the exponential of a sum over $n \le X^2$ with weights $\Lambda_X(n)$ [14, (7) and §5], have all their zeros above a fixed height on the critical line [14, Thm. 6.2]. On the line $\zeta_X(\frac12 + it) = 2e^{-i\vartheta(t)}\Re\Big(e^{i\vartheta(t)}P_X(\frac12+it)\Big)$, so these zeros are the solutions of $N_0(t)+\widetilde{S}_X(t)+\varpi(t)/\pi \in \mathbb{Z}+\frac12$, where $\pi\widetilde{S}_X(t) = \Im \log P_X(\frac12 + it)$ is the prime phase with Gonek's weights: a corrected clock equation with smoothed weights. Their closeness to the zeros of ζ is supported numerically there; under the Riemann hypothesis, a modified function ζ_X^* has, for X large depending on the interval, no zero in a closed interval between consecutive ordinates [14, Thm. 9.1].

1.4. Comparison with the cited antecedents

Table 1 compares statements about the same quantity wherever possible. A continuous t-average, a defect evaluated at a true ordinate, and the error of a point calculated from the primes are different objects; a result about one requires a transfer argument before it applies to another. "Proved here" identifies the result established by the present proof, without asserting that no earlier equivalent formulation exists. The comparison concerns the cited antecedents, not an exhaustive priority survey.

Table 1: Cited antecedents, conclusions proved here, and the scope of the difference.

Object	Established antecedent	Present result and scope
Bare midpoint clock τ_k	The RvM main term and its inversion are classical. Selberg–Korolev displacement moments give $\langle(\tau_k - \gamma_k)^2\rangle_{\mathcal{I}_T} \sim 2\log\log T/\log^2 T$; see Remark 3.17 and [24, 21].	The explicit, fully determined Lambert values are the starting point of every construction of this paper, which improves and characterizes the ordinates from them.
Prime approximation of $S(t)$	Selberg's approximation, the Fujii moment estimates, and the form recorded in [8, (3.6)] bound the residual of a short prime sum in continuous t-mean. Montgomery–Vaughan bounds its polynomial moments.	These are external inputs. They do not on their own state an error estimate for the specific output $\tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$; the sampling and transport estimates supply that step.
Exact conversion of moments and other test functions	The counting staircase and its jumps are classical. Moment and distribution formulae for Gram displacement are already available in [24, 21].	Theorems 4.1, 4.2 give the exact formula for any real C^1 model F, with both endpoint terms; no monotonicity of F is required. This is a general identity, not a claim to have discovered the displacement law.

Table 1 continued

Object	Established antecedent	Present result and scope
Defect at the true ordinate, $u_k(X)$	Prime-sum approximation controls $S(t) - S_X(t)$ in t-mean. This is an input about the counting error, not an already constructed approximation to γ_k.	Theorems 5.1, 5.12 give $\langle u_k(X)^2\rangle_{\mathcal{I}_T} = (2\pi^2)^{-1}\log(1/c) + O(1+\sqrt{\log(1/c)})$, $c = \log X/\log T$, uniformly for $2 \le X \le T^{1/2}$, including an index-dependent cutoff. The difference is the quantified statement at the ordinates and its cutoff range.
One explicit prime correction	The bare clock has the preceding $2\log\log T/\log^2 T$ baseline. Approximating the phase by primes is an established idea, not a new ingredient.	Theorem 7.3 controls the transport to τ_k. At the analytically specified cutoff, the height MSE is $(\log\log\log T + O(\sqrt{\log\log\log T}))/\log^2 T$. Its ratio to the bare-clock MSE tends to zero. Constants and onset are not evaluated.
Finite-height accounting	Expanding a squared error is elementary; it does not justify independent prime responses or a fitted gain.	Theorem 6.1 keeps the diagonal, cross term, transport, and exact nonnegative curvature term. It is an identity for a finite set, not a positive-gain guarantee at every height.
Further corrections and order	Newton-type correction and increasing rearrangement are standard operations. Rearrangement cannot increase a convex matching cost for two ordered real lists.	Theorem 8.11 gives uniform control through $\log^* T + O(1)$ prescribed corrections, with a multiplicity estimate. Corollary 8.15 gives a nondecreasing output with height MSE $O(\log^{-2} T)$. Theorem 9.6 gives the same scale without Newton iteration, with a fixed cutoff exponent, in fourth mean, with increasing points separated by at least $\pi/\log T$; its points are defined by a convex program on the whole block, not index by index. The construction and its transport control are the issue; the rate alone is not a priority claim. The cutoff constants are not yet numerical.
Tails	Moment inequalities and the conversion of moments to tail bounds are classical.	Theorem 8.3 gives the stated $2^{2-\sqrt{\lambda/C}}$ bound with its explicit admissible range of c; Corollary 8.13 treats the constructed points. These are asymptotic bounds with unspecified constants, not certified error probabilities for the available data.
Worst error of the bare clock	Bellotti–Wong [3, Thm. 1.1, Cor. 1.5] give the explicit coefficient 0.10076 for the logarithmic term of the counting and argument bounds, implying $\limsup \lvert\tau_k - \gamma_k\rvert \le 2\pi \cdot 0.10076$. Backlund's method and the exponent inputs are established tools.	Proposition 10.1 identifies the two limsups exactly. Theorem 10.2 evaluates the convex hull of the published exponent pairs of Bourgain and Heath-Brown, giving $2\pi C_{\mathcal{P}} < 0.6072889$, and 0.6060224 with the pairs of the preprints [40, 35] (Remark 10.3). This comparison improves that cited asymptotic coefficient; it is not a new explicit bound valid at a given height. No value of ζ is computed: the inputs are theorems on exponent pairs.

Table 1 continued

Object	Established antecedent	Present result and scope
Corrected clock equation, solved numerically	The equation with $\arg\zeta$ [10, (20)]; the clock equation corrected by the first N primes [26, (49)]; with smoothed weights, the equation of the zeros of Gonek's ζ_X [14, Thm. 6.2]. Accuracy shown by comparison with known zeros [26, Table I], without a proved error bound.	For $p_N \le T^{1/2}$, Theorem 5.1 gives the mean-square defect of the corrected clock equation at the true ordinates; with a fixed N the defect grows like $(2\pi^2)^{-1}\log\log T$ (Corollary 5.2), and for $N = N(T)$ it is bounded iff $\log p_N \asymp \log T$ (Corollary 5.4). Theorems 7.3–8.11 replace the root finder by explicit steps with proved mean-square error, and, for $p_N \le \sqrt{T/2}$, Theorem 9.6 by one convex regularization with a fixed cutoff exponent.
Accuracy at a specified height	Direct certified zero computations can locate individual zeros using evaluations of ζ; this is a different computational input from the prime-only clocks considered here.	No certified finite-height precision constant or individual-zero accuracy is obtained from the asymptotic O-terms here. No superiority at the available data heights is inferred from those terms.

Table 2 separates the approximations of the zeros by what has to be evaluated to produce the points. The proofs of Theorems B–E use theorems about ζ (Selberg's approximation, Fujii's moments, exponent pairs) and on Dirichlet polynomials (the Montgomery–Vaughan inequality); what is free of ζ is the construction of the points, and the numbers of Theorem E, which come from exponent pairs and not from values of ζ.

Table 2: What each approximation of the zeros evaluates, and what is proved about its error.

Approximation	Evaluated to produce the points	Error
Certified zero computations, e.g. [31, §2, p. 2]	Values of the completed zeta function on the critical line in ball arithmetic; sign changes; Turing's method.	Every zero up to the height reached is isolated rigorously.
França–LeClair, exact equation [10, (20)]	$\arg\zeta(\frac{1}{2}+it)$, principal value [10, §7.1]; root finder.	Satisfied at the zeros on the line; a unique solution for every n under the assumption that $\lim_{\delta\to0^+}\arg\zeta(\frac{1}{2}+\delta+it)$ exists for every t; accuracy shown numerically.
Clock τ_k: the Riemann–von Mangoldt main term inverted by W_0 [7]; also [10, (63)]	Nothing: a closed expression in k.	δ_k bounded; $\langle\delta_k^2\rangle \sim 2\log\log T/\log^2 T$ (Selberg, Korolev); $\limsup\lvert\delta_k\rvert$ is Theorem E.
Clock equation corrected by the first N primes, solved numerically [26, (49)]	The first N primes; root finder. No ζ, no zero.	Estimated numerically [26, Table I]; argued from $S(t) = O(1)$, which is false.

Table 2 continued

Approximation	Evaluated to produce the points	Error
Zeros of Gonek's ζ_X [14]	The prime powers $n \le X^2$ with weights $\Lambda_X(n)$; the zeros are computed numerically. No ζ, no zero.	On the critical line above a fixed height (proved); closeness to the zeros of ζ supported numerically; under RH, for a modified ζ_X^*, no zeros between consecutive ordinates for X large, without rate [14, Thm. 9.1].
Theorems B–D (this paper)	k, τ_k and the primes $p \le X(T)$; explicit Newton-type steps. No ζ, no zero.	Proved: defect at the ordinates (B), mean-square error in the ordinates, $O(1)$ mean spacings after $\log^* T + O(1)$ steps, tails (D).
Theorem D′ (this paper)	N_0 and the primes $p \le T^c$, $c < \frac{1}{2}$ fixed; one convex program on $[\frac{T}{2}, 3T]$ and linear interpolation; a dual certificate for the solver. No ζ, no zero.	Proved: $\langle \lvert\widehat{x}_k - \gamma_k\rvert^4 \rangle \ll (1 + \log \frac{1}{c})^2 / \log^4 T$ on $T < \gamma_k \le 2T$ and on $T \le \tau_k < 2T$; increasing points with gaps in $[\pi/\log T, 4\pi/\log T]$.
Theorem E (this paper)	Not a construction: a bound for $S(t)$ from exponent pairs. No value of ζ computed.	Proved, asymptotic: $\limsup \lvert\tau_k - \gamma_k\rvert \le 0.6072889$.
Bellotti–Wong [3]	Not a construction: an explicit bound for $S(t)$.	Proved for all $T \ge e$: $\lvert S(T)\rvert \le 0.10076 \log T + \ldots$; hence $\limsup \lvert\tau_k - \gamma_k\rvert \le 2\pi \cdot 0.10076$.

Meaning of the mean-square scale.. The assertion $\langle \lvert x_k - \gamma_k\rvert^2 \rangle \ll \log^{-2} T$ bounds the RMS error by a constant multiple of one mean spacing, since the local spacing is $1/\mathcal{L}(T) \sim 2\pi/\log T$. It does not assert an $o(1)$ fraction of a spacing, a numerical precision at a specified height, or a bound for each ordinate. Classical moment estimates already identify this natural scale: for example, [37, §9.25, (9.25.4)] bounds the moments of every fixed order of the gaps between consecutive ordinates on the scale $1/\log T$, and [8, (3.6)] supplies a fourth-moment prime approximation. A sublevel-integration construction is another route to this scale; its use of classical inputs does not establish priority for a particular formula. For approaching the zeros, the useful comparison is the error of the specified construction against the bare clock, together with its constants and effective range; the latter remain open when they occur only in asymptotic O-terms.

1.5. What this paper adds

Let $\mathcal{I}_T = \{k : T < \gamma_k \le 2T\}$ and let $\langle\cdot\rangle_{\mathcal{I}_T}$ be the mean over $\mathcal{I}_T$. Theorems A–E are unconditional deductions from the external inputs stated in Section 3. Their precise statements are proved here; this description does not establish historical priority. In particular, the scale $O(\log^{-2} T)$ alone is not claimed as a new result. Theorem E applies Backlund's method in the form of [38, 39, 3] to exponents of [5, 16, 18]. Table 1 separates the classical inputs from the conclusions for the specific corrected points.

A. Exact defect identity (Theorems 4.1 and 4.2). For every real C^1 model F of the counting function, every continuous g with primitive $\mathcal{G}$, and $u_k = k - \frac{1}{2} - F(\gamma_k)$, the sum of $\mathcal{G}(u_k + \frac{1}{2}) - \mathcal{G}(u_k - \frac{1}{2})$ over consecutive ordinates equals $\int g(N - F)\, dF$ up to two boundary terms. For $g(x) = x^{2j}$ this gives $\sum u_k^{2j}$ ($\frac{1}{12}$ per zero when $j = 1$); for

$g(x) = e^{i\xi x}$ it identifies the characteristic function of the empirical law of u_k with a t-integral. No hypothesis on the zeros and no monotonicity of F is used.

B. How much prime information the clock needs (Theorems 5.1 and 5.12). Uniformly for $2 \le X \le T^{1/2}$, with $c = \log X/\log T$,

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{1}{2\pi^2}\log\frac{1}{c} + O\Big(1+\sqrt{\log(1/c)}\Big), \tag{1.5}$$

and the same holds with the cutoff $X_k = \max(2, (\tau_k/2\pi)^c)$ varying with the index. A fixed set of primes leaves a mean-square defect $\sim (2\pi^2)^{-1}\log\log T$; primes up to T^c keep it bounded; boundedness holds if and only if $\liminf \log X/\log T > 0$ (Corollary 5.4). Thus merely letting the number of primes tend to infinity does not ensure a bounded ordinal defect: the logarithm of the cutoff must remain comparable with the logarithm of the height.

C. Exact accounting for the computable clock (Theorem 6.1). For the one-step clock $\tau_k^X = \tau_k - S_X(\tau_k)/\mathcal{L}(\tau_k)$, the mean-square gain over any finite set of zeros equals a diagonal prime-response term minus a cross term, with no independence assumption; the residual of τ_k^X differs from $u_k(X)$ by an explicit transport term and by the exact nonnegative term $\chi_k = \frac{\gamma_k}{2\pi}(\frac{\tau_k}{\gamma_k} - 1 - \log\frac{\tau_k}{\gamma_k})$.

D. A computable clock that approaches the zeros (Theorems 7.3, 8.6 and 8.11). For $2 \le X \le T^{1/2}$ the residual $r_k(X) = \mathcal{L}(\tau_k)(\tau_k^X - \gamma_k)$ satisfies $\langle r_k(X)^2\rangle_{\mathcal{I}_T} = \frac{1}{2\pi^2}\log\frac{1}{c} + O((1+\sqrt{\log(1/c)})(1 + c^2\log\log T))$; the term $c^2\log\log T$ comes from the bound of Lemma 7.2 for the transport between τ_k and γ_k, which is an upper bound and not the size of the transport (for $c \ge (\log\log T)^{-1/2}$ Theorem 7.5 replaces it by $1+\log\log\log T$). With $X = T^{1/\sqrt{\log\log T}}$, or with the cutoff $\tau_k^{1/\sqrt{\log\log\tau_k}}$ read off the clock itself,

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\big(\tau_k^X - \gamma_k\big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}. \tag{1.6}$$

A second Newton-type step from τ_k^X, with cutoff $T^{1/\sqrt{\log\log\log T}}$, has mean-square error $(\log\log\log\log T + O(\sqrt{\log\log\log\log T}))/\log^2 T$ (Theorem 8.6), and the m-th step, with the cutoffs (8.13) depending only on T, has error $(H_m + O(\sqrt{H_m}+1))/\log^2 T$ with absolute constants, where H_m is an iterated logarithm of T (Theorem 8.11); after $m^* = \log^* T + O(1)$ steps the error is $O(1)/\log^2 T$, that is, $O(1)$ mean spacings in mean square. The proof rests on an exact moment recursion, on a multiplicity bound for the corrected clock points of every level (Theorem 8.10, from Fujii's moments of $S(t+h)-S(t)$), and on a sampling inequality for maximal functions at points of bounded multiplicity. The second clock preserves the local order and spacing of the RvM clock outside a proportion $O(H_1^{-2})$ of the indices (Proposition 8.8). Sorting the points of the final level does not increase the unweighted mean-square error, so the final clock can be taken nondecreasing (Corollary 8.15); sorting removes inversions, not coincident points. The moments of every order of the errors are bounded in units of $\sqrt{H_j}$, which gives tails $\ll_q \lambda^{-2q}$ for the j-th clock (Corollary 8.13); and for the defect at the ordinates, with

primes up to T^c and $\exp(-(\lambda/C)^{3/2}) \le c \le 1/(C\sqrt{\lambda})$, the corrected count is off by more than λ positions on at most $2^{2-\sqrt{\lambda/C}}$ of the zeros (Theorem 8.3), where Chebyshev's inequality would give $\lambda^{-2}\log\lambda$.

D′. A regularized clock with a fixed cutoff (Theorem 9.6). Let $2 \le X \le \sqrt{T/2}$, $c = \log X/\log T$, and let $\widehat{F}$ be the best L^4 approximation of $N_0 + S_X$ on $[\frac{T}{2}, 3T]$ among the piecewise linear functions on a mesh of step at most $2\pi/\log T$ with slopes between $\frac{\log T}{4\pi}$ and $\frac{\log T}{\pi}$; it is unique and depends only on N_0 and the primes $p \le X$. The level points $\widehat{F}(\widehat{x}_k) = k - \frac{1}{2}$ are increasing, $\pi/\log T \le \widehat{x}_{k+1} - \widehat{x}_k \le 4\pi/\log T$, and

$$\frac{1}{\#\mathcal{I}_T} \sum_{k\in\mathcal{I}_T} \left|\widehat{x}_k - \gamma_k\right|^4 \ll \frac{(1+\log(1/c))^2}{\log^4 T}, \tag{1.7}$$

with absolute constants, over $T < \gamma_k \le 2T$ and equally over the window $T \le \tau_k < 2T$ defined by the clock; the tails are $\ll (1+\log\frac{1}{c})^2\lambda^{-4}$ at λ mean spacings. For fixed c this is the scale of one mean spacing, reached without Newton iteration or sorting, with a fixed cutoff exponent. If $X \le T^{1/4}$, the transport $S_X(\gamma_k) - S_X(\widehat{x}_k)$ is bounded in mean square. An approximate minimizer whose objective exceeds the minimum by at most ηT, $\eta \le 1$, gives the same bounds with $(1+\log\frac{1}{c})^2 + \eta$, and the excess is certified by an explicit dual bound computed from the data of the program (Proposition 9.7). The price is locality: $\widehat{x}_k$ depends on S_X on the whole block, through a convex program with $\asymp T\log T$ variables, whereas the Newton-type points of Theorem D are explicit index by index.

D″. The one-step clock with a fixed cutoff exponent (Theorem 7.5). Uniformly for $(\log\log T)^{-1/2} \le c \le \frac{1}{2}$,

$$\frac{1}{\#\mathcal{I}_T} \sum_{k\in\mathcal{I}_T} \left(\tau_k^{T^c} - \gamma_k\right)^2 \ll \frac{1+\log\log\log T}{\log^2 T}, \qquad \left\langle \mathfrak{t}_k(T^c)^2 \right\rangle_{\mathcal{I}_T} \ll 1+\log\log\log T, \tag{1.8}$$

by comparison with the cutoff $T^{1/\sqrt{\log\log T}}$ inside the proof; the transport bound of Theorem D, $c^2\log\log T$, is replaced by its logarithm. Whether the factor $\log\log\log T$ can be removed is not decided; a heuristic in Remark 7.6 suggests that it cannot.

E. The pointwise envelope (Proposition 10.1 and Theorem 10.2). $\limsup_k |\tau_k - \gamma_k| = 2\pi \limsup_{t\to\infty} |S(t)|/\log t$, also for the corrected clocks with $X \le (\log T)^{2-\varepsilon}$; and for every exponent profile $\mathfrak{e}$ of the Riemann zeta function on $[\frac{1}{2}, 1]$,

$$\limsup_{k\to\infty} |\tau_k - \gamma_k| \le \frac{1}{\log 2} \int_0^1 \frac{\mathfrak{e}(\sigma)\, d\sigma}{\sqrt{\sigma(2-\sigma)}}, \tag{1.9}$$

$\mathfrak{e}$ being extended to $[0, \frac{1}{2})$ by the functional equation. The lower convex hull of the exponent pairs of Bourgain [5] and Heath-Brown [16] and of their images under van der Corput's $\mathscr{A}$-process gives $C_{\mathcal{P}} = 0.0966530245541\ldots$, so $\limsup_t |S(t)|/\log t \le 0.0966531$ and $\limsup_k |\tau_k - \gamma_k| \le 2\pi C_{\mathcal{P}} < 0.6072889$, compared with 0.10076 in [3] and

0.1013 in [38]; with the exponent pairs of the preprints [40, 35] the bound is 0.6060224 (Remark 10.3). This is an asymptotic bound and does not replace the explicit bounds of [3], valid for all $T \geq e$. With the Backlund disc used here no exponent profile gives less than $(\frac{\sqrt{3}}{2} - \frac{\pi}{6})/\log 2 = 0.4940\ldots$ (the Lindelöf profile), and a numerical scan finds no better disc, although on the Lindelöf hypothesis $S(t) = o(\log t)$.

F. Simple zeros on the critical line (Lemma 11.1, Theorem 11.2, Corollary 11.3). An ordinate of multiplicity $m \geq 2$ forces $u_{k+i} = u_k + i$ along its indices, for any model F, and an ordinate of multiplicity one carries a simple zero on the critical line. Hence, for every $j \geq 1$ and every real function F, the number $N_0^s(T, 2T)$ of simple zeros on the line with $T < \gamma \leq 2T$ satisfies $N_0^s(T, 2T) \geq \#\mathcal{I}_T - \sum_{k \in \mathcal{I}_T} (2u_k)^{2j}$, and with Theorem A, for $F = F_X$ and $2 \leq X \leq T^{1/2}$,

$$\frac{N_0^s(T, 2T)}{\#\mathcal{I}_T} \geq \frac{4}{3} - 4Q_2(T, X) - o(1), \qquad Q_2(T, X) = \frac{1}{\#\mathcal{I}_T} \int_T^{2T} \Big(N - F_X\Big)^2 dF_X. \tag{1.10}$$

A positive proportion would follow from any proved bound $\limsup_T Q_2(T, T^c) < \frac{1}{3}$ at one fixed c; Theorem B gives the size of Q_2 with an absolute constant that is not computed, so no numerical proportion is claimed.

G. The finite Euler product (Theorem 12.3). Let $X \geq 2$ be fixed, let κ_k be the roots of $F_X(t) = k - \frac{1}{2}$ in the range where F_X is increasing (for the finitely many smaller indices κ_k is fixed arbitrarily), and let ζ_κ and ζ_τ be the functions of order one with the functional equation whose nontrivial zeros are $\frac{1}{2} \pm i\kappa_k$ and $\frac{1}{2} \pm i\tau_k$. Then

$$\zeta_\kappa(s) = \zeta_\tau(s)\, \frac{P_X(s)}{P_X(1)}\, e^{H_X(s)}, \qquad P_X(s) = \prod_{p \leq X} \Big(1 - p^{-s}\Big)^{-1}, \tag{1.11}$$

with H_X holomorphic in $\Re s > \frac{1}{2}$ and given explicitly by X, N_0 and the primes. The main term of $\log \zeta_\kappa$ is exactly the finite Euler product of the primes used by the construction.

Main finding. The Euler phase of the primes up to a power of the height turns the clock from an approximation of the ordinates into an approximation of the indices. For the bare clock the error in units of zeros, $s_k = \mathcal{L}(\xi_k)\delta_k$, has mean square $\sim \frac{1}{2\pi^2} \log\log T$. With $X = T^c$ the error in units of zeros of the corrected clock equation has bounded mean square (Theorem B), so outside a proportion $\ll (1 + \log(1/c))/A^2$ of the zeros it is at most A; and the computable clock, which uses only τ_k and the primes, keeps the gain up to the transport between τ_k and γ_k (Theorem D). With $X = T^{1/\sqrt{\log\log T}}$,

$$\frac{\langle (\tau_k^X - \gamma_k)^2 \rangle_{\mathcal{I}_T}}{\langle (\tau_k - \gamma_k)^2 \rangle_{\mathcal{I}_T}} \sim \frac{\log\log\log T}{2\log\log T} \longrightarrow 0, \tag{1.12}$$

and in units of zeros the mean square drops from $\frac{1}{2\pi^2} \log\log T$ to $\frac{1}{4\pi^2} \log\log\log T$; iterating the Newton-type step brings it down to $O(1)$, and the regularization of Theorem D′ does so without Newton iteration, with a fixed cutoff exponent. The gain is a statement in mean square; for the largest displacement, Proposition 10.1 shows that no cutoff $X \leq (\log T)^{2-\varepsilon}$ changes $\limsup |\tau_k - \gamma_k|$, and the bound (1.9) comes from the exponents of the Riemann zeta function, not from the primes.

1.6. Degree of proof

Theorem, *Proposition*, *Lemma* and *Corollary* are proved here unconditionally, from the external results of Section 3. No statement of this paper rests on a computation with the zeros. The only numbers that are not closed expressions are the values of the integral (10.1) for piecewise linear profiles with rational vertices, computed in closed form and in interval arithmetic by the programs named in the Data availability statement; the constants in the O-terms of Theorems B, D, D$'$ and D$''$ are absolute but not computed, and the same holds for T_0; making them explicit would require explicit forms of Selberg's and Fujii's estimates. In Theorem 8.11 the cutoffs and the number of steps depend on two such constants, $C_\star$ and H^*, whose existence is proved and whose numerical values are not; the corrected points are explicit expressions in k, τ_k and the primes once these constants are fixed. Thus a fixed number of Newton-type steps with specified cutoffs (Theorems 7.3 and 8.6) is a fully specified construction, while the uniform statement up to $\log^* T + O(1)$ steps asserts the existence of admissible constants $C_\star$, H^*.

Table 3: Main results: degree of proof, role, and whether ζ or its zeros are evaluated.

result	status	role	evaluates ζ or zeros
Theorems 4.1, 4.2 (A)	unconditional	statistics of the errors at the zeros as t-integrals	no
Theorems 5.1, 5.12 (B)	unconditional	bounded error in units of zeros iff $\log X \asymp \log T$	no (primes $p \le X$)
Theorem 6.1 (C)	unconditional, exact	finite-height balance of the gain of one correction	no
Theorems 7.3, 8.6, 8.11 (D)	unconditional	zero-free corrected points with mean-square error $O(1)$ mean spacings	no (k, τ_k, primes)
Theorem 9.6 (D$'$), Proposition 9.7	unconditional	no Newton iteration, fixed cutoff exponent, fourth mean $O(1)$ mean spacings on the clock window, separated increasing points, certified tolerance	no (N_0, primes)
Theorem 7.5 (D$''$)	unconditional	one-step clock, fixed cutoff exponent, error $\ll (1 + \log\log\log T)$ in units of zeros	no (k, τ_k, primes)
Theorem 8.3, Corollaries 8.13, 8.15	unconditional	tails of the defect and of the corrected clocks; a nondecreasing clock built without ordinates	no
Proposition 10.1, Theorem 10.2 (E)	unconditional, asymptotic	the largest displacement of the clock is the extreme size of $S(t)$	no (exponent pairs)
Lemma 11.1, Theorem 11.2, Corollary 11.3 (F)	unconditional	simple zeros on the line bounded below by the defect	no
Theorem 12.3 (G)	unconditional, exact	the zeta function of the corrected roots contains the finite Euler product exactly	no (N_0, primes $p \le X$)

1.7. Organization

Section 2 lists the notation and Section 3 the external results. Sections 4–7 prove Theorems A–D, Section 8 iterates the Newton-type step uniformly in the number of steps, Section 9 proves Theorem D′ and the tolerance certificate, Section 10 proves Theorem E, Section 11 Theorem F and Section 12 Theorem G; Theorem D″ is proved in Section 7. Section 13 summarizes the results and their scope.

2. Notation

2.1. List of symbols

The notation below fixes the meaning of the recurring symbols. Coefficients and dummy indices explicitly introduced inside a proof are local. The letter μ is not used (it is reserved for the Möbius function), π is only the number $3.14159\ldots$, and prime counts are written $\#\{p \le y\}$. Dummy variables of integration $(t,\, y)$ are local.

symbol	meaning
$\zeta(s)$	the Riemann zeta function
$\rho_k = \beta_k + i\gamma_k$	nontrivial zeros of the Riemann zeta function, $0 < \gamma_1 \le \gamma_2 \le \cdots$ with multiplicity
$\nu(\gamma)$	number of zeros of the Riemann zeta function with ordinate γ (multiplicities included)
$N(t)$, $N(t^{\pm})$	number of zeros with $0 < \gamma \le t$; one-sided limits
ϑ, Θ_0, ϖ	Riemann–Siegel theta function; $\Theta_0(t) = \frac{t}{2}\log\frac{t}{2\pi e} - \frac{\pi}{8}$; $\varpi = \vartheta - \Theta_0$
g_m	Gram points, $\vartheta(g_m) = m\pi$
N_0, $\mathcal{L}$	$N_0 = \Theta_0/\pi + 1$; $\mathcal{L} = N_0'$, $\mathcal{L}(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}$
S, S^*, G	$S(t) = \frac{1}{\pi}\arg\zeta(\frac{1}{2}+it)$; midpoint value at ordinates; $G = N - N_0 = S + \varpi/\pi$
$\mathcal{I}_T$, $\langle\cdot\rangle_I$	$\{k : T < \gamma_k \le 2T\}$; mean over k in a finite index set I
τ_k, τ_x, W_0	RvM clock (2.5); the same for real $x \ge 1$; principal Lambert function
Λ_T, J_T, L_T, h_T, t_i	$\frac{\log T}{2\pi}$; the block $[\frac{T}{2}, 3T]$; number of mesh cells, mesh step $\le \Lambda_T^{-1}$, nodes (Section 9)
$\mathcal{K}_T$	piecewise linear functions on the mesh with slopes in $[\frac{1}{2}\Lambda_T, 2\Lambda_T]$ (Section 9)
$\widehat{F}$, $\widehat{x}_k$, $\widehat{u}_k$, $t_\flat$, $t_\sharp$	regularized counting model, its level points, its defect at γ_k, the two good heights of its proof (Section 9)
$E(\mathbf{v})$, E_*, η_{reg}, B_{reg}, G, A_{reg}	objective of the convex program, its minimum, tolerance, dual lower bound, mass matrix, convexity constant: local to Proposition 9.7
s_k, δ_k, ξ_k	$s_k = k - \frac{1}{2} - N_0(\gamma_k)$; $\delta_k = \tau_k - \gamma_k$; mean-value point in $s_k = \mathcal{L}(\xi_k)\delta_k$
Δ_k	integer displacement of γ_k with respect to the Gram intervals (Theorem 3.4)
∇_k	forward difference $\nabla_k a = a_{k+1} - a_k$ of a sequence (Proposition 8.8)
$\mathfrak{d}_k$	$s_k + \Delta_k$, $\lvert\mathfrak{d}_k\rvert < 1$ (Remark 3.17)
$\mathfrak{s}_T$	$(\frac{1}{2\pi^2}\log\log T)^{1/2}$ (Remark 3.17)

symbol	meaning
Λ, a_n	von Mangoldt function; $a_n = -\Lambda(n)/(\pi\sqrt{n}\log n)$
A, A_X	set of prime powers with $\sum_{n\in A}\lvert a_n\rvert < \infty$; $A_X = \{p^m : p \le X\}$
S_A, S_X	$\sum_{n\in A} a_n \sin(t\log n)$; S_{A_X}
v_A, v_X	$\frac{1}{2}\sum_{n\in A} a_n^2$; v_{A_X}
σ_X, M_X	bounds for $\lvert S_X\rvert$ and $\lvert S_X'\rvert$, (2.11)
F, F_X	a real C^1 model of N; $F_X = N_0 + S_X$
u_k, $u_k(X)$	$k - \frac{1}{2} - F(\gamma_k)$; $k - \frac{1}{2} - F_X(\gamma_k)$
Ψ_j, κ_j	sawtooth polynomial (4.2); $\kappa_j = \frac{(2j)!}{j!(2\pi)^{2j}}$
X, c, C, T_0	prime cutoff; $c = \log X/\log T$; absolute constants
Y, Y_0, Y_k, S_Y^{pr}	polynomial length; $Y_0 = T^{1/160}$; $Y_k = T^{1/(80k)}$; $S_Y^{\mathrm{pr}}(t) = -\frac{1}{\pi}\sum_{p<Y} p^{-1/2}\sin(t\log p)$
x_{F}, a_2, κ_{F}	Fujii's length $T^{a_2/(60k)}$ and parameter (Theorem 3.5); $\kappa_{\mathrm{F}} = 1/(300\log 2)$ (Lemma 3.6)
Y_{K}, ε_{K}, c_{K}	length and constants in Korolev's lemma (Remark 3.7)
$\mathcal{J}$, V, $\mathcal{W}$, $\mathcal{W}_1$	primes between X and Y_0; $V = \sum_{p\in\mathcal{J}} p^{-1}$; $\mathcal{W} = S_{Y_0}^{\mathrm{pr}} - S_X$; its part over $\mathcal{J}$ (5.4)
E_0, E_X, $\mathcal{E}_2$	$S - S_{Y_0}^{\mathrm{pr}}$; $S - S_X$; $\int_T^{2T} E_X^2\,dt$
z_n, $\mathcal{D}$, $\mathcal{D}_1$, $\mathcal{D}_2$, η_n, η_n'	coefficients and Dirichlet polynomials in Lemmas 5.9–5.10
τ_k^X, δ_k^X, $r_k(X)$, χ_k	one-step clock; $\tau_k^X - \gamma_k$; $\mathcal{L}(\tau_k)\delta_k^X$; second-order error (6.3)
$\mathcal{R}_n^{\gamma}(I)$, $\mathcal{R}_n^{\tau}(I)$	prime-power responses at the zeros and at the clock points (6.1)
$\mathrm{Cr}_A(I)$, $D_X(I)$	cross term (6.2); diagonal term (6.5)
λ	real multiplier of an Euler phase
ς_n, $\mathbf{Q}$, $\mathbf{b}$, $\boldsymbol{\lambda}$, $\Upsilon_{\boldsymbol{\lambda}}$	$\varsigma_n(k) = \sin(\tau_k\log n)$; Gram matrix; projection vector; coefficients; trigonometric correction (Remark 6.3)
t_k^X	root of $F_X(t) = k - \frac{1}{2}$ (Remark 5.7)
ℓ_n	$\log n$
ψ, t_1, t_2	test function and endpoints in Lemma 4.7
y_0, w_T	fixed threshold; threshold tending to infinity (Corollary 5.4)
$\mathfrak{g}$, $\mathfrak{G}$, ω	continuous test function and a primitive (Theorem 4.2); real frequency variable
$\mathfrak{t}_k(X)$, y_k	transport term $S_X(\gamma_k) - S_X(\tau_k)$; one-step correction $S_X(\tau_k)/\mathcal{L}(\tau_k)$
φ, $\mathfrak{h}$, $\mathcal{M}$, $\mathfrak{b}_T$, $\mathcal{U}_k$	cutoff function; $\varphi S_X'$; centred Hardy–Littlewood maximal operator; $2\pi/\log(3T)$; sampling intervals (Lemma 7.2)
X_k, $\tilde{u}_k$, $X_\pm$, $V_\pm$	variable cutoff; its defect; extreme cutoffs in a window; $\sum_{X_-<p\le X_+} p^{-1}$ (Theorem 5.12)
$\mathcal{Q}$, $\mathfrak{w}$	maximal partial prime sum; smooth window (Lemma 5.11)
$\mathcal{I}_T^{\tau}$, τ_k^{loc}	$\{k : T \le \tau_k < 2T\}$; one-step clock with cutoff $\tau_k^{1/\sqrt{\log\log\tau_k}}$ (Corollary 7.7)
$\mathfrak{a}$, BW, BW_S	$\limsup_{t\to\infty}\lvert S(t)\rvert/\log t$; Bellotti–Wong bounds for $\lvert N(t) - \frac{t}{2\pi}\log\frac{t}{2\pi e}\rvert$ and for $\lvert S(t)\rvert$ (3.1)
$\mathfrak{e}$, (k,l)	exponent profile of the Riemann zeta function; exponent pair (Section 10)
$\sigma_\circ$, ϱ, $\varkappa$, $\sigma^\sharp$, $\delta^\sharp$	centre and radius of the Jensen disc; power in Backlund's function; parameters of Backlund's trick (Theorem 3.16)

symbol	meaning
$x_k^{(j)}$, $e_{j,k}$, $\chi_{j,k}$, $\|\cdot\|_{r,I}$	iterated Newton points (8.1); their errors; curvature terms; discrete norms (Section 8)
H_j, c_j, y_k, $\tilde{n}_j$	$H_0 = \log\log T$, $H_j = 2\log(1/c_j)$ and the cutoff law (8.13); $y_k = S_{X_1}(\tau_k)/\mathcal{L}(\tau_k)$; multiplicity functions (Lemma 8.4, Theorem 8.10)
$A_j(r)$, m^*, $C_\star$, H^*, $B_\star$, C_5, C_6	normalized moments of the errors, number of steps and constants of Theorem 8.11
$n_{\mathsf{R}}(u)$, C_1, C_2, C_3	number of ordinates of the window within $\mathsf{R}\mathfrak{b}_T$ of u; constants of Lemma 8.9 and Theorem 8.10
M, P, F, s, W, y	local names in proofs: $\#\mathcal{I}_T$; number of close pairs; g^{4r} (Lemma 8.5); window and scale (Lemma 5.11)
$\mathfrak{k}$, $\mathfrak{k}_j$	multiplicity constants: least $\mathfrak{k} \ge 1$ with $\int \tilde{n}^2 \le \mathfrak{k}\,\mathfrak{b}_T \#\mathcal{I}_T$ (Lemma 8.5, Theorem 8.11)
$y_k^{(j)}$, $\mathfrak{r}_{j,k}$, $d_{j,k}$	$\tau_k - x_k^{(j)}$ and the remainder of Lemma 8.7; gaps $x_{k+1}^{(j)} - x_k^{(j)}$ (Proposition 8.8)
g_j	$\mathcal{M}(\varphi_j S'_{X_j})$, maximal function of the derivative of the j-th Euler phase (Proposition 8.1)
$\mathcal{A}_p$	norm of the maximal operator on L^p (Theorem 3.12)
$\mathscr{A}$	van der Corput's $\mathscr{A}$-process on exponent pairs (Theorem 3.14)
$\varepsilon_\circ$, q, C_{b}, C_{d}	parameter of the Jensen disc (proof of Theorem 10.2); fixed exponent $\log n = \mathsf{q}\log T$ (Remark 6.4); constants in the proof of Lemma 8.2
$\mathfrak{f}_\varkappa$, $\mathcal{Z}_\varkappa$, $F_\circ$	Backlund's auxiliary function, its zero sum and the majorant (Theorem 3.16)

2.2. Counting functions

Let $\vartheta(t) = \Im \log\Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2}\log\pi$ for $t > 0$. For $t > 0$ not an ordinate [37, Thms. 9.3–9.4],

$$N(t) = \frac{1}{\pi}\vartheta(t) + 1 + S(t), \qquad S(t) = \frac{1}{\pi}\arg\zeta\Big(\tfrac{1}{2} + it\Big) = O(\log t), \tag{2.1}$$

with $\arg\zeta$ defined by continuous variation from 2 to $2 + it$ and then horizontally to $\frac{1}{2} + it$. At an ordinate we keep the right-continuous N and define S by (2.1); $S^*(\gamma) = \frac{1}{2}(S(\gamma^-) + S(\gamma^+))$. Put

$$\Theta_0(t) = \frac{t}{2}\log\frac{t}{2\pi e} - \frac{\pi}{8}, \qquad N_0(t) = \frac{1}{\pi}\Theta_0(t) + 1, \qquad \varpi(t) = \vartheta(t) - \Theta_0(t). \tag{2.2}$$

By Stirling's formula $\varpi(t) = \frac{1}{48t} + O(t^{-3})$ and $\varpi'(t) = O(t^{-2})$; in particular $|\varpi(t)|/\pi < \frac{1}{100}$ for $t \ge 14$. For all $t > 0$,

$$N(t) = N_0(t) + S(t) + \frac{\varpi(t)}{\pi}, \qquad \mathcal{L}(t) = N_0'(t) = \frac{1}{2\pi}\log\frac{t}{2\pi}. \tag{2.3}$$

Since $N(2T) - N(T) = N_0(2T) - N_0(T) + O(\log T)$,

$$\#\mathcal{I}_T = T\mathcal{L}(T) + O(T). \tag{2.4}$$

2.3. The RvM clock and the displacements

N_0 is C^∞ and strictly increasing on $(2\pi,\infty)$, with $N_0(2\pi)=-\frac{1}{8}$ and $N_0(t)\to\infty$. Hence for every real $x\ge 1$ the equation $N_0(t)=x-\frac{1}{2}$ has exactly one solution $\tau_x>2\pi$. Writing $t=2\pi y$, the equation is $y\log(y/e)=x-\frac{11}{8}\ge-\frac{3}{8}$; its root $y>1$ is $y=(x-\frac{11}{8})/W_0((x-\frac{11}{8})/e)$, with $y=e$ if $x=\frac{11}{8}$ [7]. Thus

$$\tau_x=\frac{2\pi\,(x-\frac{11}{8})}{W_0\Big((x-\frac{11}{8})/e\Big)},\qquad \frac{d\tau_x}{dx}=\frac{1}{\mathcal{L}(\tau_x)}=\frac{2\pi}{\log(\tau_x/2\pi)},\tag{2.5}$$

the derivative by implicit differentiation. Since $\Theta_0(\tau_k)=\pi(k-\frac{3}{2})$ and $\varpi=O(1/t)$, $\vartheta(\tau_k)=\pi(k-\frac{3}{2})+O(1/\tau_k)$, so $g_{k-2}<\tau_k<g_{k-1}$ for k large; in the phase Θ_0 the point τ_k is the midpoint of the two neighbouring levels $\pi(k-2)$ and $\pi(k-1)$.

Definition 2.1. For $k\ge1$, $s_k=k-\frac{1}{2}-N_0(\gamma_k)$ and $\delta_k=\tau_k-\gamma_k$.

Since $\gamma_1>14>2\pi$ and $\tau_k>2\pi$, the mean value theorem applied to $s_k=N_0(\tau_k)-N_0(\gamma_k)$ gives

$$s_k=\mathcal{L}(\xi_k)\,\delta_k\qquad\text{for some }\xi_k\text{ between }\gamma_k\text{ and }\tau_k.\tag{2.6}$$

Lemma 2.2. *For every $k\ge1$,*

$$N(\gamma_k^-)-N_0(\gamma_k)+\frac{1}{2}\le s_k\le N(\gamma_k^+)-N_0(\gamma_k)-\frac{1}{2},\tag{2.7}$$

and

$$\Big|s_k-\frac{\varpi(\gamma_k)}{\pi}\Big|\le\max\Big(|S(\gamma_k^-)|,|S(\gamma_k^+)|\Big)-\frac{1}{2}.\tag{2.8}$$

Proof. The indices j with $\gamma_j=\gamma_k$ are exactly $N(\gamma_k^-)+1,\dots,N(\gamma_k^+)$, so $N(\gamma_k^-)+1\le k\le N(\gamma_k^+)$; subtract $\frac{1}{2}+N_0(\gamma_k)$. By (2.3) at $\gamma_k^\pm$ and continuity of N_0 and ϖ, (2.7) reads $S^-+\frac{1}{2}\le x\le S^+-\frac{1}{2}$ with $x=s_k-\varpi(\gamma_k)/\pi$, $S^-=S(\gamma_k^-)$, $S^+=S(\gamma_k^+)$, and $S^+-S^-=\nu(\gamma_k)\ge1$. Put $\mathsf{m}=\max(|S^-|,|S^+|)$. Then $x\le S^+-\frac{1}{2}\le\mathsf{m}-\frac{1}{2}$ and $x\ge S^-+\frac{1}{2}\ge-\mathsf{m}+\frac{1}{2}$, which is (2.8). □

2.4. Euler phases and the corrected clock

For $n\ge2$ let $a_n=-\Lambda(n)/(\pi\sqrt{n}\log n)$. For a set A of prime powers with $\sum_{n\in A}|a_n|<\infty$,

$$S_A(t)=\sum_{n\in A}a_n\sin(t\log n),\qquad v_A=\frac{1}{2}\sum_{n\in A}a_n^2;\tag{2.9}$$

the series converges absolutely and uniformly in t, so S_A is continuous and bounded. For $X\ge2$, $\sum_{n\in A_X}|a_n|\le\frac{1}{\pi}\sum_{p\le X}\sum_{m\ge1}\frac{1}{mp^{m/2}}<\infty$ and, since $\arg(1-z)=-\Im\sum_{m\ge1}z^m/m$ for $|z|<1$ with the principal argument,

$$S_X(t)=-\frac{1}{\pi}\sum_{p\le X}\arg\Big(1-p^{-1/2-it}\Big)=-\frac{1}{\pi}\sum_{p\le X}\sum_{m\ge1}\frac{\sin(mt\log p)}{m\,p^{m/2}}=S_{A_X}(t).\tag{2.10}$$

Each factor has positive real part: $1-p^{-1/2}\cos(t\log p)>0$. Its argument is therefore the single continuous value $\arctan(p^{-1/2}\sin(t\log p)/(1-p^{-1/2}\cos(t\log p)))$ in $(-\pi/2,\pi/2)$. The

phase in (2.10) is the sum of these values; reducing the argument of the entire product modulo 2π would change the function. Thus no branch is selected from the zeros or by fitting. The series of derivatives converges uniformly, because $\sum_{p\le X}\log p\sum_{m\ge1}p^{-m/2}<\infty$; hence $S_X\in C^\infty(\mathbb{R})$, its derivatives may be taken termwise, and

$$|S_X(t)|\le\sigma_X:=\frac{1}{\pi}\sum_{p\le X}\sum_{m\ge1}\frac{1}{m\,p^{m/2}},\qquad |S_X'(t)|\le M_X:=\frac{1}{\pi}\sum_{p\le X}\frac{\log p}{\sqrt{p}-1},\tag{2.11}$$

with $\sigma_X\ll\sqrt{X}/\log X$ and $M_X\ll\sqrt{X}$ by the prime number theorem.

Definition 2.3. For $X\ge2$ let $F_X=N_0+S_X$ and $u_k(X)=k-\frac12-F_X(\gamma_k)=s_k-S_X(\gamma_k)$. The one-step clock and its residual are

$$\tau_k^X=\tau_k-\frac{S_X(\tau_k)}{\mathcal{L}(\tau_k)},\qquad \delta_k^X=\tau_k^X-\gamma_k,\qquad r_k(X)=\mathcal{L}(\tau_k)\,\delta_k^X.\tag{2.12}$$

Since $F_X(\tau_k)-(k-\frac12)=S_X(\tau_k)$, τ_k^X is the first step of $t\mapsto t-(F_X(t)-k+\frac12)/\mathcal{L}(t)$ from $t=\tau_k$. At a simple ordinate, (2.3) gives $N_0(\gamma_k)+S^*(\gamma_k)+\varpi(\gamma_k)/\pi=k-\frac12$, so $u_k(X)=S^*(\gamma_k)-S_X(\gamma_k)+\varpi(\gamma_k)/\pi$ is the error of the prime approximation of the argument of the Riemann zeta function at the zero.

3. External results

The results of this section are known; they are used, not reproved. Where a statement is adapted (a parameter fixed, an interval split, a prime sum lengthened), the adaptation is proved after it.

Theorem 3.1 ([37, Thms. 9.4, 9.9 and §9.12], [28])**.** *$S(t)=O(\log t)$; $\gamma_{k+1}-\gamma_k$ is bounded; and $S_1(t)=\int_0^t S(y)\,dy=O(\log t)$.*

Theorem 3.2 (Bellotti–Wong [3, Thm. 1.1, Cor. 1.5 and (1.2)])**.** *For $T\ge e$, $\bigl|N(T)-\frac{T}{2\pi}\log\frac{T}{2\pi e}\bigr|\le\mathrm{BW}(T)$ and $|S(T)|\le\mathrm{BW}_S(T)$, where*

$$\begin{aligned}\mathrm{BW}(t)&=\min\{0.10076\log t+0.24460\log\log t+8.08344,\\&\qquad\qquad 0.11200\log t+0.12567\log\log t+3.77417\},\\ \mathrm{BW}_S(t)&=0.10076\log t+\min\{0.24460\log\log t+7.20844,\ 1.68845\log\log t+1.50956\}.\end{aligned}\tag{3.1}$$

Moreover $|S(T)|\le2.5167$ for $0\le T\le30\,610\,046\,000$.

Earlier explicit versions are due to Trudgian [39] and to Hasanalizade, Shen and Wong [15]. For T sufficiently large, Trudgian [38, §5 of the preprint] proved $|S(T)|\le0.1013\log T$ with Huxley's exponent $\frac{32}{205}$ for $\zeta(\frac12+it)$. The bound $|S(T)|\le2.5167$ rests on Platt's rigorous isolation of the zeros up to that height [3, (1.2)].

Theorem 3.3 (Selberg [32]; see [11, p. 225])**.** *For every integer $j\ge1$, $\int_T^{2T}S(t)^{2j}\,dt=\kappa_j\,T(\log\log T)^j+O_j\bigl(T(\log\log T)^{j-1/2}\bigr)$.*

Fujii [11, p. 225] quotes Selberg's formula for $\int_T^{T+H}$ with $T^{1/2+a} \le H \le T$; the form above is the case $H = T$.

Theorem 3.4 (Selberg; Korolev [24], [21, Thm. 4.2, p. 73]). *Let Δ_k be the integer such that $g_{k+\Delta_k-2} < \gamma_k \le g_{k+\Delta_k-1}$ (in [21] the Gram points are $t_m = g_{m-1}$ and $t_{m-1} < \gamma_k \le t_m$ with $m = k + \Delta_k$), and let $0 < \varepsilon_0 < 10^{-3}$. For every fixed $j \ge 1$, as $n_0 \to \infty$, uniformly for $n_0^{27/82+0.9\varepsilon_0} \le h_0 \le n_0^{27/82+\varepsilon_0}$,*

$$\sum_{n_0<k\le n_0+h_0} \Delta_k^{2j} = \kappa_j h_0 (\log\log n_0)^j \big(1+o(1)\big), \qquad \Big|\sum_{n_0<k\le n_0+h_0} \Delta_k^{2j-1}\Big| \ll_j h_0 (\log\log n_0)^{j-1}. \tag{3.2}$$

Theorem 3.5 (Selberg's formula with Fujii's bounds [11, §3, pp. 232–233]). *Let a_2 be fixed in the range of Fujii's Main Theorem [11, p. 225], which contains $\frac15$. There are T_0 and C, depending only on a_2, such that for $T \ge T_0$, $T^{1/2+a_2} \le H \le T$ and every integer $k \ge 1$ with $x_{\mathrm F} = T^{a_2/(60k)} \ge 2$,*

$$\int_T^{T+H} \Big(S(t) - S^{\mathrm{pr}}_{x_{\mathrm F}^3}(t)\Big)^{2k} dt \le (Ck^2)^{2k} H. \tag{3.3}$$

For the Riemann zeta function (the principal character, $q = 1$), [11, p. 232] writes Selberg's formula [33] for $x \ge 2$ as $S(t) = S^{\mathrm{pr}}_{x^3}(t) + \sum_{i=1}^5 O(B_i(t))$, with five explicit functions B_i and absolute implied constants, and uses it with $x = T^{a_2/(60k)}$. On p. 233 it bounds the norms $(\int_T^{T+H} |B_i|^{2k})^{1/2k}$ by $k^{1/2}H^{1/2k}$ for $i = 1, 4$, by $k^2 H^{1/2k}$ for $i = 2, 3$ and by at most $k^{3/2}H^{1/2k}$ for $i = 5$, under the hypotheses of its Main Theorem, which for $q = 1$ are $T^{1/2+a_2} \le H \le T$. By the remark on p. 226, read on the page image of the original, the implied constants in that paper do not depend on q, k and h; this uniformity in k is the external input on which the moments of growing order of Lemma 3.6, and through them Theorem 8.11, depend. Fujii states no range for k; $x_{\mathrm F} \ge 2$ is the condition under which he quotes Selberg's formula. Minkowski's inequality gives (3.3). Part II [13, §§1 and 5] lists the corrections to Part I: in the Main Theorem h must be bounded and the exponent of Ak in the error term is $2k$; the remainder terms of Lemma 2 (p. 228) and of the first case of Lemma 3 (p. 230) are to be multiplied by k^2, and $k!$ in Lemma 2 replaced by $(Ak)^k$; items (iii) and (iv) of p. 234 are withdrawn. None of them concerns Selberg's formula on p. 232 or the bounds of p. 233 used above.

Lemma 3.6 (Selberg's approximation uniformly in the order). *Let $\kappa_{\mathrm F} = 1/(300\log 2)$ and $Y_k = T^{1/(80k)}$. There are absolute constants C and T_0 such that for $T \ge T_0$ and every integer k with $1 \le k \le \kappa_{\mathrm F}\log T$,*

$$\int_T^{2T} \Big(S(t) - S^{\mathrm{pr}}_{Y_k}(t)\Big)^{2k} dt \le (Ck^2)^{2k}\, T. \tag{3.4}$$

In particular $\int_T^{2T} (S - S^{\mathrm{pr}}_{Y_0})^4 \ll T$ with $Y_0 = Y_2 = T^{1/160}$.

Proof. Take $a_2 = \frac15$ and $H = T$ in Theorem 3.5; its constants are then absolute, and $x_{\mathrm F} = T^{1/(300k)} \ge 2$ because $k \le \kappa_{\mathrm F}\log T$. It remains to lengthen the prime sum from $x_{\mathrm F}^3 = T^{1/(100k)} \ge 8$ to $Y_k = T^{1/(80k)}$. Here $S^{\mathrm{pr}}_{x_{\mathrm F}^3} - S^{\mathrm{pr}}_{Y_k} = \frac{1}{\pi}\Im\mathcal{D}$ with $\mathcal{D}(t) = \sum_{p\in\mathcal{J}'} p^{-1/2-it}$, $\mathcal{J}'$ the primes in $[x_{\mathrm F}^3, Y_k)$. As $\log Y_k/\log x_{\mathrm F}^3 = \frac54$, Theorem 3.10 gives $\sum_{p\in\mathcal{J}'} p^{-1} \le \log\frac54 + O(1) \le V'$

with an absolute V'. Next, $\mathcal{D}^k = \sum_n \eta_n n^{-it}$ over products of k primes of $\mathcal{J}'$, so $n \le Y_k^k = T^{1/80}$, and $\sum_n |\eta_n|^2 \le k!\, V'^k$, since each n has at most $k!$ ordered factorizations; Theorem 3.9 on $[T, 2T]$ gives $\int_T^{2T} |\mathcal{D}|^{2k} \le k!\, V'^k (T + O(T^{1/80})) \le (C'k)^k T$. Minkowski's inequality in $L^{2k}(T, 2T)$ gives $\|S - S^{\mathrm{pr}}_{Y_k}\|_{2k} \le (Ck^2 + \sqrt{C'k}) T^{1/2k}$, which is (3.4). □

Remark 3.7 (An independent route to the uniformity in the order). Korolev [22, Lemma 3.13, p. S126] proves, under the conditions of his Lemma 3.11 (p. S124) and the conventions of pp. S13 and S123: for fixed $0 < \varepsilon_{\mathrm{K}} < 10^{-3}$, $c_0 = 2880\varepsilon_{\mathrm{K}}^{-1}$, $T_1 \ge T_0(\varepsilon_{\mathrm{K}})$, $T_1^{27/82+0.9\varepsilon_{\mathrm{K}}} \le H \le T_1^{27/82+\varepsilon_{\mathrm{K}}}$, $Y'_{\mathrm{K}} = T_1^{0.1\varepsilon_{\mathrm{K}}}$, integers $1 \le k \le c_{\mathrm{K}} \log Y'_{\mathrm{K}}$ with a sufficiently small absolute constant $c_{\mathrm{K}} > 0$ not given numerically, and $Y_{\mathrm{K}} = (Y'_{\mathrm{K}})^{1/(8k+3)}$,

$$\int_{T_1}^{T_1+H} \Big| S(t) + \frac{1}{\pi} \sum_{p \le Y_{\mathrm{K}}} \frac{\sin(t \log p)}{\sqrt{p}} \Big|^{2k} dt < \frac{c_0}{56} (c_0 k)^{2k} H. \tag{3.5}$$

(The quotation in [23, Lemma 12, p. 16] omits the factor k^{2k}.) For a fixed ε_{K} this gives (3.4), even with $(Ck)^{2k}$ in place of $(Ck^2)^{2k}$, for $1 \le k \le 0.1 c_{\mathrm{K}} \varepsilon_{\mathrm{K}} \log T$, after three adaptations. (i) *Cutoff.* $S + \frac{1}{\pi} \sum_{p \le Y_{\mathrm{K}}} p^{-1/2} \sin(t \log p) = S - S^{\mathrm{pr}}_{Y_k} - \frac{1}{\pi} \sum_{Y_{\mathrm{K}} < p < Y_k} p^{-1/2} \sin(t \log p)$, and $1 < \log Y_k / \log Y_{\mathrm{K}} = \frac{(8k+3)\log T}{8k\, \varepsilon_{\mathrm{K}} \log T_1} \le \frac{11}{8\varepsilon_{\mathrm{K}}}$ for $T \le T_1 \le 3T$; so the primes between Y_{K} and Y_k have $\sum p^{-1} \le \log \frac{11}{8\varepsilon_{\mathrm{K}}} + O(1)$, and Theorem 3.9 on $[T_1, T_1 + H]$, applicable since $Y_k^k = T^{1/80} \le H$, bounds their sum in $L^{2k}(T_1, T_1 + H)$ by $C_{\varepsilon_{\mathrm{K}}} \sqrt{k}\, H^{1/2k}$, as in the proof of Lemma 3.6. (ii) *Short intervals.* Cover $[T, 2T]$ by consecutive intervals $[T_1, T_1 + H]$ with the common length $H = T^{27/82+0.95\varepsilon_{\mathrm{K}}}$, which is admissible for every $T_1 \in [T, 3T]$ once $T \ge T_0(\varepsilon_{\mathrm{K}})$. Each interval has its own $Y_{\mathrm{K}} = Y_{\mathrm{K}}(T_1)$, and (i) is applied on each interval before summing, so that the sum is over the same polynomial $S^{\mathrm{pr}}_{Y_k}$. (iii) *Range of orders.* The condition $k \le c_{\mathrm{K}} \log Y'_{\mathrm{K}} = 0.1 c_{\mathrm{K}} \varepsilon_{\mathrm{K}} \log T_1$ follows from $k \le 0.1 c_{\mathrm{K}} \varepsilon_{\mathrm{K}} \log T$. Along this route κ_{F} is replaced by the unspecified absolute constant $0.1 c_{\mathrm{K}} \varepsilon_{\mathrm{K}}$; Theorems 8.3 and 8.11 below need only some absolute constant in that place, with 300 replaced by $1/(0.1 c_{\mathrm{K}} \varepsilon_{\mathrm{K}} \log 2)$ in the choice of C_8. Uniform bounds of the same shape at the Gram points t_n, for $\sum_n (S(t_n + 0) + \frac{1}{\pi} \sum_{p<y} p^{-1/2} \sin(t_n \log p))^{2k}$ and for the moments of $S(t_n + 0) - S(t_{n-1} + 0)$, are [21, Lemma 2.4, p. 60, and Thm. 3.2, p. 67]; they concern sums over Gram points and are not used here.

Theorem 3.8 (Fujii [13, (α'), p. 33])**.** *There are absolute constants A and T_0 such that for every integer $k \ge 1$, $T \ge T_0$ and $0 < h \le 1$,*

$$\int_T^{2T} \Big(S(t + h) - S(t) \Big)^{2k} dt \ll (Ak)^k T \Big(\log\Big((h \log T + 3)\, e^k \Big) \Big)^k,$$

with an absolute implied constant; the interval $(T, 2T)$ may be replaced by $(0, T)$ [13, p. 34, footnote].

This is Fujii's corrected form of the moment estimate of [11, Main Theorem, p. 225], announced in [12, Lemma, p. 140]; for bounded h it is also stated in [37, (9.25.3), p. 246], there with $0 \le h \le \frac{1}{2}T$, whereas [13, §5-1] requires h to be bounded in the asymptotic formula (α) from which (α') is derived; (α') is printed with $h \wedge 1 = \min\{h, 1\}$ in place of h and without an explicit bound on h, but for unbounded h its right side would be independent of h, and with $k = 1$ that would bound $\int_H^{2H} S^2$ by $O_T(H)$ for every H, against Theorem 3.3;

so it is stated here for $0<h\le 1$ only. Only $0<h\le 1$ is used: with $k=4$ in Lemma 9.4 (for $h>1$ that lemma uses Theorem 3.3 instead), and with $k=1,2$ in Lemma 8.9, in the form

$$\int_{T-1}^{2T}\Big(S(t+h)-S(t)\Big)^{2k}dt \le C\,T\Big(\log(2+h\log T)\Big)^{k} \qquad (k=1,2,\ 0<h\le 1,\ T\ge T_0), \tag{3.6}$$

which follows from the theorem on $(T/2,T)$ and $(T,2T)$, since $\log((h\log T+3)e^k)\le k+\log(3+h\log T)\le (2k+2)\log(2+h\log T)$.

Theorem 3.9 (Montgomery–Vaughan [30, Cor. 3]). *For complex z_n, real t_0 and $H>0$, $\int_{t_0}^{t_0+H}\Big|\sum_{n\le Y} z_n n^{-it}\Big|^2dt=\sum_{n\le Y}|z_n|^2\Big(H+O(n)\Big)$ with an absolute implied constant.*

(Replace z_n by $z_n n^{-it_0}$ to pass from $[0,H]$ to $[t_0,t_0+H]$.)

Theorem 3.10 (Mertens [29]; see [36, Ch. I.1, Thms. 1.8 and 1.10]). *For $y\ge 2$, $\sum_{p\le y}\frac{\log p}{p}=\log y+O(1)$ and $\sum_{p\le y}\frac{1}{p}=\log\log y+O(1)$.*

Theorem 3.11 (Selberg [32]; Tsang [41]; see [8, (1.1)]). *For fixed real $y_1<y_2$, $\frac{1}{T}\operatorname{meas}\{t\in[T,2T]: y_1<S(t)/\mathfrak{s}_T\le y_2\}\to\frac{1}{\sqrt{2\pi}}\int_{y_1}^{y_2}e^{-y^2/2}dy$, where $\mathfrak{s}_T=(\frac{1}{2\pi^2}\log\log T)^{1/2}$.*

In [8] the statement is that $\log\zeta(\frac12+it)/\sqrt{\log\log T}$ tends to a standard complex Gaussian for t uniform in $[T,2T]$; its imaginary part is $\pi S(t)/\sqrt{\log\log T}$, of variance $\frac12$.

Theorem 3.12 (Hardy–Littlewood [34, Thm. I.1], as quoted in [19, (1.2)]; Kinnunen [19, Thm. 1.4]). *Let $\mathcal{M}f(x)=\sup_{y>0}\frac{1}{2y}\int_{x-y}^{x+y}|f|$ be the centred maximal function of $f\in L^1_{\mathrm{loc}}(\mathbb{R})$. For $1<p<\infty$ and $f\in L^p(\mathbb{R})$, $\|\mathcal{M}f\|_p\le\mathcal{A}_p\|f\|_p$, where $\mathcal{A}_p$ is bounded for $p\ge 2$. If f and f' belong to $L^p(\mathbb{R})$, $1<p<\infty$, then $\mathcal{M}f$ has a locally absolutely continuous representative, $(\mathcal{M}f)'\in L^p(\mathbb{R})$, and $|(\mathcal{M}f)'|\le\mathcal{M}(f')$ almost everywhere.*

Theorem 3.13 (Carleson [6], Hunt [17]; on $\mathbb{R}$ see [27, (1.1) and p. 1]). *For $f\in L^2(\mathbb{R})\cap L^4(\mathbb{R})$ with Fourier transform $\mathcal{F}f(\omega)=\int_{\mathbb{R}}f(t)e^{-i\omega t}dt$, let $\mathcal{C}f(x)=\sup_{\omega_1\in\mathbb{R}}\Big|\frac{1}{2\pi}\int_{-\infty}^{\omega_1}\mathcal{F}f(\omega)e^{i\omega x}d\omega\Big|$. Then $\|\mathcal{C}f\|_4\ll\|f\|_4$.*

In [27, p. 1] the Carleson–Hunt theorem is stated on $\mathbb{R}$ in the form that $f\mapsto\sup_{\xi\in\mathbb{R}}|\mathcal{H}(e^{i\xi\cdot}f)|$, $\mathcal{H}$ the Hilbert transform, is bounded on $L^p(\mathbb{R})$ for $1<p<\infty$; since the partial Fourier integral up to ω_1 is $e^{i\omega_1 x}\cdot\frac12(\mathrm{Id}\pm i\mathcal{H})(e^{-i\omega_1\cdot}f)(x)$, the operator $\mathcal{C}f$ is at most $\frac12(|f|+\sup_\xi|\mathcal{H}(e^{i\xi\cdot}f)|)$.

Theorem 3.14 ([18, Lemma 2.8 and (7.57)]; Bourgain [5, §5, Thm. 6]; Heath-Brown [16, Thm. 2]). *(a) If (k,l) is an exponent pair, so is $\mathscr{A}(\mathsf{k},\mathsf{l})=\Big(\frac{\mathsf{k}}{2\mathsf{k}+2},\frac12+\frac{\mathsf{l}}{2\mathsf{k}+2}\Big)$. (b) If (k,l) is an exponent pair with $\mathsf{l}-\mathsf{k}\ge\frac12$, then $\zeta(\mathsf{l}-\mathsf{k}+it)\ll|t|^{\mathsf{k}}\log|t|$ for $|t|\ge 2$. (c) For every $\varepsilon>0$, $(\frac{13}{84}+\varepsilon,\frac{55}{84}+\varepsilon)$ is an exponent pair. (d) For every integer $m\ge 3$ and every $\varepsilon>0$, $\Big(\frac{2}{(m-1)^2(m+2)},1-\frac{3m-2}{m(m-1)(m+2)}+\varepsilon\Big)$ is an exponent pair.*

For (b): [18, (7.57)] gives $\zeta(\sigma+it)\ll|t|^{(\mathsf{k}+\mathsf{l}-\sigma)/2}\log|t|$ for $\frac12\le\sigma\le\mathsf{l}-\mathsf{k}$; take $\sigma=\mathsf{l}-\mathsf{k}$.

Theorem 3.15 (Phragmén–Lindelöf principle in a strip [3, Lemma 2.5], [39, Lemma 3])**.** *Let $\sigma_1 < \sigma_2$ and $e_1 \ge e_2 \ge 0$. If $|\zeta(\sigma_i + it)| \le C_i |t|^{e_i} \log|t|$ for $|t| \ge 2$ and $i = 1, 2$, then $|\zeta(\sigma+it)| \ll |t|^{e(\sigma)} \log|t|$ uniformly for $\sigma_1 \le \sigma \le \sigma_2$ and $|t| \ge 2$, where e is the linear function with $e(\sigma_i) = e_i$.*

(Apply [3, Lemma 2.5] to the entire function $(s-1)\zeta(s)$ of order 1, with exponents $1 + e_i$ and one power of the logarithm on the two lines.)

Theorem 3.16 (Backlund's method [3, Lemmas 2.1, 2.3 and 2.4], after [39, 15])**.** *Let T be large, $\varkappa \in \mathbb{N}$, $\mathfrak{f}_\varkappa(s) = \frac{1}{2}\Big[((s+iT-1)\zeta(s+iT))^\varkappa + ((s-iT-1)\zeta(s-iT))^\varkappa\Big]$, and for a disc $D(\sigma_\circ, \varrho)$ let $\mathcal{Z}_\varkappa(\sigma_\circ, \varrho) = \frac{1}{\varkappa}\sum_z \log\frac{\varrho}{|z-\sigma_\circ|}$ over the zeros z of $\mathfrak{f}_\varkappa$ in $D(\sigma_\circ, \varrho)$. Let $1 < \sigma_\circ < \varrho$, $\sigma^\sharp = \sigma_\circ + (\sigma_\circ - \frac{1}{2})^2/\varrho$, $\delta^\sharp = 2\sigma_\circ - \sigma^\sharp - \frac{1}{2}$, and assume $0 < \delta^\sharp < \frac{1}{2}$.*

(a) If $F_\circ : [-\pi, \pi] \to \mathbb{R}$ is even and $F_\circ(\theta) \ge \frac{1}{\varkappa}\log|\mathfrak{f}_\varkappa(\sigma_\circ + \varrho e^{i\theta})|$ for all $\varkappa$ and θ, then there are $\varkappa_m \to \infty$ with $\limsup_m \mathcal{Z}_{\varkappa_m}(\sigma_\circ, \varrho) \le \log\frac{\zeta(\sigma_\circ)}{\sqrt{(\sigma_\circ-1)^2+T^2}\,\zeta(2\sigma_\circ)} + \frac{1}{\pi}\int_0^\pi F_\circ(\theta)\,d\theta$.

(b) For every $\varkappa$, the change of $\arg\Big((\sigma+iT-1)\zeta(\sigma+iT)\Big)$ by continuous variation from $\sigma = \sigma^\sharp$ to $\sigma = \frac{1}{2}$ is at most, in absolute value, $\frac{\pi\mathcal{Z}_\varkappa(\sigma_\circ,\varrho)}{2\log(\varrho/(\sigma_\circ-\frac{1}{2}))} + \frac{\pi}{100} + \frac{3\pi}{2\varkappa} + \frac{\pi}{4}$.

In (b) the term $\frac{\pi}{100}$ bounds $\frac{1}{2}E(T, \delta^\sharp)$ of [3, Lemma 2.3], by the monotonicity in $\delta^\sharp$ and the bound for $E(T, \frac{1}{2})$ in [3, Lemma 2.4]; (a) is [3, Lemma 2.1], which also requires $\sigma_\circ - \varrho < \frac{1}{2} < 1 < \sigma_\circ < \sigma^\sharp < \sigma_\circ + \varrho$, a consequence of $1 < \sigma_\circ < \varrho$.

Remark 3.17 (Known consequences for the clock). (a) *Moments and law of s_k.* Since $\vartheta(\gamma_k)/\pi \in (k + \Delta_k - 2, k + \Delta_k - 1]$ and $\vartheta/\pi = N_0 - 1 + \varpi/\pi$, one has $s_k + \Delta_k = k + \Delta_k - \frac{1}{2} - N_0(\gamma_k) \in [-\frac{1}{2} + \frac{\varpi(\gamma_k)}{\pi}, \frac{1}{2} + \frac{\varpi(\gamma_k)}{\pi})$, so $s_k = -\Delta_k + \mathfrak{d}_k$ with $|\mathfrak{d}_k| < 1$. The index set $\mathcal{I}_T$ is the integer interval $(N(T), N(2T)]$, of length $\asymp T\log T$. Cover it by consecutive windows $(n_0, n_0 + h_0]$ with $h_0 = \lfloor n_0^{27/82 + 0.95\varepsilon_0}\rfloor$, the last one shifted to end at $N(2T)$, so that it overlaps the previous one in at most $h_0 = O(T^{-1/2}\#\mathcal{I}_T)$ indices; in each window $\log\log n_0 = \log\log T + o(1)$. Summing (3.2) over the windows, and bounding the doubly counted indices by one window sum (for odd powers, by Hölder's inequality and the even moments), gives $\langle\Delta_k^{2j}\rangle_{\mathcal{I}_T} = \kappa_j(\log\log T)^j(1 + o(1))$ and $\langle\Delta_k^{2j-1}\rangle_{\mathcal{I}_T} = O_j((\log\log T)^{j-1})$. Expanding $s_k^r = (-\Delta_k + \mathfrak{d}_k)^r$ and bounding $\langle|\Delta_k|^{r-i}\rangle_{\mathcal{I}_T} \le \langle\Delta_k^{2\lceil r/2\rceil}\rangle_{\mathcal{I}_T}^{(r-i)/2\lceil r/2\rceil} \ll (\log\log T)^{(r-i)/2}$ for $1 \le i \le r$, one obtains for every fixed $j \ge 1$

$$\langle s_k^{2j}\rangle_{\mathcal{I}_T} = \kappa_j(\log\log T)^j\Big(1 + o(1)\Big), \qquad \langle s_k^{2j-1}\rangle_{\mathcal{I}_T} = O_j\Big((\log\log T)^{j-1}\Big). \tag{3.7}$$

With $\mathfrak{s}_T^2 = \frac{\log\log T}{2\pi^2}$, so that $\kappa_j(\log\log T)^j = \frac{(2j)!}{j!2^j}\mathfrak{s}_T^{2j}$, the normalized moments $\langle(s_k/\mathfrak{s}_T)^r\rangle_{\mathcal{I}_T}$ therefore tend to the moments of a standard normal variable Z for every $r \ge 1$: the even ones by the first relation, the odd ones to 0 because $(\log\log T)^{j-1}/\mathfrak{s}_T^{2j-1} \ll_j (\log\log T)^{-1/2}$. The even moments alone determine only the law of $|s_k|$; it is the odd-moment bound in (3.2) that gives the sign. The normal law is determined by its moments, so the method of moments [4, Thm. 30.2] shows that the empirical law of $s_k/\mathfrak{s}_T$ on $\mathcal{I}_T$ tends to that of Z. Only consequences for $|s_k|$ are used in this paper: $\langle s_k^2\rangle_{\mathcal{I}_T} \sim \frac{1}{2\pi^2}\log\log T$, $\limsup_k |s_k| = \infty$, and $\#\{k \in \mathcal{I}_T : |s_k| \le y_0\} = o(\#\mathcal{I}_T)$ for every fixed $y_0 > 0$; for these the even moments suffice.

(b) *Bound for δ_k.* Theorem 3.2, applied as $T \to \gamma_k^{\pm}$ (both sides have one-sided limits), gives $|N(\gamma_k^{\pm}) - N_0(\gamma_k) + \frac{7}{8}| \le \mathrm{BW}(\gamma_k)$, and (2.7) gives $-\mathrm{BW}(\gamma_k) - \frac{3}{8} \le s_k \le \mathrm{BW}(\gamma_k) - \frac{11}{8}$. As $\mathcal{L}$ is positive and increasing on $(2\pi, \infty)$ and $s_k = \int_{\gamma_k}^{\tau_k} \mathcal{L}$,

$$-\frac{\mathrm{BW}(\gamma_k) + 3/8}{\mathcal{L}(\tau_k)} \le \delta_k \le \frac{\mathrm{BW}(\gamma_k) - 11/8}{\mathcal{L}(\gamma_k)} \qquad (k \ge 1). \tag{3.8}$$

Hence $\delta_k = O(1)$, $\mathcal{L}(\tau_k)/\mathcal{L}(\gamma_k) \to 1$, and $\limsup_k |\delta_k| \le 2\pi \cdot 0.10076 = 0.6331\ldots$, an immediate consequence of [3].

4. Theorem A: an exact defect identity at the zeros

Theorem 4.1. *Let $1 \le k_1 < k_2$ be indices, F a real C^1 function on an open interval containing $[\gamma_{k_1}, \gamma_{k_2}]$, and $u_k = k - \frac{1}{2} - F(\gamma_k)$ for $k_1 \le k \le k_2$. For every integer $j \ge 1$,*

$$\int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(N(t) - F(t)\Big)^{2j} F'(t)\, dt = \sum_{k=k_1+1}^{k_2-1} \Psi_j(u_k) + \frac{(\frac{1}{2} + u_{k_1})^{2j+1} + (\frac{1}{2} - u_{k_2})^{2j+1}}{2j+1}, \tag{4.1}$$

where

$$\Psi_j(u) = \frac{(\frac{1}{2}+u)^{2j+1} + (\frac{1}{2}-u)^{2j+1}}{2j+1} = \sum_{l=0}^{j} \binom{2j+1}{2l} \frac{4^{l-j}}{2j+1}\, u^{2l} = u^{2j} + \sum_{l<j} \binom{2j+1}{2l} \frac{4^{l-j}}{2j+1} u^{2l}. \tag{4.2}$$

In particular, since $\Psi_1(u) = u^2 + \frac{1}{12}$,

$$\sum_{k=k_1+1}^{k_2-1} u_k^2 = \int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(N(t) - F(t)\Big)^2 F'(t)\, dt - \frac{k_2 - k_1 - 1}{12} - \frac{(\frac{1}{2}+u_{k_1})^3 + (\frac{1}{2}-u_{k_2})^3}{3}. \tag{4.3}$$

Proof. Existence. On $[\gamma_{k_1}, \gamma_{k_2}]$ the function N is a step function with finitely many jumps (Theorem 3.1) and F, F' are continuous; the integrand is bounded and piecewise continuous, hence Riemann integrable.

One gap. Fix k with $k_1 \le k < k_2$. If $\gamma_k < \gamma_{k+1}$, then $N(t) = k$ for $t \in (\gamma_k, \gamma_{k+1})$, because exactly k ordinates counted with multiplicity are $\le t$. The function $t \mapsto -(k - F(t))^{2j+1}/(2j+1)$ is C^1 with derivative $(k - F(t))^{2j} F'(t)$ (chain rule; no monotonicity of F is needed), so

$$\begin{aligned} \int_{\gamma_k}^{\gamma_{k+1}} \Big(k - F(t)\Big)^{2j} F'(t)\, dt &= \frac{(k - F(\gamma_k))^{2j+1} - (k - F(\gamma_{k+1}))^{2j+1}}{2j+1} \\ &= \frac{(\frac{1}{2} + u_k)^{2j+1} + (\frac{1}{2} - u_{k+1})^{2j+1}}{2j+1}, \end{aligned} \tag{4.4}$$

because $k - F(\gamma_k) = \frac{1}{2} + u_k$ and $k - F(\gamma_{k+1}) = -\frac{1}{2} + u_{k+1}$. If $\gamma_k = \gamma_{k+1}$, the left side of (4.4) is 0; then $u_{k+1} = u_k + 1$, and the right side equals $((\frac{1}{2} + u_k)^{2j+1} + (-\frac{1}{2} - u_k)^{2j+1})/(2j+1) = 0$. So (4.4) holds for every k, and the values of N at the finitely many ordinates do not affect the integrals.

Summation and boundary terms. Summing (4.4) over $k_1 \le k < k_2$ (a finite sum), each k with $k_1 < k < k_2$ contributes $(\frac{1}{2} + u_k)^{2j+1}$ from the gap starting at γ_k and $(\frac{1}{2} - u_k)^{2j+1}$ from the gap ending at γ_k, that is $\Psi_j(u_k)$; the unmatched terms are $(\frac{1}{2} + u_{k_1})^{2j+1}$ and $(\frac{1}{2} - u_{k_2})^{2j+1}$. In the binomial expansion of $(\frac{1}{2}+u)^{2j+1} + (\frac{1}{2}-u)^{2j+1}$ the odd powers of u cancel and u^{2l} has coefficient $2\binom{2j+1}{2l} 2^{2l-2j-1}$, which gives (4.2); the leading coefficient is $\binom{2j+1}{2j}/(2j+1) = 1$. $\square$

Theorem 4.2. *Let $k_1 < k_2$, F and u_k be as in Theorem 4.1, and let $\mathfrak{g} : \mathbb{R} \to \mathbb{C}$ be continuous with a primitive $\mathfrak{G}$. Then*

$$\int_{\gamma_{k_1}}^{\gamma_{k_2}} \mathfrak{g}\Big(N(t)-F(t)\Big)F'(t)\,dt = \sum_{k=k_1+1}^{k_2-1}\Big[\mathfrak{G}\Big(u_k+\tfrac{1}{2}\Big)-\mathfrak{G}\Big(u_k-\tfrac{1}{2}\Big)\Big]+\mathfrak{G}\Big(u_{k_1}+\tfrac{1}{2}\Big)-\mathfrak{G}\Big(u_{k_2}-\tfrac{1}{2}\Big). \quad (4.5)$$

Proof. The integrand is bounded and piecewise continuous, as in the proof of Theorem 4.1. If $\gamma_k < \gamma_{k+1}$, then $N = k$ on (γ_k, γ_{k+1}) and $t \mapsto -\mathfrak{G}(k - F(t))$ is C^1 with derivative $\mathfrak{g}(k - F(t))F'(t)$, so the integral over $[\gamma_k, \gamma_{k+1}]$ is $\mathfrak{G}(k - F(\gamma_k)) - \mathfrak{G}(k - F(\gamma_{k+1})) = \mathfrak{G}(u_k + \frac{1}{2}) - \mathfrak{G}(u_{k+1} - \frac{1}{2})$. If $\gamma_k = \gamma_{k+1}$, the integral is 0 and $u_{k+1} = u_k + 1$, so the same formula holds. Summing over $k_1 \le k < k_2$, each k with $k_1 < k < k_2$ contributes $\mathfrak{G}(u_k + \frac{1}{2})$ from the gap that starts at γ_k and $-\mathfrak{G}(u_k - \frac{1}{2})$ from the gap that ends there. □

For $\mathfrak{g}(x) = x^{2j}$ this is (4.1), since $\mathfrak{G}(u + \frac{1}{2}) - \mathfrak{G}(u - \frac{1}{2}) = \Psi_j(u)$.

Corollary 4.3. *For real $\omega \neq 0$,*

$$\frac{2\sin(\omega/2)}{\omega}\sum_{k=k_1+1}^{k_2-1} e^{i\omega u_k} = \int_{\gamma_{k_1}}^{\gamma_{k_2}} e^{i\omega(N(t)-F(t))}F'(t)\,dt - \frac{e^{i\omega(u_{k_1}+1/2)} - e^{i\omega(u_{k_2}-1/2)}}{i\omega}. \quad (4.6)$$

Proof. Theorem 4.2 with $\mathfrak{g}(x) = e^{i\omega x}$ and $\mathfrak{G}(x) = e^{i\omega x}/(i\omega)$, for which $\mathfrak{G}(u + \frac{1}{2}) - \mathfrak{G}(u - \frac{1}{2}) = e^{i\omega u} \cdot 2\sin(\omega/2)/\omega$. □

Corollary 4.4 (Signed law of s_k from the t-law of S)**.** *For fixed real ξ, $\langle e^{i\xi s_k/\mathfrak{s}_T}\rangle_{\mathcal{I}_T} \to e^{-\xi^2/2}$. Hence the empirical law of $s_k/\mathfrak{s}_T$ on $\mathcal{I}_T$ tends to the standard normal law.*

Proof. Let $\xi \neq 0$, $\omega = \xi/\mathfrak{s}_T$, $k_1 = N(T)$ and $k_2 = N(2T)+1$, so that $\{k_1+1, \dots, k_2-1\} = \mathcal{I}_T$. Apply (4.6) with $F = N_0$, so $u_k = s_k$ and $N - F = G$. By Theorem 3.1 the intervals $[\gamma_{k_1}, T]$ and $[2T, \gamma_{k_2}]$ have bounded length, so the integral equals $\int_T^{2T} e^{i\omega G}\mathcal{L}\,dt + O(\log T)$; the boundary term is $O(1/|\omega|) = O(\mathfrak{s}_T)$; and $2\sin(\omega/2)/\omega = 1 + O(\omega^2)$. Divide by $\#\mathcal{I}_T = \int_T^{2T} \mathcal{L}\,dt + O(\log T)$. On $[T, 2T]$, $|e^{i\omega G} - e^{i\omega S}| \le |\omega||\varpi|/\pi \ll (\mathfrak{s}_T T)^{-1}$ and $\mathcal{L}(t) = \mathcal{L}(T)(1 + O(1/\log T))$, whence

$$\langle e^{i\omega s_k}\rangle_{\mathcal{I}_T} = \frac{1}{T}\int_T^{2T} e^{i\xi S(t)/\mathfrak{s}_T}dt + O\Big(\omega^2 + \frac{1}{\log T}\Big).$$

By Theorem 3.11 and the definition of convergence in distribution, applied to the bounded continuous function $y \mapsto e^{i\xi y}$, the integral tends to $e^{-\xi^2/2}$, and $\omega \to 0$. The last statement is Lévy's continuity theorem [4, Thm. 26.3]. □

Remark 4.5. $\Psi_j(u) = \int_{u-1/2}^{u+1/2} x^{2j}dx$ is the mean of x^{2j} over the unit interval centred at u; in particular $\Psi_1(u) = u^2 + \frac{1}{12}$ and $\frac{1}{12} = \int_{-1/2}^{1/2} x^2 dx$. The constant comes from the unit jump of $N - F$ at each zero, through the primitive in Theorem 4.2; when F is not monotone, $N - F$ need not run through an interval of length one between consecutive zeros, and the identities remain exact with the integral counted with sign. With $F = N_0$, Theorem 3.3 and induction on j through the lower-degree terms of Ψ_j recover the even moments in (3.7) without Gram intervals; Corollary 4.4 gives the sign without Korolev's odd moments. The main purpose of Theorem 4.1 is Theorem 5.1, where $F = F_X$ need not be monotone and no Gram-interval structure is available.

Corollary 4.6. *For $X \geq 2$ and $1 \leq k_1 < k_2$,*

$$\sum_{k=k_1+1}^{k_2-1} u_k(X)^2 = \int_{\gamma_{k_1}}^{\gamma_{k_2}} \Big(S - S_X + \frac{\varpi}{\pi}\Big)^2 \big(\mathcal{L} + S_X'\big)dt \\ - \frac{k_2 - k_1 - 1}{12} - \frac{(\frac{1}{2} + u_{k_1}(X))^3 + (\frac{1}{2} - u_{k_2}(X))^3}{3}. \tag{4.7}$$

Proof. F_X is C^∞ on $(0,\infty)$ (Section 2.4), $F_X' = \mathcal{L} + S_X'$, and $N - F_X = S - S_X + \varpi/\pi$ by (2.3); apply Theorem 4.1 with $j = 1$. □

Lemma 4.7 (First-moment identity)**.** *Let $2\pi < t_1 < t_2$ be points which are not ordinates and $\psi \in C^1[t_1, t_2]$, real or complex. Then*

$$\sum_{t_1 < \gamma_k \leq t_2} s_k \psi(\gamma_k) = \int_{t_1}^{t_2} G(t)\psi(t)\mathcal{L}(t)\,dt + \frac{1}{2}\Big[\psi\, G^2\Big]_{t_1}^{t_2} - \frac{1}{2}\int_{t_1}^{t_2} G(t)^2\psi'(t)\,dt. \tag{4.8}$$

Proof. On $[t_1, t_2]$ there are finitely many ordinates, none at t_1 or t_2. The function $G = N - N_0$ is right-continuous and of bounded variation, C^∞ with $G' = -\mathcal{L}$ between ordinates, and it jumps by $\nu(\gamma)$ at an ordinate γ. Let $G^*(\gamma) = \frac{1}{2}(G(\gamma^-) + G(\gamma^+))$. The indices with $\gamma_k = \gamma$ are $N(\gamma^-) + 1, \ldots, N(\gamma^-) + \nu(\gamma)$, so $\sum_{\gamma_k=\gamma}(k - \frac{1}{2}) = \nu(\gamma)N(\gamma^-) + \frac{\nu(\gamma)^2}{2} = \nu(\gamma)\frac{N(\gamma^-)+N(\gamma^+)}{2}$ and, N_0 being continuous, $\sum_{\gamma_k=\gamma} s_k = \nu(\gamma)G^*(\gamma)$. The left side of (4.8) therefore equals the finite sum $\sum_{t_1<\gamma<t_2} \nu(\gamma)G^*(\gamma)\psi(\gamma)$ over distinct ordinates. As Lebesgue–Stieltjes measures on $(t_1, t_2]$,

$$d(G^2) = -2G\mathcal{L}\,dt + \sum_{t_1<\gamma<t_2} 2G^*(\gamma)\nu(\gamma)\,\delta_\gamma, \tag{4.9}$$

because $(G^2)' = 2GG' = -2G\mathcal{L}$ between ordinates and $G(\gamma^+)^2 - G(\gamma^-)^2 = (G(\gamma^+) + G(\gamma^-))\nu(\gamma) = 2G^*(\gamma)\nu(\gamma)$. Integrating the bounded Borel function ψ against (4.9) gives $\sum_{t_1<\gamma<t_2} \nu(\gamma)G^*(\gamma)\psi(\gamma) = \frac{1}{2}\int_{(t_1,t_2]} \psi\, d(G^2) + \int_{t_1}^{t_2} G\psi\mathcal{L}\,dt$. Since ψ is C^1 and G^2 is right-continuous of bounded variation and continuous at t_1 and t_2, integration by parts for Lebesgue–Stieltjes integrals [9, §3.5] gives $\int_{(t_1,t_2]} \psi\, d(G^2) = [\psi G^2]_{t_1}^{t_2} - \int_{t_1}^{t_2} G^2\psi'\,dt$. All integrands are bounded and measurable, so all integrals exist. □

5. Theorem B: how much prime information the clock needs

Theorem 5.1. *There are absolute constants T_0 and C such that for $T \geq T_0$ and every X with $2 \leq X \leq T^{1/2}$, writing $c = \log X/\log T$,*

$$\Big|\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 - \frac{1}{2\pi^2}\log\frac{1}{c}\Big| \leq C\Big(1 + \sqrt{\log(1/c)}\Big). \tag{5.1}$$

Here $\log(1/c) = \log\log T - \log\log X \in [\log 2, \log\log T - \log\log 2]$. For fixed c both the main term and the error are bounded; the theorem does not identify a limit at fixed c. It concerns the defect at the true ordinate and does not select a root of $F_X(t) = k - \frac{1}{2}$.

Corollary 5.2 (fixed cutoff)**.** *For fixed $X \geq 2$, $\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{1}{2\pi^2}\log\log T + O_X(\sqrt{\log\log T})$, and $\#\{k \in \mathcal{I}_T : |u_k(X)| \leq y_0\} = o(\#\mathcal{I}_T)$ for every fixed $y_0 > 0$.*

Proof. The first statement is (5.1). Since $|u_k(X) - s_k| \le \sigma_X$, $|u_k(X)| \le y_0$ implies $|s_k| \le y_0 + \sigma_X$; apply Remark 3.17(a). □

Corollary 5.3 (power cutoff)**.** *For fixed $0 < c \le \frac{1}{2}$ and $X = T^c$,*

$$\limsup_{T\to\infty} \frac{1}{\#\mathcal{I}_T} \sum_{k\in\mathcal{I}_T} u_k(T^c)^2 \le \frac{1}{2\pi^2}\log\frac{1}{c} + C\Big(1+\sqrt{\log(1/c)}\Big), \tag{5.2}$$

and $\#\{k \in \mathcal{I}_T : |u_k(T^c)| > w_T\} = o(\#\mathcal{I}_T)$ whenever $w_T \to \infty$. The same holds for $X = N(T)^c$ with fixed $0 < c < \frac{1}{2}$, since then $X = T^{c'}$ with $c' = c + O(\log\log T/\log T)$.

Proof. (5.1) and Chebyshev's inequality. □

Corollary 5.4 (dichotomy)**.** *Let $X = X(T)$ with $2 \le X \le T^{1/2}$ for $T \ge T_0$. Then*

$$\sup_{T\ge T_0} \frac{1}{\#\mathcal{I}_T} \sum_{k\in\mathcal{I}_T} u_k(X(T))^2 < \infty \quad \textit{if and only if} \quad \liminf_{T\to\infty} \frac{\log X}{\log T} > 0.$$

For fixed $X_0 \ge 2$ and $c \in (0, \frac{1}{2}]$,

$$\frac{1}{\#\mathcal{I}_T} \sum_{k\in\mathcal{I}_T} \Big(u_k(X_0)^2 - u_k(T^c)^2\Big) = \frac{1}{2\pi^2}\log\log T + O\Big(\sqrt{\log\log T}\Big). \tag{5.3}$$

Proof. Both directions follow from (5.1) (on a bounded range of T the means are bounded, since $|u_k(X)| \ll \log T + \sqrt{X}$), because $y \mapsto \frac{y}{2\pi^2} \pm C(1+\sqrt{y})$ is bounded on bounded sets and tends to infinity; (5.3) is the difference of the two cases of (5.1). □

Proposition 5.5 (Rounding the corrected count misassigns a positive proportion)**.** *There are absolute constants $c_* > 0$, $\kappa_0 > 0$ and T_0 such that for $T \ge T_0$ and $2 \le X \le T^{c_*}$,*

$$\frac{1}{T}\operatorname{meas}\Big\{t \in [T, 2T] : \Big\lfloor F_X(t) + \tfrac{1}{2}\Big\rfloor \ne N(t)\Big\} \ge \kappa_0.$$

Proof. If $\lfloor F_X(t) + \frac{1}{2}\rfloor = N(t)$ then $|N(t) - F_X(t)| \le \frac{1}{2}$, i.e. $|E| \le \frac{1}{2}$ with $E = E_X + \varpi/\pi$. Let $\mathfrak{m}$ be the measure of $\{t \in [T, 2T] : |E| > \frac{1}{2}\}$. Then $\int_T^{2T} E^2 \le \frac{T}{4} + (\int_T^{2T} E^4)^{1/2}\mathfrak{m}^{1/2}$. By (5.10) and (5.11) (with $|\varpi| \ll 1/T$), $\int_T^{2T} E^2 \ge \frac{T}{2\pi^2}V - CT(1+\sqrt{V})$ and $\int_T^{2T} E^4 \le CT(1+V)^2$, so $\mathfrak{m}/T \ge (\frac{V}{2\pi^2} - C(1+\sqrt{V}) - \frac{1}{4})_+^2/(C(1+V)^2)$, which is $\ge \kappa_0 > 0$ once $V \ge V_0$, that is $c \le c_*$ by Lemma 5.8. □

Rounding the corrected count to the nearest integer therefore fails to reproduce $N(t)$, for small cutoffs, on a set of t of positive density: the obstruction is the size of $S - S_X$, not the sawtooth $N - N_0$. The proposition concerns the measure of a set of t; it does not give, without a further argument on the values at the γ_k, a positive proportion of ordinates to which rounding assigns a wrong index. For c near $\frac{1}{2}$ the answer depends on the constant in (5.1), and for $c > \frac{1}{2}$ nothing is proved.

Remark 5.6. Since $\log\#\{p \le X\} = \log X + O(\log\log X)$, the condition of Corollary 5.4 says that the logarithm of the number of primes must grow proportionally to $\log T$. A fixed number of primes leaves almost all zeros violating the corrected equation by more than any fixed amount; growth of the number of primes alone is necessary but not sufficient. The range $c > \frac{1}{2}$ is not covered. In Theorem 5.1 the cutoff is constant on each window $\mathcal{I}_T$; Theorem 5.12 below allows the cutoff $X_k = (\tau_k/2\pi)^c$, which varies at every ordinate.

Remark 5.7 (Localization of the ordinates). If $M_X \le \frac{1}{2}\mathcal{L}(T)$, then $F_X' \ge \mathcal{L} - M_X \ge \frac{1}{2}\mathcal{L}(T)$ on $[T, 2T]$, so F_X is strictly increasing there. For $k \in \mathcal{I}_T$ with $F_X(T) \le k - \frac{1}{2} \le F_X(2T)$ the equation $F_X(t) = k - \frac{1}{2}$ has exactly one root $t_k^X \in [T, 2T]$ (intermediate value theorem), and $u_k(X) = F_X(t_k^X) - F_X(\gamma_k)$ gives $|\gamma_k - t_k^X| \le 2|u_k(X)|/\mathcal{L}(T)$ (mean value theorem). As $M_X \asymp \sqrt{X}$, this applies only for $X \ll \log^2 T$, and in that range the localization gains nothing in order (Proposition 7.8). For $X = T^c$, F_X need not be monotone, and Theorem 5.1 controls the equation at the true ordinates without selecting a root; Theorem 7.3 replaces the root by one Newton-type step.

5.1. Lemmas on prime polynomials

Fix $T \ge T_0$ and $2 \le X \le T^{1/2}$, let $Y_0 = T^{1/160}$, and split $S_X = S_X^{(1)} + S_X^{(2)} + S_X^{(\ge 3)}$ according to $m = 1$, $m = 2$, $m \ge 3$ in (2.10); the three series converge absolutely and uniformly. Let $\mathcal{J}$ be the set of primes in (X, Y_0) if $X < Y_0$ and in $[Y_0, X]$ if $X \ge Y_0$, and put

$$\begin{aligned} \mathcal{W}_1(t) &= -\frac{1}{\pi} \sum_{p \in \mathcal{J}} \frac{\sin(t \log p)}{\sqrt{p}}, \qquad V = \sum_{p \in \mathcal{J}} \frac{1}{p}, \\ \mathcal{W} &= S_{Y_0}^{\mathrm{pr}} - S_X = \pm \mathcal{W}_1 - S_X^{(2)} - S_X^{(\ge 3)}, \end{aligned} \tag{5.4}$$

with $+$ if $X < Y_0$ and $-$ if $X \ge Y_0$. The last identity holds because $S_X^{(1)}$ is the sum of $-\frac{1}{\pi} p^{-1/2} \sin(t \log p)$ over the primes $p \le X$, while $S_{Y_0}^{\mathrm{pr}}$ is the same sum over $p < Y_0$.

Lemma 5.8. $V = \log(1/c) + O(1)$ *uniformly for* $\log 2/\log T \le c \le \frac{1}{2}$.

Proof. By Theorem 3.10, $V = |\log\log Y_0 - \log\log X| + O(1) = |\log \frac{1}{160c}| + O(1)$. For $c \le \frac{1}{160}$ this is $\log \frac{1}{c} - \log 160 + O(1)$; for $\frac{1}{160} < c \le \frac{1}{2}$ both V and $\log \frac{1}{c}$ are $O(1)$. □

Lemma 5.9. *For real* z_n, $2 \le n \le Y$,

$$\int_T^{2T} \Big(\sum_{n \le Y} z_n \sin(t \log n) \Big)^2 dt = \frac{T}{2} \sum_{n \le Y} z_n^2 + O\Big(\sum_{n \le Y} n z_n^2 + \Big(\sum_{n \le Y} |z_n| \Big)^2 \Big), \tag{5.5}$$

and the same with cos *in place of* sin.

Proof. With the finite sum $\mathcal{D}(t) = \sum z_n n^{-it}$, $\sum z_n \sin(t \log n) = -\Im \mathcal{D}$ and $(\Im \mathcal{D})^2 = \frac{1}{2}(|\mathcal{D}|^2 - \Re \mathcal{D}^2)$. Theorem 3.9 evaluates $\int |\mathcal{D}|^2$, and $|\int_T^{2T} (n_1 n_2)^{-it} dt| \le 2/\log(n_1 n_2) \le 2/\log 4$ gives $|\int \mathcal{D}^2| \le (2/\log 4)(\sum |z_n|)^2$. For cos use $\Re \mathcal{D}$ and $(\Re \mathcal{D})^2 = \frac{1}{2}(|\mathcal{D}|^2 + \Re \mathcal{D}^2)$. □

Lemma 5.10. *With absolute implied constants,*

$$\int_T^{2T} \mathcal{W}^2 \, dt = \frac{T}{2\pi^2} V + O\Big(T(1 + \sqrt{V}) \Big), \tag{5.6}$$

$$\int_T^{2T} \mathcal{W}^4 \, dt \ll T(1 + V)^2, \tag{5.7}$$

$$\int_T^{2T} S_X'(t)^2 \, dt \ll T(1 + \log X)^2. \tag{5.8}$$

Proof. All primes involved are $\le \max(X, Y_0) \le T^{1/2}$. Lemma 5.9 with $z_p = \frac{1}{\pi\sqrt{p}}$ ($p \in \mathcal{J}$) gives $\int \mathcal{W}_1^2 = \frac{T}{2\pi^2}V + O(T^{1/2} + (\sum_{p\le T^{1/2}} p^{-1/2})^2) = \frac{T}{2\pi^2}V + O(T^{1/2})$. For $S_X^{(2)}(t) = -\frac{1}{2\pi}\sum_{p\le X} p^{-1}\sin(t\log p^2)$ the same lemma gives $\int (S_X^{(2)})^2 \ll T\sum_p p^{-2} + \#\{p \le X\} + (\sum_{p\le X} p^{-1})^2 \ll T$. Finally $|S_X^{(\ge 3)}| \le \frac{1}{\pi}\sum_p \sum_{m\ge 3} \frac{1}{mp^{m/2}} \le \frac{1}{\pi}\sum_p \frac{p^{-3/2}}{1-p^{-1/2}} < \infty$, uniformly in X. Expanding $\mathcal{W}^2$ and bounding the cross terms by Cauchy–Schwarz gives (5.6).

For (5.7) use $(y_1+y_2+y_3)^4 \le 27(y_1^4+y_2^4+y_3^4)$. Let $\mathcal{D}_1(t) = \sum_{p\in\mathcal{J}} p^{-1/2}p^{-it}$, so that $|\mathcal{W}_1| \le |\mathcal{D}_1|/\pi$ and $\mathcal{D}_1^2 = \sum_n \eta_n n^{-it}$ with $\eta_{p^2} = p^{-1}$, $\eta_{p_1p_2} = 2(p_1p_2)^{-1/2}$ for $p_1 < p_2$, all $n \le T$. Theorem 3.9 gives $\int |\mathcal{D}_1|^4 = \sum |\eta_n|^2 (T + O(n)) \ll T\sum |\eta_n|^2 \le T(1+2V^2)$. For $S_X^{(2)}$ let $\mathcal{D}_2(t) = \sum_{p\le X} p^{-1}(p^2)^{-it}$, so that $|S_X^{(2)}| \le |\mathcal{D}_2|/2\pi$ and $\mathcal{D}_2^2 = \sum_n \eta'_n n^{-it}$ over $n = (p_1p_2)^2$ with $|\eta'_n| \le 2(p_1p_2)^{-1}$; then $\sum |\eta'_n|^2 \ll 1$ and $\sum n|\eta'_n|^2 \le 4\#\{p \le X\}^2 \le 4X^2 \le 4T$. $S_X^{(\ge 3)}$ is bounded.

For (5.8), $S'_X(t) = \sum_{n\in A_X} a_n \log n \cos(t\log n)$ (termwise differentiation, Section 2.4). The terms $m = 1$ have $|z_p| = \frac{\log p}{\pi\sqrt{p}}$; Lemma 5.9 and Theorem 3.10 give $\ll T\log X \sum_{p\le X}\frac{\log p}{p} + \sum_{p\le X}\log^2 p + (\sum_{p\le X}\frac{\log p}{\sqrt{p}})^2 \ll T(1+\log X)^2 + X\log X$, and $X \le T^{1/2}$. The terms $m = 2$ have $|z_{p^2}| = \frac{\log p}{\pi p}$ and contribute $\ll T + X\log X + \log^2 X$. The terms $m \ge 3$ are bounded by $\frac{1}{\pi}\sum_p \log p \sum_{m\ge 3} p^{-m/2} < \infty$. Combine with $(y_1+y_2+y_3)^2 \le 3(y_1^2+y_2^2+y_3^2)$. □

5.2. Proof of Theorem 5.1

Proof. Integrability. On $[T, 2T]$, S is bounded (Theorem 3.1) and piecewise continuous, and S_X, S'_X, ϖ are continuous; all integrals below exist.

Second and fourth moments of E_X. Let $E_0 = S - S_{Y_0}^{\mathrm{pr}}$ and $E_X = S - S_X = E_0 + \mathcal{W}$. Lemma 3.6 with $k = 2$ and Hölder's inequality give

$$\int_T^{2T} E_0^4\,dt \ll T, \qquad \int_T^{2T} E_0^2\,dt \le T^{1/2}\Big(\int_T^{2T} E_0^4\,dt\Big)^{1/2} \ll T. \tag{5.9}$$

Let $\mathcal{E}_2 = \int_T^{2T} E_X^2\,dt$. From (5.6), (5.9) and Cauchy–Schwarz,

$$\mathcal{E}_2 = \frac{T}{2\pi^2}V + O\Big(T(1+\sqrt{V})\Big), \tag{5.10}$$

and from (5.7), (5.9) and $(y_1+y_2)^4 \le 8(y_1^4+y_2^4)$,

$$\int_T^{2T} E_X^4\,dt \ll T(1+V)^2. \tag{5.11}$$

Transfer to the ordinates. Let k_1, k_2 be the smallest and largest indices in $\mathcal{I}_T$ ($\mathcal{I}_T \ne \emptyset$ for $T \ge T_0$) and apply (4.7). By Lemma 2.2 and (2.11), $|u_k(X)| \ll \log T + \sqrt{X} \ll T^{1/4}$, so the boundary terms and $u_{k_1}(X)^2 + u_{k_2}(X)^2$ are $O(T^{3/4})$. The intervals $[T, \gamma_{k_1}]$ and $[\gamma_{k_2}, 2T]$ have bounded length (Theorem 3.1), and on them $E_X^2(\mathcal{L} + |S'_X|) \ll T^{1/2}\cdot T^{1/4}$. Since $|\varpi| \ll 1/T$, replacing $(E_X + \varpi/\pi)^2$ by E_X^2 costs $\ll T^{-1}\int_T^{2T}(|E_X|+1)(\mathcal{L}+|S'_X|)\,dt \ll (1+V)^{1/2}\log T$ by Cauchy–Schwarz, (5.10) and (5.8). Hence

$$\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \int_T^{2T} E_X^2\mathcal{L}\,dt + \int_T^{2T} E_X^2 S'_X\,dt - \frac{\#\mathcal{I}_T}{12} + O(T^{3/4}). \tag{5.12}$$

Main term and derivative term. Since $0 \le \mathcal{L}(t) - \mathcal{L}(T) \le \frac{\log 2}{2\pi}$ on $[T, 2T]$, the first integral is $\mathcal{L}(T)\mathcal{E}_2 + O(\mathcal{E}_2)$. By Cauchy–Schwarz, (5.11) and (5.8),

$$\Big|\int_T^{2T} E_X^2 S_X'\, dt\Big| \le \Big(\int_T^{2T} E_X^4\Big)^{1/2}\Big(\int_T^{2T} S_X'^2\Big)^{1/2} \ll T(1+V)(1+\log X). \tag{5.13}$$

Dividing by $\#\mathcal{I}_T = T\mathcal{L}(T) + O(T)$ and using (5.10),

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T} u_k(X)^2 = \frac{V}{2\pi^2} + O\Big(1 + \sqrt{V} + \frac{1+V}{\log T} + (1+V)\,c\Big), \tag{5.14}$$

because $\log X = c\log T$. By Lemma 5.8, $(1+V)c \ll c(1+\log\frac{1}{c}) \ll 1$ on $(0, \frac12]$; $V \ll \log\log T$ since $c \ge \log 2/\log T$, so $(1+V)/\log T = o(1)$. A second use of Lemma 5.8 replaces $\frac{V}{2\pi^2}$ by $\frac{1}{2\pi^2}\log\frac{1}{c} + O(1)$ and $\sqrt{V}$ by $\sqrt{\log(1/c)} + O(1)$, which proves (5.1). □

5.3. A cutoff varying with the index

Let $\mathcal{I}_T^\tau = \{k : T \le \tau_k < 2T\}$; by (2.5), $\#\mathcal{I}_T^\tau = N_0(2T) - N_0(T) + O(1) \asymp T\log T$.

Lemma 5.11. *Let $T \ge T_0$ and $2 \le X_- < X_+ \le (T/3)^{1/2}$ with $X_+ \le 4X_-$; let $V_\pm = \sum_{X_-<p\le X_+} p^{-1}$ and*

$$\mathcal{Q}(t) = \max_{X_-\le y\le X_+}\Big|\sum_{X_-<p\le y} p^{-1/2-it}\Big|.$$

Then $\mathcal{Q}$ is Lipschitz, $|\mathcal{Q}'(t)| \le \max_{X_-\le y\le X_+}|\sum_{X_-<p\le y}\log p\, p^{-1/2-it}|$ for almost every t, and

$$\int_{T-3}^{2T+3}\mathcal{Q}^4\,dt \ll TV_\pm^2, \qquad \int_{T-3}^{2T+3}|\mathcal{Q}'|^4\,dt \ll T(\log X_+)^4V_\pm^2, \tag{5.15}$$

$$\langle\mathcal{Q}(\gamma_k)^2\rangle_{\mathcal{I}_T} \ll V_\pm\Big(1 + \frac{\log X_+}{\log T}\sqrt{\log\log T}\Big) + \frac{X_+}{T}, \qquad \langle\mathcal{Q}(\tau_k)^2\rangle_{\mathcal{I}_T^\tau} \ll V_\pm. \tag{5.16}$$

Proof. If $(X_-, X_+]$ contains no prime, $\mathcal{Q} = 0$. Otherwise let $p_1 < \cdots < p_J$ be its primes and $P_j(t) = \sum_{i\le j} p_i^{-1/2-it}$ ($P_0 = 0$), so that $\mathcal{Q} = \max_{0\le j\le J}|P_j|$, a maximum of finitely many Lipschitz functions; hence $\mathcal{Q}$ is Lipschitz and differentiable almost everywhere. At a point t where $\mathcal{Q}$ and all $|P_j|$ are differentiable and $\mathcal{Q}(t) = |P_j(t)|$, the function $\mathcal{Q} - |P_j| \ge 0$ has a minimum, so $\mathcal{Q}'(t) = |P_j|'(t)$, and $\big||P_j|'\big| \le |P_j'|$ wherever $|P_j|$ is differentiable. This gives the bound for $\mathcal{Q}'$.

Window. Fix $\mathsf{s} \in C_c^\infty(-\frac12, \frac12)$, real, even, nonnegative and not identically 0, let $\check{\mathsf{s}}(x) = \int \mathsf{s}(\omega)e^{i\omega x}d\omega$ and $\mathsf{W} = |\check{\mathsf{s}}|^2/\check{\mathsf{s}}(0)^2$. Then $\mathsf{W} \ge 0$ is a Schwartz function with $\mathsf{W}(0) = 1$, and $\mathcal{F}W$ is supported in $[-1,1]$. Choose $0 < \mathsf{y} \le \frac14$ with $\mathsf{W} \ge \frac12$ on $[-\mathsf{y},\mathsf{y}]$ and put $\mathfrak{w}(t) = \mathsf{W}(\mathsf{y}(t - \frac32 T)/(\frac{T}{2}+3))$. Then $\mathfrak{w} \ge \frac12$ on $[T-3, 2T+3]$, $\|\mathfrak{w}\|_4^4 \asymp T$, and $\mathcal{F}\mathfrak{w}$ and $\mathcal{F}(\mathfrak{w}^4)$ are supported in $|\omega| \le 2\mathsf{y}/T$ and $|\omega| \le 8\mathsf{y}/T$.

Proof of (5.15). Let $f_j = \mathfrak{w}P_j$. Then $\mathcal{F}f_j(\omega) = \sum_{i\le j} p_i^{-1/2}\mathcal{F}\mathfrak{w}(\omega + \log p_i)$, a sum of packets supported in $|\omega + \log p_i| \le 2\mathsf{y}/T$. Since $\log p_{i+1} - \log p_i \ge \log(1 + 1/X_+) \ge (3/T)^{1/2}/2 > 4\mathsf{y}/T$ for $T \ge T_0$, the packets are disjoint, and f_j is obtained from f_J by removing the packets of $p_{j+1}, \dots, p_J$, which lie below a point ω_j separating them from

the others: $f_j(t) = f_J(t) - \frac{1}{2\pi}\int_{-\infty}^{\omega_j} \mathcal{F}f_J(\omega)e^{i\omega t}d\omega$. Hence $\max_j |f_j| \le |f_J| + \mathcal{C}f_J$, and on $[T-3, 2T+3]$, $\mathcal{Q} \le 2\max_j |f_j|$. By Theorem 3.13,

$$\int_{T-3}^{2T+3} \mathcal{Q}^4 \le 16 \int_{\mathbb{R}} \Big(|f_J| + \mathcal{C}f_J\Big)^4 \ll \int_{\mathbb{R}} \mathfrak{w}^4 |P_J|^4 \, dt.$$

Write $P_J^2 = \sum_n \eta_n n^{-it}$ with $\eta_{p^2} = p^{-1}$ and $\eta_{p_i p_{i'}} = 2(p_i p_{i'})^{-1/2}$ for $i < i'$, all $n \le X_+^2 \le T/3$ (unique factorization makes the representation unique). Expanding, $\int \mathfrak{w}^4 |P_J^2|^2 = \sum_{n,n'} \eta_n \eta_{n'} \mathcal{F}(\mathfrak{w}^4)(\log(n/n'))$, and $|\log(n/n')| \ge 1/\max(n,n') \ge 3/T > 8\mathfrak{y}/T$ for $n \ne n'$, so only the diagonal remains: $\int \mathfrak{w}^4 |P_J|^4 = \|\mathfrak{w}\|_4^4 \sum_n \eta_n^2 \ll T\Big(\sum_p p^{-2} + 4V_\pm^2\Big) \ll TV_\pm^2$, because $\sum_{X_- < p \le X_+} p^{-2} \le V_\pm/X_- \le 4V_\pm/X_+ \le 4V_\pm^2$. The same argument with the polynomials $\sum_{i \le j} \log p_i \, p_i^{-1/2-it}$, whose maximum dominates $|\mathcal{Q}'|$, gives the second bound, with η_n multiplied by $\log p_i \log p_{i'} \le (\log X_+)^2$.

At the ordinates. With the right-continuous N, the atoms of dN in $(T, 2T]$ are exactly the ordinates of $\mathcal{I}_T$, with their multiplicities, whether or not T and $2T$ are ordinates. As Lebesgue–Stieltjes measures on $(T, 2T]$, $dN = \mathcal{L}\,dt + dG$, with $G = N - N_0$ right-continuous and of bounded variation on $[T, 2T]$; since $\mathcal{Q}^2$ is continuous (indeed Lipschitz), it has no jump in common with G, and integration by parts on $(T, 2T]$ [9, §3.5] gives, with no restriction on the endpoints,

$$\sum_{k \in \mathcal{I}_T} \mathcal{Q}(\gamma_k)^2 = \int_T^{2T} \mathcal{Q}^2 \mathcal{L}\, dt + \Big[\mathcal{Q}^2 G\Big]_T^{2T} - 2\int_T^{2T} G\, \mathcal{Q}\mathcal{Q}'\, dt, \tag{5.17}$$

multiplicities included. By Cauchy–Schwarz and (5.15) the first term is $\ll \log T \cdot T^{1/2}(TV_\pm^2)^{1/2} = TV_\pm \log T$. The boundary term is $\ll X_+ \log T$, since $\mathcal{Q} \le \sum_{p \le X_+} p^{-1/2} \ll X_+^{1/2}$ and $G \ll \log T$. By Hölder's inequality with exponents $2, 4, 4$, Theorem 3.3 ($\int_T^{2T} G^2 \ll T \log\log T$) and (5.15), the last term is $\ll (T\log\log T)^{1/2}(TV_\pm^2)^{1/4}(T(\log X_+)^4 V_\pm^2)^{1/4} = T\sqrt{\log\log T}\,(\log X_+)V_\pm$. Divide by $\#\mathcal{I}_T \gg T\log T$.

At the clock points. For consecutive indices, $\tau_{k+1} - \tau_k = \int_k^{k+1} dx/\mathcal{L}(\tau_x) \ge 2\pi/\log(3T) = \mathfrak{b}_T$ while $\tau_{k+1} \le 3T$, so the intervals $\mathcal{U}_k = [\tau_k, \tau_k + \mathfrak{b}_T]$, $k \in \mathcal{I}_T^\tau$, are disjoint and contained in $[T, 2T+1]$. For an absolutely continuous g on an interval $\mathcal{U}$ of length $\mathfrak{b}$ and $x \in \mathcal{U}$, $|g(x)| \le \frac{1}{\mathfrak{b}}\int_{\mathcal{U}} |g| + \int_{\mathcal{U}} |g'|$, hence $g(x)^2 \le \frac{2}{\mathfrak{b}}\int_{\mathcal{U}} g^2 + 2\mathfrak{b}\int_{\mathcal{U}} g'^2$. With $g = \mathcal{Q}$ and summing, $\sum_{k \in \mathcal{I}_T^\tau} \mathcal{Q}(\tau_k)^2 \le \frac{2}{\mathfrak{b}_T}\int_T^{2T+1} \mathcal{Q}^2 + 2\mathfrak{b}_T \int_T^{2T+1} \mathcal{Q}'^2 \ll TV_\pm \log T + \frac{T(\log X_+)^2 V_\pm}{\log T}$, by Cauchy–Schwarz and (5.15). Divide by $\#\mathcal{I}_T^\tau \gg T\log T$ and use $\log X_+ \le \log T$. □

Theorem 5.12. *For $\frac{\log 2}{\log T} \le c \le \frac{1}{2}$ let $X_k = \max\{2, (\tau_k/2\pi)^c\}$ and $\tilde{u}_k = s_k - S_{X_k}(\gamma_k)$. Uniformly in c,*

$$\left|\langle \tilde{u}_k^2 \rangle_{\mathcal{I}_T} - \frac{1}{2\pi^2}\log\frac{1}{c}\right| \ll 1 + \sqrt{\log(1/c)}. \tag{5.18}$$

For fixed c, $\langle (\tilde{u}_k - u_k(X_-))^2 \rangle_{\mathcal{I}_T} \to 0$, with X_- as in the proof.

Proof. For $T \ge T_0$ and $k \in \mathcal{I}_T$, $|\delta_k| \le 1$ by (3.8), so $\tau_k \in [T-1, 2T+1]$ and $X_- \le X_k \le X_+$ with $X_\pm$ the values of $\max\{2, (y/2\pi)^c\}$ at $y = T-1$ and $y = 2T+1$. Then $X_+ \le ((2T+1)/2\pi)^{1/2} \le (T/3)^{1/2}$ and $X_+/X_- \le ((2T+1)/(T-1))^{1/2} \le 2$. The prime powers p^m with $X_- < p \le X_k$ are those in $S_{X_k} - S_{X_-}$, so by (2.10)

$$|S_{X_k}(\gamma_k) - S_{X_-}(\gamma_k)| \le \frac{1}{\pi}\mathcal{Q}(\gamma_k) + \frac{1}{\pi}\sum_{X_- < p \le X_+} \sum_{m \ge 2} \frac{1}{mp^{m/2}} \le \frac{1}{\pi}\mathcal{Q}(\gamma_k) + 1.1\, V_\pm,$$

since the terms $m=1$ are $\frac{1}{\pi}\Im\sum_{X_-<p\le X_k}p^{-1/2-i\gamma_k}$ and $\sum_{m\ge2}p^{-m/2}/m\le p^{-1}(1-2^{-1/2})^{-1}$. By partial summation from the first estimate of Theorem 3.10, written $\sum_{p\le y}\frac{\log p}{p}=\log y+R(y)$ with $|R|\le C_{\mathrm M}$, one has $V_\pm=\int_{X_-}^{X_+}\frac{d(\log y+R(y))}{\log y}=\log\frac{\log X_+}{\log X_-}+\Big[\frac{R(y)}{\log y}\Big]_{X_-}^{X_+}+\int_{X_-}^{X_+}\frac{R(y)\,dy}{y\log^2 y}=\log\frac{\log X_+}{\log X_-}+O\Big(\frac{1}{\log X_-}\Big)$, hence $V_\pm\ll\frac{1}{\log X_-}\ll\frac{1}{1+c\log T}$ (and $V_\pm=0$ if $X_-=X_+=2$). Lemma 5.11 gives

$$\langle(\tilde u_k-u_k(X_-))^2\rangle_{\mathcal I_T}\ll V_\pm+\frac{(1+c\log T)\sqrt{\log\log T}}{(1+c\log T)\log T}+T^{-1/2}+V_\pm^2\ll1,$$

which tends to 0 for fixed c. Theorem 5.1 with $X=X_-$, for which $\log(1/c_-)=\log(1/c)+O(1)$ with $c_-=\log X_-/\log T$, gives (5.1) for $u_k(X_-)$; (5.18) follows from $|\langle\tilde u^2\rangle-\langle u^2\rangle|\le 2\langle u^2\rangle^{1/2}\langle(\tilde u-u)^2\rangle^{1/2}+\langle(\tilde u-u)^2\rangle$. □

6. Theorem C: exact accounting for the one-step clock

For a finite nonempty set I of indices and a prime power n, define the response statistics at the zeros and at the clock points,

$$\mathcal R_n^\gamma(I)=\frac{2}{a_n}\langle s_k\sin(\gamma_k\log n)\rangle_I,\qquad \mathcal R_n^\tau(I)=\frac{2}{a_n}\langle s_k\sin(\tau_k\log n)\rangle_I,\tag{6.1}$$

and, for a set A of prime powers with $\sum_{n\in A}|a_n|<\infty$, the cross term

$$\mathrm{Cr}_A(I)=\langle S_A(\tau_k)^2\rangle_I-v_A.\tag{6.2}$$

For a prime p, $2/a_p=-2\pi\sqrt p$. These are finite-window statistics of the zeros of the Riemann zeta function, defined without any hypothesis; Cr_A contains the finite-window diagonal discrepancies and all off-diagonal products and has no fixed sign.

Theorem 6.1 (Finite-window accounting)**.** *Let I be a finite nonempty set of indices and A a set of prime powers with $\sum_{n\in A}|a_n|<\infty$.*

(a) For every k and $X\ge2$,

$$\begin{aligned}&r_k(X)=u_k(X)+\Big(S_X(\gamma_k)-S_X(\tau_k)\Big)+\chi_k,\\&\chi_k=\frac{\gamma_k}{2\pi}\Big(\frac{\tau_k}{\gamma_k}-1-\log\frac{\tau_k}{\gamma_k}\Big),\qquad\frac{\delta_k^2}{4\pi\max(\gamma_k,\tau_k)}\le\chi_k\le\frac{\delta_k^2}{4\pi\min(\gamma_k,\tau_k)},\\&|S_X(\gamma_k)-S_X(\tau_k)|\le M_X|\delta_k|.\end{aligned}\tag{6.3}$$

(b) For every real λ,

$$\Big\langle s_k^2-(s_k-\lambda S_A(\tau_k))^2\Big\rangle_I=\sum_{n\in A}a_n^2\Big(\lambda\,\mathcal R_n^\tau(I)-\frac{\lambda^2}{2}\Big)-\lambda^2\,\mathrm{Cr}_A(I).\tag{6.4}$$

In particular, for the full local factors,

$$\begin{aligned}&\langle s_k^2-(s_k-S_X(\tau_k))^2\rangle_I=D_X(I)-\mathrm{Cr}_X(I),\\&D_X(I)=\sum_{n\in A_X}a_n^2\Big(\mathcal R_n^\tau(I)-\frac12\Big)=2\langle s_kS_X(\tau_k)\rangle_I-v_X,\end{aligned}\tag{6.5}$$

with $\mathrm{Cr}_X=\mathrm{Cr}_{A_X}$.

(c) In units of zeros, the gain of the one-step clock over the bare clock is

$$\begin{aligned}&\Big\langle(\mathcal{L}(\tau_k)\delta_k)^2 - r_k(X)^2\Big\rangle_I = D_X(I) - \mathrm{Cr}_X(I) + 2\langle\chi_k S_X(\tau_k)\rangle_I,\\ &|\langle\chi_k S_X(\tau_k)\rangle_I| \le \sigma_X \max_{k\in I}\chi_k.\end{aligned} \tag{6.6}$$

Proof. (a) By (2.12), $r_k(X) = \mathcal{L}(\tau_k)\delta_k - S_X(\tau_k)$. Put $\chi_k = \mathcal{L}(\tau_k)\delta_k - s_k$; since $s_k = N_0(\tau_k) - N_0(\gamma_k) = \int_{\gamma_k}^{\tau_k}\mathcal{L}$,

$$\chi_k = \int_{\gamma_k}^{\tau_k}\Big(\mathcal{L}(\tau_k) - \mathcal{L}(y)\Big)dy = \frac{1}{2\pi}\int_{\gamma_k}^{\tau_k}\log\frac{\tau_k}{y}\,dy = \frac{\gamma_k}{2\pi}\Big(\frac{\tau_k}{\gamma_k} - 1 - \log\frac{\tau_k}{\gamma_k}\Big),$$

the last step by $\int\log y\,dy = y\log y - y$. If $\gamma_k < \tau_k$, then $\frac{\tau_k-y}{\tau_k} \le \log\frac{\tau_k}{y} \le \frac{\tau_k-y}{y} \le \frac{\tau_k-y}{\gamma_k}$ for $y\in[\gamma_k,\tau_k]$, and integration gives $\frac{\delta_k^2}{4\pi\tau_k} \le \chi_k \le \frac{\delta_k^2}{4\pi\gamma_k}$; if $\tau_k < \gamma_k$, write $\chi_k = \frac{1}{2\pi}\int_{\tau_k}^{\gamma_k}\log\frac{y}{\tau_k}\,dy$ and use $\frac{y-\tau_k}{\gamma_k} \le \frac{y-\tau_k}{y} \le \log\frac{y}{\tau_k} \le \frac{y-\tau_k}{\tau_k}$; if $\tau_k = \gamma_k$, $\chi_k = 0$. Insert $s_k = u_k(X) + S_X(\gamma_k)$. The last bound is the mean value theorem with (2.11).

(b) The series $S_A(\tau_k) = \sum_n a_n\sin(\tau_k\log n)$ converges absolutely, and so does the double series for $S_A(\tau_k)^2$; since I is finite, the mean over I may be exchanged with these sums. Expanding, the left side is $2\lambda\langle s_k S_A(\tau_k)\rangle_I - \lambda^2\langle S_A(\tau_k)^2\rangle_I$, with $\langle s_k S_A(\tau_k)\rangle_I = \sum_n a_n\langle s_k\sin(\tau_k\log n)\rangle_I = \frac{1}{2}\sum_n a_n^2\mathcal{R}_n^\tau(I)$ and $\langle S_A(\tau_k)^2\rangle_I = v_A + \mathrm{Cr}_A(I)$. Taking $\lambda = 1$ and $A = A_X$ gives (6.5).

(c) By (a), $\mathcal{L}(\tau_k)\delta_k = s_k + \chi_k$ and $r_k(X) = s_k + \chi_k - S_X(\tau_k)$, so $(\mathcal{L}(\tau_k)\delta_k)^2 - r_k(X)^2 = 2s_kS_X(\tau_k) - S_X(\tau_k)^2 + 2\chi_kS_X(\tau_k)$; average and use (6.5) and $|S_X| \le \sigma_X$. □

Remark 6.2. (6.5) is an exact sign criterion for the gain of the full Euler phase at the clock points: the gain is positive exactly when the diagonal responses outweigh the cross term. No independence of the phases $\tau_k\log p$ is assumed. A threshold $\mathcal{R}_p^\tau = \frac{1}{2}$ describes a single diagonal prime contribution and does not determine the optimal cutoff when Cr_X and the prime powers matter.

Remark 6.3 (Optimal trigonometric correction). For a finite set A of frequencies let $\varsigma_n(k) = \sin(\tau_k\log n)$, $\mathbf{Q} = (\langle\varsigma_{n_1}\varsigma_{n_2}\rangle_I)_{n_1,n_2\in A}$, $\mathbf{b} = (\langle s_k\varsigma_n(k)\rangle_I)_{n\in A}$ and $\Upsilon_{\boldsymbol{\lambda}} = \sum_n\lambda_n\varsigma_n$. Ordinary least squares gives $\langle(s_k - \Upsilon_{\boldsymbol{\lambda}}(k))^2\rangle_I = \langle s_k^2\rangle_I - 2\boldsymbol{\lambda}^{\mathsf{T}}\mathbf{b} + \boldsymbol{\lambda}^{\mathsf{T}}\mathbf{Q}\boldsymbol{\lambda}$, minimized by $\boldsymbol{\lambda} = \mathbf{Q}^\dagger\mathbf{b}$ with gain $\mathbf{b}^{\mathsf{T}}\mathbf{Q}^\dagger\mathbf{b}$; for prime powers $b_n = \frac{1}{2}a_n\mathcal{R}_n^\tau(I)$. The Euler phase is the choice $\lambda_n = a_n$, and (6.4) measures what it loses through $\mathcal{R}_n^\tau \ne 1$ and $\mathrm{Cr}_A \ne 0$.

Remark 6.4 (Clock point against zero). For a prime power n, $\sin(\gamma_k\ell_n) - \sin(\tau_k\ell_n) = \int_0^1\frac{d}{dy}\sin((\tau_k - y\delta_k)\ell_n)\,dy$, whence, exchanging the finite mean and the integral,

$$\mathcal{R}_n^\gamma(I) - \mathcal{R}_n^\tau(I) = -\frac{2\ell_n}{a_n}\int_0^1\Big\langle s_k\delta_k\cos\Big((\tau_k - y\delta_k)\ell_n\Big)\Big\rangle_I dy. \tag{6.7}$$

For fixed n and $I = \mathcal{I}_T$ the right side is $O(\langle|s_k\delta_k|\rangle_{\mathcal{I}_T}) = O(\log\log T/\log T)$ by (2.6) and Remark 3.17; for $\log n = \mathsf{q}\log T$ with fixed $0 < \mathsf{q} < 1$ the phase change $\delta_k\ell_n$ has mean square $\mathsf{q}^2\log^2 T\,\langle\delta_k^2\rangle_{\mathcal{I}_T} \sim 2\mathsf{q}^2\log\log T$, so it is not small, and the transport integral need not be small; no asymptotic statement about this difference is proved.

Remark 6.5 (Fixed prime sets). For fixed X, $|\delta_k^X - \delta_k| \le \sigma_X/\mathcal{L}(\tau_k) \to 0$ and $|u_k(X) - s_k| \le \sigma_X$. A fixed set of primes changes neither $\limsup_k|\delta_k|$ nor the unboundedness of the defect.

7. Theorem D: a computable clock that approaches the zeros

The one-step clock $\tau_k^X = \tau_k - y_k$, $y_k = S_X(\tau_k)/\mathcal{L}(\tau_k)$, uses only τ_k and the primes $p \le X$. By Theorem 6.1(a),

$$r_k(X) = u_k(X) + \mathfrak{t}_k(X) + \chi_k, \qquad \mathfrak{t}_k(X) = S_X(\gamma_k) - S_X(\tau_k), \tag{7.1}$$

where Theorem 5.1 controls $u_k(X)$ and $0 \le \chi_k \ll 1/T$ on $\mathcal{I}_T$. This section controls the transport term $\mathfrak{t}_k(X)$ in mean square, with no assumption on the joint behaviour of δ_k and the phases $\tau_k \log p$.

Lemma 7.1. *For $T \ge T_0$ and $2 \le X \le T^{1/2}$,*

$$\int_{T-3}^{2T+3} S_X'(t)^4\,dt \ll T(1+\log X)^4, \qquad \int_{T-3}^{2T+3} S_X''(t)^4\,dt \ll T(1+\log X)^8.$$

Proof. By Section 2.4, the series

$$S_X'(t) = \sum_{n\in A_X} a_n \log n \cos(t\log n), \qquad S_X''(t) = -\sum_{n\in A_X} a_n \log^2 n \sin(t\log n)$$

converge absolutely and uniformly. Split according to $m=1$, $m=2$, $m\ge 3$ in $n=p^m$ and use $(y_1+y_2+y_3)^4 \le 27(y_1^4+y_2^4+y_3^4)$. For $\nu\in\{1,2\}$ the terms $m\ge 3$ of the ν-th derivative are bounded by $\frac{1}{\pi}\sum_p\sum_{m\ge3}\frac{(m\log p)^\nu}{mp^{m/2}} < \infty$, uniformly in X and t. The terms $m=1$ are bounded by $\frac{1}{\pi}|\mathcal{D}_\nu(t)|$ with $\mathcal{D}_\nu(t) = \sum_{p\le X}\log^\nu p\; p^{-1/2-it}$; $\mathcal{D}_\nu^2 = \sum_n \eta_n n^{-it}$ over $n = pp' \le X^2 \le T$ with $\sum_n \eta_n^2 \le 2(\sum_{p\le X}\log^{2\nu}p/p)^2 \ll (1+\log X)^{4\nu}$ (Theorem 3.10 and partial summation), so Theorem 3.9 on an interval of length $T+6$ gives $\int|\mathcal{D}_\nu|^4 = \sum\eta_n^2(T+6+O(n)) \ll T(1+\log X)^{4\nu}$. The terms $m=2$ are bounded by $\frac{1}{2\pi}|\mathcal{D}_\nu'(t)|$ with $\mathcal{D}_\nu'(t) = \sum_{p\le X}(2\log p)^\nu p^{-1-2it}$; here $\mathcal{D}_\nu'^2 = \sum_n \eta_n' n^{-it}$ over $n=(pp')^2 \le X^4$, with $\sum_n\eta_n'^2 \ll 1$ and $\sum_n n\eta_n'^2 \ll (\sum_{p\le X}\log^{2\nu}p)^2 \ll X^2(1+\log X)^{4\nu} \le T(1+\log X)^{4\nu}$, so again $\int|\mathcal{D}_\nu'|^4 \ll T(1+\log X)^{4\nu}$. □

Lemma 7.2 (Transport)**.** *For $T\ge T_0$ and $2\le X\le T^{1/2}$,*

$$\big\langle \mathfrak{t}_k(X)^2\big\rangle_{\mathcal{I}_T} \ll \frac{\log\log T\,(1+\log X)^2}{\log^2 T}. \tag{7.2}$$

Proof. Fourth moment of δ_k. For $T\ge T_0$ and $k\in\mathcal{I}_T$, $|\delta_k|\le 1$ by (3.8); so the point ξ_k of (2.6) is $\ge T-1$ and $\mathcal{L}(\xi_k)\gg\log T$. By (3.7) with $j=2$,

$$\langle\delta_k^4\rangle_{\mathcal{I}_T} = \langle s_k^4\mathcal{L}(\xi_k)^{-4}\rangle_{\mathcal{I}_T} \ll \frac{(\log\log T)^2}{\log^4 T}. \tag{7.3}$$

Maximal function. Let $\varphi\in C^\infty(\mathbb{R})$ with $0\le\varphi\le 1$, $\varphi=1$ on $[T-2,2T+2]$, $\varphi=0$ outside $[T-3,2T+3]$ and $|\varphi'|\le 2$, and let $\mathfrak{h}=\varphi S_X'$. For $k\in\mathcal{I}_T$ the segment between γ_k and τ_k lies in $[T-1,2T+1]$, where $\mathfrak{h}=S_X'$, so

$$|\mathfrak{t}_k(X)| = \Big|\int_{\tau_k}^{\gamma_k}\mathfrak{h}(y)\,dy\Big| \le \int_{\tau_k-|\delta_k|}^{\tau_k+|\delta_k|}|\mathfrak{h}| \le 2|\delta_k|\,\mathcal{M}\mathfrak{h}(\tau_k) \tag{7.4}$$

(trivially if $\delta_k = 0$). The function $\mathfrak{h}$ is C^∞ with compact support. By Theorem 3.12, $g = \mathcal{M}\mathfrak{h}$ is locally absolutely continuous with $\|g\|_4 \ll \|\mathfrak{h}\|_4$ and $\|g'\|_4 \le \|\mathcal{M}(\mathfrak{h}')\|_4 \ll \|\mathfrak{h}'\|_4$; by Lemma 7.1, $\|\mathfrak{h}\|_4^4 \ll T(1+\log X)^4$ and, since $\mathfrak{h}' = \varphi' S_X' + \varphi S_X''$, $\|\mathfrak{h}'\|_4^4 \ll T(1+\log X)^8$.

Sampling at the clock points. As in the proof of Lemma 5.11, the intervals $\mathcal{U}_k = [\tau_k, \tau_k + \mathfrak{b}_T]$, $k \in \mathcal{I}_T$, are disjoint and contained in $[T-1, 2T+2]$. For $x \in \mathcal{U}$, an interval of length $\mathfrak{b}$, $|g(x)| \le \frac{1}{\mathfrak{b}} \int_{\mathcal{U}} |g| + \int_{\mathcal{U}} |g'|$; by Hölder's inequality and $(y_1+y_2)^4 \le 8(y_1^4+y_2^4)$, $g(x)^4 \le \frac{8}{\mathfrak{b}} \int_{\mathcal{U}} g^4 + 8\mathfrak{b}^3 \int_{\mathcal{U}} |g'|^4$. Summing over $k \in \mathcal{I}_T$ and dividing by $\#\mathcal{I}_T \gg T \log T$,

$$\langle g(\tau_k)^4 \rangle_{\mathcal{I}_T} \ll \frac{1}{T}\int_{\mathbb{R}} g^4 + \frac{1}{T\log^4 T}\int_{\mathbb{R}} |g'|^4 \ll (1+\log X)^4 + \frac{(1+\log X)^8}{\log^4 T} \ll (1+\log X)^4.$$

Conclusion. By (7.4) and Cauchy–Schwarz, $\langle \mathfrak{t}_k^2 \rangle \le 4\langle \delta_k^2 g(\tau_k)^2 \rangle \le 4\langle \delta_k^4 \rangle^{1/2} \langle g(\tau_k)^4 \rangle^{1/2}$; insert (7.3). □

Theorem 7.3 (Theorem D)**.** *There are absolute constants T_0 and C such that for $T \ge T_0$ and $2 \le X \le T^{1/2}$, with $c = \log X / \log T$,*

$$\left| \left\langle r_k(X)^2 \right\rangle_{\mathcal{I}_T} - \frac{1}{2\pi^2} \log \frac{1}{c} \right| \le C\Big(1 + \sqrt{\log(1/c)}\Big)\Big(1 + c^2 \log\log T\Big). \tag{7.5}$$

Proof. By Theorem 6.1(a), $0 \le \chi_k \le \delta_k^2/(4\pi(T-1)) \le 1/T$ on $\mathcal{I}_T$. Since $1 + \log X \le (1 + \frac{1}{\log 2}) c \log T$, Lemma 7.2 gives $\langle \mathfrak{t}_k^2 \rangle \ll c^2 \log\log T$, so $d_k = r_k(X) - u_k(X) = \mathfrak{t}_k + \chi_k$ has $\langle d_k^2 \rangle \ll c^2 \log\log T + T^{-2}$. Theorem 5.1 gives $\langle u_k(X)^2 \rangle \ll 1 + \log(1/c)$. Then $|\langle r_k^2 \rangle - \langle u_k^2 \rangle| \le 2\langle u_k^2 \rangle^{1/2} \langle d_k^2 \rangle^{1/2} + \langle d_k^2 \rangle \ll (1 + \sqrt{\log(1/c)})(c\sqrt{\log\log T} + T^{-1}) + c^2 \log\log T$, which is bounded by the right side of (7.5) because $y \le 1 + y^2$. Combine with (5.1). □

Corollary 7.4. *Let $X = T^{1/\sqrt{\log\log T}}$. Then*

$$\langle r_k(X)^2 \rangle_{\mathcal{I}_T} = \frac{1}{4\pi^2} \log\log\log T + O\Big(\sqrt{\log\log\log T}\Big), \tag{7.6}$$

$$\frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}_T} \Big(\tau_k^X - \gamma_k\Big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}, \tag{7.7}$$

while $\frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}_T} (\tau_k - \gamma_k)^2 = (2 + o(1)) \log\log T / \log^2 T$; the ratio is asymptotic to $\log\log\log T / (2 \log\log T)$.

Proof. Here $c = (\log\log T)^{-1/2}$, $2 \le X \le T^{1/2}$ for $T \ge T_0$, $\log(1/c) = \frac{1}{2}\log\log\log T$ and $c^2 \log\log T = 1$, so (7.5) gives (7.6). On $\mathcal{I}_T$, $\tau_k \in [T-1, 2T+1]$ and $\mathcal{L}(\tau_k) = \frac{\log T}{2\pi}(1 + O(\frac{1}{\log T}))$; divide r_k^2 by $\mathcal{L}(\tau_k)^2$. For the bare clock, $\delta_k = s_k / \mathcal{L}(\xi_k)$ with $\mathcal{L}(\xi_k) = \frac{\log T}{2\pi}(1 + O(\frac{1}{\log T}))$, and $\langle s_k^2 \rangle \sim \frac{1}{2\pi^2} \log\log T$ by (3.7). □

Theorem 7.5 (Theorem D″: the one-step clock with a fixed cutoff exponent)**.** *There are absolute constants T_0 and C such that for $T \ge T_0$ and $(\log\log T)^{-1/2} \le c \le \frac{1}{2}$, with $X = T^c$,*

$$\begin{gathered} \left\langle r_k(X)^2 \right\rangle_{\mathcal{I}_T} \le C(1 + \log\log\log T), \qquad \left\langle \mathfrak{t}_k(X)^2 \right\rangle_{\mathcal{I}_T} \le C(1 + \log\log\log T), \\ \frac{1}{\#\mathcal{I}_T} \sum_{k \in \mathcal{I}_T} \Big(\tau_k^X - \gamma_k\Big)^2 \le \frac{C(1 + \log\log\log T)}{\log^2 T}. \end{gathered} \tag{7.8}$$

In particular, for every fixed $0 < c \le \frac{1}{2}$ and $T \ge T_0(c)$ the one-step clock $\tau_k^{T^c}$ has mean-square error $\ll (1+\log\log\log T)/\log^2 T$ in the ordinates, where Lemma 7.2 alone gives $\ll (1+c^2\log\log T)/\log^2 T$.

Proof. An auxiliary cutoff. Let $Y = T^{c_Y}$ with $c_Y = (\log\log T)^{-1/2}$, so that $2 \le Y \le X \le T^{1/2}$ for $T \ge T_0$. Corollary 7.4 gives $\langle r_k(Y)^2\rangle_{\mathcal{I}_T} \ll 1+\log\log\log T$. The cutoff Y appears only in this proof.

The exact relation between the two residuals. By (2.12), $r_k(X) = \mathcal{L}(\tau_k)(\tau_k-\gamma_k) - S_X(\tau_k)$, so

$$r_k(X) = r_k(Y) - \mathfrak{D}(\tau_k), \qquad \mathfrak{D} = S_X - S_Y.$$

Moments of $\mathfrak{D}$. The function $\mathfrak{D}$ is the part of S_X with primes $Y < p \le X$. Writing it as in Section 2.4, its terms $m=1$ are $-\frac{1}{\pi}\sum_{Y<p\le X} p^{-1/2}\sin(t\log p)$; Lemma 5.9 on $[T-1,2T-2]$ and on $[T+1,2T+2]$, whose union contains $[T-1,2T+2]$, with $z_p = \frac{1}{\pi\sqrt{p}}$ gives

$$\int_{T-1}^{2T+2}\Big(\frac{1}{\pi}\sum_{Y<p\le X}\frac{\sin(t\log p)}{\sqrt{p}}\Big)^2 dt \ll T\sum_{Y<p\le X}\frac{1}{p} + X + \Big(\sum_{p\le X} p^{-1/2}\Big)^2 \ll T\Big(1+\log\frac{\log X}{\log Y}\Big),$$

by Theorem 3.10 and $X \le T^{1/2}$; the terms $m=2$ contribute $\ll T$ as in the proof of Lemma 5.10, and the terms $m \ge 3$ are bounded. Hence $\int_{T-1}^{2T+2}\mathfrak{D}^2 \ll T(1+\log(c/c_Y)) = T(1+\log(c\sqrt{\log\log T}))$. Likewise $\mathfrak{D}' = S_X' - S_Y'$ and (5.8) on the same two intervals give $\int_{T-1}^{2T+2}\mathfrak{D}'^2 \ll T(1+\log X)^2$ (the proof of (5.8) uses $X \le T^{1/2}$ only through $X\log X \ll T$, so it applies with $T \mp 1$ in place of T).

Sampling at the clock points. The intervals $\mathcal{U}_k = [\tau_k, \tau_k + \mathfrak{b}_T]$, $k \in \mathcal{I}_T$, are disjoint and contained in $[T-1,2T+2]$ (proof of Lemma 7.2). For $x \in \mathcal{U}_k$ and $y \in \mathcal{U}_k$, $\mathfrak{D}(x) = \mathfrak{D}(y) + \int_y^x \mathfrak{D}'$; averaging over y and using Cauchy–Schwarz, $\mathfrak{D}(\tau_k)^2 \le \frac{2}{\mathfrak{b}_T}\int_{\mathcal{U}_k}\mathfrak{D}^2 + 2\mathfrak{b}_T\int_{\mathcal{U}_k}\mathfrak{D}'^2$. Summing over $k \in \mathcal{I}_T$ and dividing by $\#\mathcal{I}_T \gg T\log T$, with $\mathfrak{b}_T \asymp 1/\log T$,

$$\begin{aligned}\big\langle \mathfrak{D}(\tau_k)^2\big\rangle_{\mathcal{I}_T} &\ll \frac{1}{T}\int_{T-1}^{2T+2}\mathfrak{D}^2 + \frac{1}{T\log^2 T}\int_{T-1}^{2T+2}\mathfrak{D}'^2 \\ &\ll 1+\log\big(c\sqrt{\log\log T}\big) + \frac{(1+\log X)^2}{\log^2 T} \ll 1+\log\log\log T,\end{aligned}$$

since $c \le \frac{1}{2}$ and $X \le T^{1/2}$.

Conclusion. $\langle r_k(X)^2\rangle \le 2\langle r_k(Y)^2\rangle + 2\langle \mathfrak{D}(\tau_k)^2\rangle \ll 1+\log\log\log T$, the first bound. By (7.1), $\mathfrak{t}_k(X) = r_k(X) - u_k(X) - \chi_k$, with $\langle u_k(X)^2\rangle \ll 1+\log(1/c) \le 1+\frac{1}{2}\log\log\log T$ (Theorem 5.1) and $0 \le \chi_k \le 1/T$ (Theorem 6.1(a)); this gives the second bound. The third follows from the first as in the proof of Corollary 7.4. □

Remark 7.6 (What is not proved for a fixed cutoff exponent). Theorem 7.5 replaces the factor $c^2\log\log T$ of Lemma 7.2 by $1+\log\log\log T$, the better of the two when $c^2\log\log T \ge \log\log\log T$, in particular for fixed c; but it does not give $\langle r_k(T^c)^2\rangle_{\mathcal{I}_T} = O_c(1)$, which by Remark 7.9 is equivalent to bounded mean-square transport from τ_k. The following heuristic suggests that the factor cannot be removed. For a fixed displacement h, Lemma 5.9 gives $\frac{1}{T}\int_T^{2T}(S_X(t+h)-S_X(t))^2dt \approx \frac{1}{\pi^2}\sum_{p\le X}\frac{1-\cos(h\log p)}{p} \approx \frac{1}{\pi^2}\log(h\log X)$ when $h\log X \gg 1$. The transport is such an increment with $h = |\delta_k|$, and $|\delta_k|\log T$ is of order $\sqrt{\log\log T}$ for most k

(Remark 3.17(a)), so $h\log X \approx c\sqrt{\log\log T}$; the primes that drive the increment are those between $\exp(\log T/\sqrt{\log\log T})$ and X, while δ_k is governed mainly by the small primes. If the displacement δ_k and these large-prime phases could be treated as independent, the transport would have mean square $\approx \frac{1}{2\pi^2}\log(c^2\log\log T)$, of the order of the upper bound (7.8). The step from the fixed displacement to δ_k, which depends on the same phases and is sampled at the ordinates, is not justified; no lower bound for the transport is proved here. For the same reason the scale $c \asymp (\log\log T)^{-1/2}$ of Corollary 7.4 is the one allowed by the available estimates, not a proved minimizer of the true error. The regularized points of Theorem 9.6 avoid the transport from τ_k altogether.

Corollary 7.7 (A cutoff read off the clock)**.** *For $\tau_k > e^e$ let $X_k^{\mathrm{loc}} = \tau_k^{1/\sqrt{\log\log\tau_k}}$ and $\tau_k^{\mathrm{loc}} = \tau_k - S_{X_k^{\mathrm{loc}}}(\tau_k)/\mathcal{L}(\tau_k)$. Then*

$$\frac{1}{\#\mathcal{I}_T^\tau}\sum_{k\in\mathcal{I}_T^\tau}\Big(\tau_k^{\mathrm{loc}}-\gamma_k\Big)^2 = \frac{\log\log\log T + O(\sqrt{\log\log\log T})}{\log^2 T}.$$

Proof. Let $X_0 = T^{1/\sqrt{\log\log T}}$ and $X_0' = (2T)^{1/\sqrt{\log\log 2T}}$. The function $y\mapsto y/\sqrt{\log y}$ is increasing for $y > e^{1/2}$, so $X_0 \le X_k^{\mathrm{loc}} \le X_0'$ for $k\in\mathcal{I}_T^\tau$, and $\log X_0' - \log X_0 \le \log 2/\sqrt{\log\log T} = o(1)$; hence $X_0' \le 2X_0$ and $X_0' \le (T/3)^{1/2}$ for $T \ge T_0$. As in the proof of Theorem 5.12, $|S_{X_k^{\mathrm{loc}}}(\tau_k) - S_{X_0}(\tau_k)| \le \frac{1}{\pi}\mathcal{Q}(\tau_k) + 1.1V_\pm$ with $X_\pm$ replaced by X_0, X_0', and Lemma 5.11 gives a mean square over $\mathcal{I}_T^\tau$ that is $\ll V_\pm \ll 1/\log X_0 = \sqrt{\log\log T}/\log T$. The sets $\mathcal{I}_T$ and $\mathcal{I}_T^\tau$ differ by $O(\log T)$ indices, those with γ_k or τ_k within distance 1 of T or $2T$, and for them $|r_k(X_0)| \le |s_k| + \chi_k + \sigma_{X_0} \ll \log T + X_0^{1/2}$ (Lemma 2.2); so the means of $r_k(X_0)^2$ over the two sets differ by $O(T^{-1/2})$, and (7.6) holds over $\mathcal{I}_T^\tau$. Finally $\mathcal{L}(\tau_k)(\tau_k^{\mathrm{loc}} - \gamma_k) = r_k(X_0) - (S_{X_k^{\mathrm{loc}}}(\tau_k) - S_{X_0}(\tau_k))$; conclude with Cauchy–Schwarz and $\mathcal{L}(\tau_k) = \frac{\log T}{2\pi}(1 + O(\frac{1}{\log T}))$. □

Proposition 7.8 (The root of the corrected equation in the monotone regime)**.** *Let $X = X(T) \ge 2$ with $M_X \le \frac12\mathcal{L}(T)$, and let $\mathcal{I}_T'$ be the set of $k\in\mathcal{I}_T$ with $F_X(T) \le k - \frac12 \le F_X(2T)$, so that the root $t_k^X \in [T, 2T]$ of Remark 5.7 exists. Then $X \ll \log^2 T$, $\#(\mathcal{I}_T \setminus \mathcal{I}_T') \ll \log T$, and*

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T'}\Big(t_k^X-\gamma_k\Big)^2 \ge \frac{4}{9\mathcal{L}(2T)^2}\Big(\frac{1}{2\pi^2}\log\log T - O\Big(\sqrt{\log\log T}\Big)\Big) \gg \frac{\log\log T}{\log^2 T}.$$

Proof. $M_X \ge \frac{1}{\pi}\sum_{p\le X}\log p/\sqrt{p} \gg \sqrt{X}$ by Chebyshev's estimate, so $X \ll \mathcal{L}(T)^2 \ll \log^2 T$ and $c = \log X/\log T \le (2\log\log T + O(1))/\log T$. On $[T, 2T]$, $0 < F_X' \le \mathcal{L} + M_X \le \frac32\mathcal{L}(2T)$, so $|u_k(X)| = |F_X(t_k^X) - F_X(\gamma_k)| \le \frac32\mathcal{L}(2T)|t_k^X - \gamma_k|$ for $k\in\mathcal{I}_T'$. An index $k\in\mathcal{I}_T$ is missing from $\mathcal{I}_T'$ only if $k - \frac12 < F_X(T)$ or $k - \frac12 > F_X(2T)$; since $N(T) < k \le N(2T)$ and $|F_X - N| \le |G| + \sigma_X \ll \log T$ at T and $2T$, there are $O(\log T)$ such k, each with $|u_k(X)| \ll \log T$ (Lemma 2.2). Hence $\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T'} u_k(X)^2 = \langle u_k(X)^2\rangle_{\mathcal{I}_T} + O(\frac{\log^3 T}{T\log T})$, and Theorem 5.1 with $\log(1/c) \ge \log\log T - \log(2\log\log T + O(1))$ bounds it from below. □

Thus, in the regime $M_X \le \frac12\mathcal{L}(T)$ where the corrected equation is monotone on the whole window, its root gains nothing in order: the error of t_k^X in the ordinates has mean square of the same order $\log\log T/\log^2 T$ as that of τ_k. The proposition says nothing about roots at larger cutoffs selected by a local criterion. Theorem 7.3 avoids roots and takes one Newton-type step instead.

Remark 7.9 (The Newton term and the restriction on c). Let ξ_k^X be a point between γ_k and τ_k with $\mathfrak{t}_k(X) = -S'_X(\xi_k^X)\delta_k$ (mean value theorem). Since $\delta_k = (r_k(X) + S_X(\tau_k))/\mathcal{L}(\tau_k)$ by (2.12), (7.1) becomes the exact relation

$$r_k(X)\Big(1 + \frac{S'_X(\xi_k^X)}{\mathcal{L}(\tau_k)}\Big) = u_k(X) + \chi_k - y_k\, S'_X(\xi_k^X). \tag{7.9}$$

The term $y_k S'_X(\xi_k^X)$ is the second-order term of Newton's method. For fixed c, Theorem 5.1 gives $\|u(X)\|_{2,\mathcal{I}_T} = O_c(1)$, and $\|\chi\|_{2,\mathcal{I}_T} \ll 1/T$; by (7.1) and the triangle inequality, $\|r(X)\|_{2,\mathcal{I}_T} = O_c(1)$ holds if and only if $\|\mathfrak{t}(X)\|_{2,\mathcal{I}_T} = O_c(1)$. A transport of unbounded mean square cannot be compensated by the defect, which is bounded in mean square. Lemma 7.2 gives the upper bound $\langle \mathfrak{t}_k(X)^2\rangle_{\mathcal{I}_T} \ll c^2 \log\log T$, and Theorem 7.5 below gives $\ll 1 + \log\log\log T$ for $c \geq (\log\log T)^{-1/2}$, the better bound for fixed c; whether the transport is bounded for fixed c is not decided here (Remark 7.6). The denominator in (7.9) can vanish only when $S'_X(\xi_k^X) = -\mathcal{L}(\tau_k)$; its sign cannot be decided from $F'_X(\tau_k)$ alone, because the derivative here is evaluated at the intermediate point ξ_k^X. The construction (2.12) does not divide by this denominator or select a root of F_X.

8. Iterating the Newton-type step

Theorem 7.3 takes one Newton-type step from τ_k. Its restriction $c \leq (\log\log T)^{-1/2}$ comes from the available bound (Lemma 7.2) for the transport over the interval between τ_k and γ_k, whose length is that of δ_k. After one step the error is smaller, so a second step can use a larger cutoff. This section proves the second step and then every further step, with constants uniform in the number of steps (Section 8.5).

8.1. The exact recursion

For a sequence of cutoffs $X_1, X_2, \dots \geq 2$ define $x_k^{(0)} = \tau_k$ and

$$x_k^{(j)} = x_k^{(j-1)} - \frac{N_0(x_k^{(j-1)}) + S_{X_j}(x_k^{(j-1)}) - (k - \frac{1}{2})}{\mathcal{L}(x_k^{(j-1)})}, \qquad e_{j,k} = x_k^{(j)} - \gamma_k, \tag{8.1}$$

so that $x_k^{(1)} = \tau_k^{X_1}$ and $e_{0,k} = \delta_k$. For a finite index set I and $r \geq 1$ let $\|a\|_{r,I} = \langle |a_k|^r\rangle_I^{1/r}$.

Proposition 8.1 (Exact recursion)**.** *Let $j \geq 1$ and suppose $x_k^{(j-1)} > 2\pi$. Then*

$$\mathcal{L}(x_k^{(j-1)})\, e_{j,k} = u_k(X_j) + \Big(S_{X_j}(\gamma_k) - S_{X_j}(x_k^{(j-1)})\Big) + \chi_{j,k}, \qquad 0 \leq \chi_{j,k} \leq \frac{e_{j-1,k}^2}{4\pi \min(\gamma_k, x_k^{(j-1)})}. \tag{8.2}$$

Let $\varphi_j \in C^\infty(\mathbb{R})$, $0 \leq \varphi_j \leq 1$, be 1 on an interval containing all segments between γ_k and $x_k^{(j-1)}$, $k \in I$, and $\mathfrak{g}_j = \mathcal{M}(\varphi_j S'_{X_j})$. Then for every $r \geq 1$,

$$\begin{aligned} &\Big\|\mathcal{L}(x^{(j-1)})e_j\Big\|_{2r,I} \leq \|u(X_j)\|_{2r,I} + 2\,\|e_{j-1}\|_{4r,I}\,\|\mathfrak{g}_j(x^{(j-1)})\|_{4r,I} + \|\chi_j\|_{2r,I}, \\ &\|\chi_j\|_{2r,I} \leq \frac{\|e_{j-1}\|_{4r,I}^2}{4\pi \min_{k\in I} \min(\gamma_k, x_k^{(j-1)})}. \end{aligned} \tag{8.3}$$

Proof. Write $x = x_k^{(j-1)}$. By (8.1), $\mathcal{L}(x)e_{j,k} = \mathcal{L}(x)(x-\gamma_k) - N_0(x) - S_{X_j}(x) + k - \frac12$. As in the proof of Theorem 6.1(a), $N_0(x) - N_0(\gamma_k) = \int_{\gamma_k}^x \mathcal{L} = \mathcal{L}(x)(x-\gamma_k) - \chi_{j,k}$ with $\chi_{j,k} = \int_{\gamma_k}^x (\mathcal{L}(x)-\mathcal{L}(y))\,dy$, which lies between $(x-\gamma_k)^2/(4\pi\max(\gamma_k,x))$ and $(x-\gamma_k)^2/(4\pi\min(\gamma_k,x))$ by the same estimates of log. Hence $\mathcal{L}(x)e_{j,k} = k - \frac12 - N_0(\gamma_k) - S_{X_j}(x) + \chi_{j,k} = u_k(X_j) + S_{X_j}(\gamma_k) - S_{X_j}(x) + \chi_{j,k}$. For (8.3), $|S_{X_j}(\gamma_k) - S_{X_j}(x)| \le \int_{x-|e_{j-1,k}|}^{x+|e_{j-1,k}|} |\varphi_j S'_{X_j}| \le 2|e_{j-1,k}|\,\mathtt{g}_j(x)$, and Minkowski's and Hölder's inequalities in $\ell^{2r}(I)$. □

The recursion (8.3) is exact and uses no independence between the errors and the prime phases. It reduces the error of each step to three quantities: the moments of the defect $u_k(X_j)$ at the true ordinates (Theorem 5.1 and its higher-moment versions), the moments of the previous error, and the moments of the maximal function $\mathtt{g}_j$ sampled at the computable points $x_k^{(j-1)}$. The last is the only new difficulty: the points $x_k^{(j-1)}$ are not separated like the τ_k.

8.2. Moments of order $2r$ of the defect

Lemma 8.2. *There is an absolute constant C such that for every integer $r \ge 1$, every $T \ge T_0$ with $2r \le \kappa_{\mathrm F}\log T$ and every $2 \le X \le T^{1/(160r)}$, with V as in Lemma 5.8 computed with $Y_{2r} = T^{1/(160r)}$ in place of Y_0,*

$$\int_T^{2T} E_X^{4r}\,dt \le \Big(C(r^2+\sqrt{r(1+V)})\Big)^{4r}T, \qquad \langle u_k(X)^{2r}\rangle_{\mathcal{I}_T} \le \Big(C(r^2+\sqrt{r(1+V)})\Big)^{2r},$$

and for every $r \ge 1$, $2 \le X \le T^{1/(2r)}$ and every interval $\mathcal{J} \subset [\frac{T}{2}, 3T]$ of length at most $2T$ (in particular $[T-3, 2T+3]$ and $\mathcal{J}_T^\sharp = [T-3T^{1/2}, 2T+3T^{1/2}]$),

$$\int_{\mathcal{J}} |S_X^{(\nu)}(t)|^{4r}dt \le (Cr)^{2r}\,T(1+\log X)^{4r\nu} \qquad (\nu = 1,2).$$

Proof. The first bound: $E_X = E_0 + \mathcal{W}$ with $E_0 = S - S^{\mathrm{pr}}_{Y_{2r}}$, $\|E_0\|_{L^{4r}(T,2T)} \le C(2r)^2T^{1/4r}$ by Lemma 3.6 with $k = 2r$ (admissible since $2r \le \kappa_{\mathrm F}\log T$), and $\mathcal{W}$ as in Lemma 5.10: $|\mathcal{W}_1| \le |\mathcal{D}_1|/\pi$, $\mathcal{D}_1^{2r} = \sum_n \eta_n n^{-it}$ over products of $2r$ primes $\le \max(X, Y_{2r})$, so $n \le T^{1/80}$, with $\sum|\eta_n|^2 \le (2r)!\,(V+1)^{2r}$ (each n has at most $(2r)!$ ordered factorizations, so $\sum_n|\eta_n|^2 \le (2r)!\sum_{p_1,\dots,p_{2r}}\prod p_i^{-1}$); Theorem 3.9 gives $\int|\mathcal{D}_1|^{4r} \le 2T\sum|\eta_n|^2$, whence $\|\mathcal{W}_1\|_{L^{4r}} \le C\sqrt{r}\sqrt{1+V}\,T^{1/4r}$ by $((2r)!)^{1/4r} \le \sqrt{2r}$. The term $m=2$ is at most $C\sqrt{r}\,T^{1/4r}$ in the same norm (products of $2r$ squares of primes, $\sum|\eta_n'|^2 \le (2r)!\,(\sum_p p^{-2})^{2r} \le (2r)!$, and $((2r)!)^{1/4r} \le \sqrt{2r}$), and the terms $m \ge 3$ form a bounded function, $O(T^{1/4r})$ in that norm; the factor $\sqrt{r}$ is absorbed by Cr^2. Minkowski's inequality in L^{4r} gives the first bound. The second: Theorem 4.1 with $j = r$ and $F = F_X$, $\Psi_r(u) \ge u^{2r}$, gives $\sum_{k\in\mathcal{I}_T}u_k^{2r} \le \int_T^{2T}(|E_X|+T^{-1})^{2r}(\mathcal{L}+|S'_X|)\,dt + C_{\mathrm b}\Big(4^rT^{1/100}+(2\log T)^{2r+1}\Big)$ with an absolute $C_{\mathrm b}$, the last term bounding the boundary terms and the end pieces, since $|u_k(X)| \le \log T + \sqrt{X}$ (Lemma 2.2 and (2.11)) and $X^{r+1/2} \le T^{1/100}$. Since $r \le \log T/(160\log 2)$, $4^rT^{1/100} \le T^{1/80+1/100} \le T^{1/2}$. For $(2\log T)^{2r+1}$ there are two cases: if $r \ge \sqrt{2\log T/C}$ then $2\log T \le Cr^2$ and $(2\log T)^{2r+1} \le (Cr^2)^{2r}\,2\log T \le (Cr^2)^{2r}\,T\log T$; if $r < \sqrt{2\log T/C}$ then $(2\log T)^{2r+1} \le \exp\Big(3\sqrt{2\log T/C}\,\log(2\log T)\Big) \le T^{1/2}$ for $T \ge T_0$. In both cases the boundary term is at

most $C_{\mathrm{b}}(Cr^2)^{2r}(T^{1/2}+T\log T)\le 9\pi C_{\mathrm{b}}(Cr^2)^{2r}\,\#\mathcal{I}_T$ for $T\ge T_0$. With $\mathcal{E}_{4r}=\int_T^{2T}E_X^{4r}$, Hölder's inequality gives $\int_T^{2T}E_X^{2r}\le T^{1/2}\mathcal{E}_{4r}^{1/2}$ and, by (5.8), $\int_T^{2T}E_X^{2r}|S_X'|\le \mathcal{E}_{4r}^{1/2}C_{\mathrm{d}}T^{1/2}(1+\log X)$, C_{d} the constant of (5.8); so the integral is at most $(\mathcal{L}(2T)+C_{\mathrm{d}}(1+\log X))T^{1/2}(\mathcal{E}_{4r}^{1/2}+T^{-2r+1/2})$. Dividing by $\#\mathcal{I}_T\ge\frac12 T\mathcal{L}(T)$ and using $\mathcal{L}(2T)\le 2\mathcal{L}(T)$ and $1+\log X\le 2\pi\mathcal{L}(T)$, $\langle u_k(X)^{2r}\rangle_{\mathcal{I}_T}\le (4+4\pi C_{\mathrm{d}})T^{-1/2}\mathcal{E}_{4r}^{1/2}+(9\pi C_{\mathrm{b}}+1)(Cr^2)^{2r}$, and the first bound gives the second with $(9\pi C_{\mathrm{b}}+2)C$ in place of C, since $(9\pi C_{\mathrm{b}}+2)^{2r}\ge 9\pi C_{\mathrm{b}}+2$. The third: the proof of Lemma 7.1 on $\mathcal{J}$ in place of $[T-3,2T+3]$ (Theorem 3.9 on an interval of length $H\le 2T$ gives $\sum|\eta_n|^2(H+O(n))\le(2T+O(T))\sum|\eta_n|^2$ for $n\le T$), with products of $2r$ primes, $n\le X^{2r}\le T$, and $\sum\eta_n^2\le(2r)!\,(\sum_{p\le X}\log^{2\nu}p/p)^{2r}\le(Cr)^{2r}(1+\log X)^{4r\nu}$, by Theorem 3.10 and partial summation; the terms $m=2$ and $m\ge 3$ as in Lemma 7.1. □

The dependence on r matters in Theorem 8.11, where r grows with the number of Newton-type steps; for fixed r the three bounds read $\ll_r T(1+V)^{2r}$, $\ll_r (1+V)^r$ and $\ll_r T(1+\log X)^{4r\nu}$.

The moments of all orders give a tail bound for the defect that no second-moment argument can give: the proportion of zeros whose corrected index is off by more than λ positions decays exponentially in $\sqrt{\lambda}$, provided the cutoff is chosen in a range that depends on λ.

Theorem 8.3 (Exponential tails of the ordinal defect)**.** *There are absolute constants $C_8\ge 1$, λ_0 and T_0 such that, for $T\ge T_0$, every $\lambda\ge\lambda_0$ and every $X=T^c\ge 2$ with*

$$\exp\bigl(-(\lambda/C_8)^{3/2}\bigr)\le c\le\frac{1}{C_8\sqrt{\lambda}},\tag{8.4}$$

$$\#\{k\in\mathcal{I}_T:|u_k(X)|>\lambda\}\le 2^{\,2-\sqrt{\lambda/C_8}}\,\#\mathcal{I}_T.\tag{8.5}$$

Proof. Let C be the constant of Lemma 8.2 and $r=\lfloor\sqrt{\lambda/(4C)}\rfloor$, so $r\ge 1$ for $\lambda\ge 4C$ and $Cr^2\le\lambda/4$. The lemma applies (we may take $C\ge 1$). First, $X\le T^{1/(160r)}$ because $c\le 1/(C_8\sqrt{\lambda})\le 1/(160r)$ once $C_8\ge 80$, since $160r\le 160\sqrt{\lambda/(4C)}\le 80\sqrt{\lambda}$. Second, $2r\le\kappa_{\mathrm{F}}\log T$: indeed $2r\le\sqrt{\lambda/C}\le\sqrt{\lambda}$, while $X=T^c\ge 2$ gives $\log T\ge\log 2/c\ge C_8\log 2\,\sqrt{\lambda}\ge\sqrt{\lambda}/\kappa_{\mathrm{F}}$ once $C_8\ge 300=1/(\kappa_{\mathrm{F}}\log 2)$. With $Y_{2r}=T^{1/(160r)}\ge X$, Lemma 5.8 gives $V=\log(1/(160rc))+O(1)\le\log(1/c)+C_9\le(\lambda/C_8)^{3/2}+C_9$, with an absolute constant $C_9\ge 0$, by (8.4), so $1+V\le 2(\lambda/C_8)^{3/2}$ for $\lambda\ge C_8(C_9+1)^{2/3}$. Then $r(1+V)\le\sqrt{\lambda/(4C)}\cdot 2\lambda^{3/2}C_8^{-3/2}=\lambda^2/(\sqrt{C}\,C_8^{3/2})$ and $C\sqrt{r(1+V)}\le C^{3/4}C_8^{-3/4}\lambda\le\lambda/4$ once $C_8\ge 7C$. Hence $C(r^2+\sqrt{r(1+V)})\le\lambda/2$, and Chebyshev's inequality with the second bound of the lemma gives

$$\#\{k\in\mathcal{I}_T:|u_k(X)|>\lambda\}\le\lambda^{-2r}\sum_{k\in\mathcal{I}_T}u_k(X)^{2r}\le 2^{-2r}\,\#\mathcal{I}_T\le 2^{\,2-\sqrt{\lambda/C}}\,\#\mathcal{I}_T,$$

since $2r\ge 2\sqrt{\lambda/(4C)}-2$. Take $C_8=\max(300,7C)$ and $\lambda_0=\max\bigl(4C,C_8(C_9+1)^{2/3}\bigr)$; then every condition used above holds for $\lambda\ge\lambda_0$, and $2^{-\sqrt{\lambda/C}}\le 2^{-\sqrt{\lambda/C_8}}$. □

For the cutoff $X = T^{1/(C_8\sqrt{\lambda})}$, Theorem 5.1 gives $\langle u_k(X)^2\rangle_{\mathcal{I}_T} = \frac{1}{4\pi^2}\log\lambda + O(\sqrt{\log\lambda})$, so Chebyshev's inequality alone bounds the proportion in (8.5) by $\lambda^{-2}\log\lambda$; the moments of order $2r \asymp \sqrt{\lambda}$ replace this by $2^{2-\sqrt{\lambda/C_8}}$. The bound (8.5) is for cutoffs that depend on λ; for a fixed cutoff exponent c it applies only to $\lambda \le (C_8 c)^{-2}$, and no exponential tail is claimed beyond that. The range (8.4) is the price of the moment order: Lemma 8.2 needs $X \le T^{1/(160r)}$, and Lemma 3.6 with $2r$ in place of k needs the prime polynomial of length $T^{1/(160r)}$ and the order $2r \le \kappa_{\mathrm{F}}\log T$.

8.3. Multiplicity of the one-step clock points

Let $X_1 = T^{c_1}$ with $c_1 = (\log\log T)^{-1/2}$, $x_k^{(1)} = \tau_k^{X_1} = \tau_k - y_k$ with $y_k = S_{X_1}(\tau_k)/\mathcal{L}(\tau_k)$, $\mathfrak{b}_T = 2\pi/\log(3T)$ as in Lemma 5.11, and

$$\tilde{n}(u) = \#\{k \in \mathcal{I}_T :\ u - \mathfrak{b}_T \le x_k^{(1)} \le u\} \qquad (u \in \mathbb{R}).$$

Note that $|y_k| \le \sigma_{X_1}/\mathcal{L}(\tau_k) \ll T^{c_1/2}$, so $x_k^{(1)} \in [T - T^{1/2}, 2T + T^{1/2}]$ for $T \ge T_0$.

Lemma 8.4 (Multiplicity)**.** $\displaystyle\int_{\mathbb{R}} \tilde{n}(u)^2\,du = \sum_{j,k\in\mathcal{I}_T} \Big(\mathfrak{b}_T - |x_j^{(1)} - x_k^{(1)}|\Big)_+ \ll \mathfrak{b}_T\,\#\mathcal{I}_T.$

Proof. The identity: $\tilde{n} = \sum_k \mathbf{1}_{[x_k^{(1)}, x_k^{(1)}+\mathfrak{b}_T]}$, and the measure of the intersection of two such intervals is $(\mathfrak{b}_T - |x_j^{(1)} - x_k^{(1)}|)_+$. Write $b = \mathfrak{b}_T$ and $\mathsf{M} = \#\mathcal{I}_T$. The pairs with $|j-k| \le 1$ contribute at most $3b\mathsf{M}$. Let $j = k+m$ with $m \ge 2$ and $|x_{k+m}^{(1)} - x_k^{(1)}| \le b$. Since $\tau_{k+m} - \tau_k \ge mb$ (proof of Lemma 5.11), this forces

$$|y_{k+m} - y_k| \ge (m-1)b. \tag{8.6}$$

Two consequences of (8.6) are counted separately; a close pair at distance m satisfies both.

Variation. $y = S_{X_1}/\mathcal{L}$ is C^∞ with $y' = S'_{X_1}/\mathcal{L} - S_{X_1}\mathcal{L}'/\mathcal{L}^2$, and on $J_{k,m} = [\tau_k, \tau_{k+m}]$, of length $|J_{k,m}| \le mb(1+o(1))$,

$$|y_{k+m} - y_k| \le \frac{1+o(1)}{\mathcal{L}(T)}\int_{J_{k,m}} |S'_{X_1}| + \frac{\sigma_{X_1}|J_{k,m}|}{2\pi(T-1)\mathcal{L}(T)^2},$$

and the last term is $\ll T^{c_1/2-1}mb/\log^2 T = o((m-1)b)$ uniformly in $m \ge 2$. As $b\mathcal{L}(T) = 1 - o(1)$, (8.6) gives $\frac{1}{|J_{k,m}|}\int_{J_{k,m}} |S'_{X_1}| \ge \frac{1}{3}\mathcal{L}(T)$ for $T \ge T_0$. Each t lies in at most m of the intervals $J_{k,m}$ (k varying), and $|J_{k,m}| \ge mb$; by Chebyshev's inequality and Jensen's inequality,

$$\begin{aligned}\#\{k : \text{variation}\} &\le \frac{9}{\mathcal{L}(T)^2}\sum_k \frac{1}{|J_{k,m}|}\int_{J_{k,m}} |S'_{X_1}|^2 \\ &\le \frac{9}{\mathcal{L}(T)^2\,b}\int_{T-1}^{2T+1} |S'_{X_1}|^2 \ll T\log T\,c_1^2 \asymp \mathsf{M}\,c_1^2.\end{aligned}$$

by (5.8), uniformly in m.

Size. (8.6) gives $\max(|y_k|, |y_{k+m}|) \ge \frac{m-1}{2}b$, hence $\max(|S_{X_1}(\tau_k)|, |S_{X_1}(\tau_{k+m})|) \ge \frac{m-1}{3}$ for $T \ge T_0$. Sampling at the clock points as in the proof of Lemma 5.11 (Sobolev on the disjoint intervals $[\tau_k, \tau_k + b]$), $\langle S_{X_1}(\tau_k)^4\rangle_{\mathcal{I}_T} \ll \frac{1}{T}\int S_{X_1}^4 + \frac{1}{T\log^4 T}\int |S'_{X_1}|^4 \ll (1+\log\log T)^2$, the first

integral by Theorem 3.9 with products of two primes ($\sum_{p\le X_1} p^{-1} \ll \log\log T$) and the second by Lemma 7.1. By Chebyshev's inequality of order 4, $\#\{k : \text{size}\} \ll \mathsf{M}(\log\log T)^2/(m-1)^4$.

Sum. With $H_0 = \log\log T$ and $c_1^2 = H_0^{-1}$, the number of close ordered pairs at distance $m \ge 2$ is $\ll \mathsf{M}\min(H_0^{-1}, H_0^2(m-1)^{-4})$, and

$$\sum_{m\ge 2} \min\Big(H_0^{-1}, \frac{H_0^2}{(m-1)^4}\Big) \le \sum_{m-1\le H_0^{3/4}} H_0^{-1} + \sum_{m-1> H_0^{3/4}} \frac{H_0^2}{(m-1)^4} \ll H_0^{-1/4}.$$

Hence the number of pairs (j,k) with $|x_j^{(1)} - x_k^{(1)}| \le b$ is $\ll \mathsf{M}$, and each contributes at most b. □

Lemma 8.5 (Sampling at points of bounded multiplicity)**.** *Let x_k, $k \in \mathcal{I}_T$, be real points of $[T - 2T^{1/2}, 2T + 2T^{1/2}]$ with multiplicity function $\tilde n(u) = \#\{k \in \mathcal{I}_T : u - \mathfrak{b}_T \le \mathsf{x}_k \le u\}$ satisfying $\int_{\mathbb{R}} \tilde n^2 \le \mathfrak{k}\,\mathfrak{b}_T \#\mathcal{I}_T$ for some $\mathfrak{k} \ge 1$. Let $r \ge 1$ be an integer, $2 \le X \le T^{1/(4r)}$, $\varphi^\sharp \in C^\infty(\mathbb{R})$ with $0 \le \varphi^\sharp \le 1$, $\varphi^\sharp = 1$ on $[T - 2T^{1/2}, 2T + 2T^{1/2}]$, $\varphi^\sharp = 0$ outside $\mathcal{J}_T^\sharp = [T - 3T^{1/2}, 2T + 3T^{1/2}]$ and $|\varphi^{\sharp\prime}| \le 2$ (a cutoff different from the one of Lemma 7.2, which equals 1 only on $[T-2, 2T+2]$), and $g = \mathcal{M}(\varphi^\sharp S_X')$. Then, with an absolute constant C,*

$$\|g(\mathsf{x}_k)\|_{4r,\mathcal{I}_T} \le C\,\mathfrak{k}^{1/(8r)}\sqrt{r}\,(1+\log X).$$

In particular $\langle g(x_k^{(1)})^4\rangle_{\mathcal{I}_T} \ll (1+\log X)^4$ at the one-step clock points $x_k^{(1)}$, for $2 \le X \le T^{1/4}$.

Proof. Let $\mathsf{F} = g^{4r}$, locally absolutely continuous with $\mathsf{F}' = 4rg^{4r-1}g'$ (Theorem 3.12). For $x \in [u-b, u]$, $\mathsf{F}(x) \le \frac{1}{b}\int_{u-b}^u \mathsf{F} + \int_{u-b}^u |\mathsf{F}'|$. Averaging over $u \in [\mathsf{x}_k, \mathsf{x}_k + b]$ and summing over k,

$$\sum_{k\in\mathcal{I}_T} \mathsf{F}(\mathsf{x}_k) \le \frac{1}{b}\int_{\mathbb{R}} \tilde n(u)\Big(\frac{1}{b}\int_{u-b}^u \mathsf{F} + \int_{u-b}^u |\mathsf{F}'|\Big)du \le \frac{1}{b}\|\tilde n\|_2\Big(\|\mathsf{F}\|_2 + b\|\mathsf{F}'\|_2\Big),$$

by Cauchy–Schwarz and because the averaging operator $h \mapsto \frac{1}{b}\int_{u-b}^u h$ has norm ≤ 1 on $L^2(\mathbb{R})$. Here $\|\varphi^\sharp S_X'\|_{8r} \le \|S_X'\|_{L^{8r}(\mathcal{J}_T^\sharp)}$ and $\|(\varphi^\sharp S_X')'\|_{8r} \le 2\|S_X'\|_{L^{8r}(\mathcal{J}_T^\sharp)} + \|S_X''\|_{L^{8r}(\mathcal{J}_T^\sharp)}$, both bounded by the third estimate of Lemma 8.2 on $\mathcal{J} = \mathcal{J}_T^\sharp$; so $\|\mathsf{F}\|_2 = \|g\|_{8r}^{4r} \le (\mathcal{A}_{8r}\|\varphi^\sharp S_X'\|_{8r})^{4r} \le (C\sqrt{r}(1+\log X))^{4r}T^{1/2}$ and $\|\mathsf{F}'\|_2 \le 4r\|g\|_{8r}^{4r-1}\|g'\|_{8r} \le 4r(1+\log X)\,(C\sqrt{r}(1+\log X))^{4r}T^{1/2}$, by Theorem 3.12 ($p = 8r$, $\mathcal{A}_p$ bounded) and Lemma 8.2 (third bound with $2r$ in place of r, which needs $X^{4r} \le T$). Since $\log X \ge \log 2$ and $r\log X \le \frac14 \log T$, uniformly in r and X, $b\cdot 4r(1+\log X) = \frac{8\pi r(1+\log X)}{\log(3T)} \le 8\pi\Big(1 + \frac{1}{\log 2}\Big)\frac{r\log X}{\log(3T)} \le 2\pi\Big(1+\frac{1}{\log 2}\Big)\frac{\log T}{\log(3T)} < 16$, and we get $\sum_k \mathsf{F}(\mathsf{x}_k) \le 17\,b^{-1}(\mathfrak{k}b\mathsf{M})^{1/2}T^{1/2}(C\sqrt{r}(1+\log X))^{4r} \le C'\sqrt{\mathfrak{k}}\,\mathsf{M}\,(C\sqrt{r}(1+\log X))^{4r}$, since $b^{-1}T \asymp \mathsf{M} = \#\mathcal{I}_T$; take the $4r$-th root. For the $x_k^{(1)}$, $\mathfrak{k} \ll 1$ by Lemma 8.4. □

8.4. The second step

Theorem 8.6 (Second Newton-type step)**.** *Let $H_0 = \log\log T$, $H_1 = \log H_0$, $H_2 = \log H_1$, $X_1 = T^{H_0^{-1/2}}$, $X_2 = T^{H_1^{-1/2}}$, and let $x_k^{(2)}$ be given by* (8.1)*. Then*

$$\begin{aligned}
&\Big\langle \mathcal{L}(x_k^{(1)})^2 e_{2,k}^2\Big\rangle_{\mathcal{I}_T} = \frac{1}{4\pi^2}H_2 + O\Big(1+\sqrt{H_2}\Big),\\
&\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(x_k^{(2)} - \gamma_k\Big)^2 = \frac{\log\log\log\log T + O\Big(\sqrt{\log\log\log\log T}\Big)}{\log^2 T}.
\end{aligned} \tag{8.7}$$

Proof. Fourth moment of e_1. By (8.2) with $j=1$ (that is (7.1)), $\mathcal{L}(\tau_k)e_{1,k} = u_k(X_1)+\mathfrak{t}_k(X_1)+\chi_k$. Lemma 8.2 with $r=2$ gives $\langle u_k(X_1)^4\rangle \ll (1+\log(1/c_1))^2 \ll H_1^2$. For the transport, $|\mathfrak{t}_k| \le 2|\delta_k|\mathfrak{g}_1(\tau_k)$ with $\mathfrak{g}_1 = \mathcal{M}(\varphi S'_{X_1})$, so $\langle \mathfrak{t}_k^4\rangle \le 16\langle\delta_k^8\rangle^{1/2}\langle \mathfrak{g}_1(\tau_k)^8\rangle^{1/2}$; here $\langle\delta_k^8\rangle \ll H_0^4/\log^8 T$ by (3.7) with $j=4$, and $\langle \mathfrak{g}_1(\tau_k)^8\rangle \ll (c_1\log T)^8$ by the sampling argument of Lemma 7.2 in L^8 (Sobolev on the intervals $\mathcal{U}_k$ with g^8, Theorem 3.12 with $p=8$, Lemma 8.2 with $r=2$). Hence $\langle\mathfrak{t}_k^4\rangle \ll (c_1^2H_0)^2 = 1$, and $\chi_k \ll 1/T$. Dividing by $\mathcal{L}(\tau_k)^4 \asymp \log^4 T$,

$$\langle e_{1,k}^4\rangle_{\mathcal{I}_T} \ll \frac{H_1^2}{\log^4 T}. \tag{8.8}$$

Transport of the second step. The interval centred at $x_k^{(1)}$ of radius $|e_{1,k}|$ contains γ_k and lies in $[T-2T^{1/2}, 2T+2T^{1/2}]$, so $|S_{X_2}(\gamma_k) - S_{X_2}(x_k^{(1)})| \le 2|e_{1,k}|\mathfrak{g}_2(x_k^{(1)})$ with $\mathfrak{g}_2 = \mathcal{M}(\varphi^\sharp S'_{X_2})$, and by Cauchy–Schwarz, (8.8) and Lemma 8.5 (applicable since $X_2 \le T^{1/8}$),

$$\Big\langle (S_{X_2}(\gamma_k) - S_{X_2}(x_k^{(1)}))^2\Big\rangle \le 4\langle e_{1,k}^4\rangle^{1/2}\langle \mathfrak{g}_2(x_k^{(1)})^4\rangle^{1/2} \ll \frac{H_1}{\log^2 T}(c_2\log T)^2 = c_2^2 H_1 = 1.$$

Conclusion. By (8.2) with $j=2$, $0 \le \chi_{2,k} \le e_{1,k}^2/(2\pi T)$, and $\langle\chi_{2,k}^2\rangle \ll T^{-2}$. Theorem 5.1 with $X = X_2$ (allowed, $2 \le X_2 \le T^{1/2}$) gives $\langle u_k(X_2)^2\rangle = \frac{1}{2\pi^2}\log\frac{1}{c_2} + O(1+\sqrt{\log(1/c_2)})$ with $\log\frac{1}{c_2} = \frac12 H_2$. Combining as in the proof of Theorem 7.3 gives the first statement of (8.7), and $\mathcal{L}(x_k^{(1)}) = \frac{\log T}{2\pi}(1+O(\frac{1}{\log T}))$ on $\mathcal{I}_T$, since $x_k^{(1)} \in [T-T^{1/2}, 2T+T^{1/2}]$, gives the second. □

8.5. All further steps: the exact identity, the local deformation, and the multiplicity

This subsection proves the Newton-type steps $m \ge 3$ with constants uniform in m, up to $m = \log^* T + O(1)$ steps, where the error becomes $O(1)$ in units of zeros.

Lemma 8.7 (The second-level identity)**.** *For $j \ge 1$, with $y_k^{(j)} = \tau_k - x_k^{(j)}$ and $\chi''_{j,k} = \int_{x_k^{(j-1)}}^{\tau_k}(\mathcal{L}(\tau_k)-\mathcal{L}(y))\,dy$,*

$$\begin{aligned} y_k^{(j)} &= \Big(1 - \frac{\mathcal{L}(\tau_k)}{\mathcal{L}(x_k^{(j-1)})}\Big)y_k^{(j-1)} + \frac{S_{X_j}(x_k^{(j-1)}) + \chi''_{j,k}}{\mathcal{L}(x_k^{(j-1)})}, \\ \textit{hence}\qquad \mathcal{L}(\tau_k)y_k^{(j)} &= S_{X_j}(x_k^{(j-1)}) + \mathfrak{r}_{j,k}, \end{aligned} \tag{8.9}$$

with $|\mathfrak{r}_{j,k}| \ll T^{c_j-1}$ if $|y_k^{(j-1)}| \le T^{c_{j-1}/2}$ and $c_{j-1} \le c_j$ (true by (2.11) and induction): the three contributions are $\mathcal{L}(\tau_k)(1-\mathcal{L}(\tau_k)/\mathcal{L}(x))y_k^{(j-1)} \ll (y_k^{(j-1)})^2/T$, $(\mathcal{L}(\tau_k)/\mathcal{L}(x)-1)S_{X_j}(x) \ll |y_k^{(j-1)}|\sigma_{X_j}/(T\log T)$ and $\chi''_{j,k} \le (y_k^{(j-1)})^2/(4\pi(T-T^{1/2}))$. Newton's step does not accumulate the corrections of the previous levels: it replaces the previous one by the phase of the new cutoff evaluated at the previous point.

Proof. $N_0(x_k^{(j-1)}) - (k-\frac12) = N_0(x_k^{(j-1)}) - N_0(\tau_k) = -\mathcal{L}(\tau_k)y_k^{(j-1)} + \chi''_{j,k}$, with $0 \le \chi''_{j,k} \le (y_k^{(j-1)})^2/(4\pi(T-T^{1/2}))$; insert in (8.1) and use $1-\mathcal{L}(\tau_k)/\mathcal{L}(x) = O(|y_k^{(j-1)}|/(T\log T))$ and $|S_{X_j}| \le \sigma_{X_j}$. □

Proposition 8.8 (Local deformation of the second clock)**.** *Let $X_1 = T^{c_1}$, $X_2 = T^{c_2}$ with $c_1 = H_0^{-1/2}$, $c_2 = H_1^{-1/2}$, and $d_{j,k} = x_{k+1}^{(j)} - x_k^{(j)}$ ($d_{0,k} = \tau_{k+1} - \tau_k$). Then, for $T \geq T_0$,*

$$\Big\langle \Big| \frac{d_{2,k} - d_{0,k}}{d_{0,k}} \Big|^4 \Big\rangle_{\mathcal{I}_T} \ll c_2^4 = \frac{1}{H_1^2}, \qquad \frac{1}{\#\mathcal{I}_T} \#\Big\{ k \in \mathcal{I}_T : \Big| \frac{d_{2,k}}{d_{0,k}} - 1 \Big| > \alpha \Big\} \ll \frac{1}{\alpha^4 H_1^2} \quad (0 < \alpha \leq 1), \tag{8.10}$$

in particular $\#\{k \in \mathcal{I}_T : d_{2,k} \leq 0\} \ll \#\mathcal{I}_T/(\log\log\log T)^2$: the second clock preserves the order and a spacing comparable to that of the RvM clock outside a proportion $O(H_1^{-2})$ of the indices.

Proof. By (8.9) with $j = 2$, $d_{2,k} - d_{0,k} = -\nabla_k\big(S_{X_2}(x^{(1)})/\mathcal{L}(\tau)\big) - \nabla_k(\mathfrak{r}_2/\mathcal{L}(\tau))$, where $\mathfrak{r}_2$ is the sequence $\mathfrak{r}_{2,k}$ of Lemma 8.7, $\nabla_k a = a_{k+1} - a_k$ is the forward difference (Δ_k is reserved for the Gram displacement); by Lemma 8.7 with $j = 2$ the last term is $\ll T^{c_2-1}/\log T$, and $\nabla_k(1/\mathcal{L}(\tau)) \ll 1/(T\log^3 T)$; divided by $d_{0,k} \asymp 1/\log T$, these terms and the term $O(T^{c_1/2-1})$ below contribute $O(T^{-1/2}\log^3 T)$ to the $\ell^4(\mathcal{I}_T)$ norm of $(d_2 - d_0)/d_0$, since $c_1, c_2 \leq \frac{1}{2}$, $|S_{X_2}| \leq \sigma_{X_2}$ and $\mathsf{g}_2 \leq M_{X_2} \ll T^{c_2/2}$ (2.11), so it suffices to bound $\nabla_k S_{X_2}(x^{(1)})/(d_{0,k}\mathcal{L}(\tau_k))$ in $\ell^4(\mathcal{I}_T)$. Let $\mathsf{g}_j = \mathcal{M}(\varphi^\sharp S'_{X_j})$ with the cutoff $\varphi^\sharp$ of Lemma 8.5. Two applications of $|S(a) - S(b)| \leq 2|a-b|\,\mathcal{M}(S')(a)$:

$$|x_{k+1}^{(1)} - x_k^{(1)}| \leq d_{0,k} + |y_{k+1} - y_k| \leq d_{0,k}\Big(1 + \frac{2\mathsf{g}_1(\tau_k)}{\mathcal{L}(\tau_k)}\Big) + O\Big(T^{c_1/2-1}\Big),$$

$$|\nabla_k S_{X_2}(x^{(1)})| \leq 2|x_{k+1}^{(1)} - x_k^{(1)}|\,\mathsf{g}_2(x_k^{(1)}).$$

Hence, with $\mathcal{L}(\tau_k) \asymp \log T$,

$$\Big\| \frac{\nabla S_{X_2}(x^{(1)})}{d_0\,\mathcal{L}(\tau)} \Big\|_{4,\mathcal{I}_T} \ll \Big\| \frac{\mathsf{g}_2(x^{(1)})}{\log T} \Big\|_{4,\mathcal{I}_T} + \Big\| \frac{\mathsf{g}_2(x^{(1)})}{\log T} \Big\|_{8,\mathcal{I}_T} \Big\| \frac{\mathsf{g}_1(\tau)}{\log T} \Big\|_{8,\mathcal{I}_T} + O\Big(T^{-1/2}\log^3 T\Big).$$

Sampling at the τ_k in L^8 (Sobolev on the disjoint intervals $\mathcal{U}_k$ as in Lemma 7.2, with Lemma 8.2 for $\int |S'_{X_1}|^8$, $\int |S''_{X_1}|^8$ and Theorem 3.12 with $p = 8$) gives $\|\mathsf{g}_1(\tau)/\log T\|_8 \ll c_1$. Sampling at the points $x_k^{(1)}$ in L^8 is Lemma 8.5 with $\mathsf{x}_k = x_k^{(1)}$ and $\mathsf{F} = \mathsf{g}_2^8$: $\|\mathsf{F}\|_2 = \|\mathsf{g}_2\|_{16}^8$ and $\|\mathsf{F}'\|_2 \leq 8\|\mathsf{g}_2\|_{16}^7 \|\mathsf{g}_2'\|_{16}$, which need Theorem 3.9 with products of 8 primes ($X_2^{16} \leq T$, true for $T \geq T_0$ since $c_2 \to 0$) and Theorem 3.12 with $p = 16$; the result is $\|\mathsf{g}_2(x^{(1)})/\log T\|_8 \ll c_2$. Hence

$$\Big\| \frac{d_2 - d_0}{d_0} \Big\|_{4,\mathcal{I}_T} \ll c_2 + c_1 c_2 + T^{-1/2}\log^3 T \ll c_2,$$

because $c_2 = H_1^{-1/2} \gg T^{-1/2}\log^3 T$; the first inequality of (8.10) follows, and the second is Markov's inequality; the last statement is the case $\alpha = \frac{1}{2}$, since $d_{2,k} \leq 0$ gives $|d_{2,k}/d_{0,k} - 1| \geq 1 > \frac{1}{2}$ (with $\alpha = 1$ the strict inequality would miss $d_{2,k} = 0$). □

Proposition 8.8 controls consecutive differences; the multiplicity of the points $x_k^{(j)}$ at scale $\mathfrak{b}_T$, which Lemma 8.5 needs, is obtained for every j by comparing $x_k^{(j)}$ with γ_k through two unconditional counts of near pairs and near triples of ordinates. The ordinates enter the proof of the bound, as in Theorem 5.1, and not the construction of the points.

Lemma 8.9 (Near pairs and triples of ordinates)**.** *Let $b = \mathfrak{b}_T$, $\mathsf{M} = \#\mathcal{I}_T$ and, for $\mathsf{R} \ge 1$ and real u, $n_{\mathsf{R}}(u) = \#\{l \in \mathcal{I}_T : |\gamma_l - u| \le \mathsf{R}b\}$. There are absolute constants C_1, C_2 such that for $T \ge T_0$ and every $\mathsf{R} \ge 1$,*

$$\#\{(k,l) \in \mathcal{I}_T^2 : |\gamma_k - \gamma_l| \le \mathsf{R}b\} \le C_1 \mathsf{R}\mathsf{M}, \qquad \sum_{k \in \mathcal{I}_T} n_{\mathsf{R}}(\gamma_k)^2 \le C_2 \mathsf{R}^2 \mathsf{M}. \tag{8.11}$$

Proof. If $4\mathsf{R}b > 1$, then $N(u+h)-N(u) \le Ch\log T$ for $1 \le h \le T$ and $u \in [T-1, 2T]$, by (2.3) and Theorem 3.1. Only $l \in \mathcal{I}_T$ are counted. If $2\mathsf{R}b \ge T$, then $n_{\mathsf{R}}(\gamma_k) \le \#\mathcal{I}_T \le CT\log T \le 2C\mathsf{R}b\log T \le 4\pi C\mathsf{R}$. Otherwise $h = \max(1, 2\mathsf{R}b)$ satisfies $1 \le h \le T$ and, since $4\mathsf{R}b > 1$, $h \le 4\mathsf{R}b$; the ordinates counted by $n_{\mathsf{R}}(\gamma_k)$ lie in $(T, 2T] \cap [\gamma_k - \mathsf{R}b, \gamma_k + \mathsf{R}b] \subset [u_k, u_k + h]$ with $u_k = \max(T-1, \gamma_k - \mathsf{R}b) \in [T-1, 2T]$, because $h \ge 2\mathsf{R}b$; since $N(u_k+h) - N(u_k)$ counts $(u_k, u_k + h]$, we bound the count by $N(u_k+h) - N(u_k - 1)$, an interval of length $h + 1 \le 2h$ with $u_k - 1 \in [T-2, 2T]$, which changes only the constant, so $n_{\mathsf{R}}(\gamma_k) \le Ch\log T \le 4C\mathsf{R}b\log T \le 8\pi C\mathsf{R}$. Both bounds follow. Let $4\mathsf{R}b \le 1$ and $h \in \{2\mathsf{R}b, 4\mathsf{R}b\}$. For $u \in [T-1, 2T]$, $N(u+h) - N(u) = h\mathcal{L}(u) + S(u+h) - S(u) + O(1/T)$ with $h\mathcal{L}(u) \le 5\mathsf{R}$, so (3.6) and $\log(2 + h\log T) \le \log(30\mathsf{R}) \le 30\mathsf{R}$ give

$$\int_{T-1}^{2T} \big(N(u+h) - N(u)\big)^{2k} du \le C\mathsf{R}^{2k} T \qquad (k = 1, 2).$$

A pair (k,l) with $|\gamma_k - \gamma_l| \le \mathsf{R}b$ lies in $[u, u + 2\mathsf{R}b]$ for every $u \in [\min(\gamma_k, \gamma_l) - \mathsf{R}b, \min(\gamma_k, \gamma_l)]$, a set of measure $\mathsf{R}b$ contained in $[T-1, 2T]$, because $\min(\gamma_k, \gamma_l) \in (T, 2T]$ and $\mathsf{R}b \le \frac{1}{4}$; hence the number of pairs is at most $\frac{1}{\mathsf{R}b}\int_{T-1}^{2T} (N(u+2\mathsf{R}b) - N(u))^2 du \le C\mathsf{R}T/b \le C_1\mathsf{R}\mathsf{M}$, as $\mathsf{M} \asymp T/b$. A triple (k,l,l') with $|\gamma_l - \gamma_k|, |\gamma_{l'} - \gamma_k| \le \mathsf{R}b$ lies in $[u, u + 4\mathsf{R}b]$ for $u \in [\gamma_k - 3\mathsf{R}b, \gamma_k - \mathsf{R}b]$, a set of measure $2\mathsf{R}b$ contained in $[T-1, 2T]$ since $\gamma_k \in (T, 2T]$ and $4\mathsf{R}b \le 1$; by Hölder's inequality the number of triples is at most $\frac{1}{2\mathsf{R}b}\int_{T-1}^{2T} (N(u + 4\mathsf{R}b) - N(u))^3 du \le \frac{T^{1/4}(C\mathsf{R}^4 T)^{3/4}}{2\mathsf{R}b} \le C_2\mathsf{R}^2\mathsf{M}$; the triples with $l, l' \in \mathcal{I}_T$ are among them. □

Theorem 8.10 (Multiplicity of the corrected clock points)**.** *Let $j \ge 0$ and let $x_k^{(j)}$ be given by* (8.1) *with cutoffs $X_i = T^{c_i}$, $c_i \le \frac{1}{2}$, so that $x_k^{(j)} \in [T - 2T^{1/2}, 2T + 2T^{1/2}]$ for $k \in \mathcal{I}_T$ and $T \ge T_0$ (Lemma 8.7). Let $\tilde{n}_j(u) = \#\{k \in \mathcal{I}_T : u - \mathfrak{b}_T \le x_k^{(j)} \le u\}$, and let $\mathsf{A} \ge 1$, $\mathsf{H} \ge 1$ be such that $\|\mathcal{L}(\tau_k) e_{j,k}\|_{4, \mathcal{I}_T} \le \mathsf{A}\sqrt{\mathsf{H}}$. Then, with an absolute constant C_3,*

$$\int_{\mathbb{R}} \tilde{n}_j(u)^2 \, du \le C_3 \, \mathsf{A}\sqrt{\mathsf{H}} \; \mathfrak{b}_T \, \#\mathcal{I}_T. \tag{8.12}$$

Proof. Write $b = \mathfrak{b}_T$, $\mathsf{M} = \#\mathcal{I}_T$, $x_k = x_k^{(j)}$, $e_k = e_{j,k}$. As in Lemma 8.4, $\int \tilde{n}_j^2 = \sum_{k,l \in \mathcal{I}_T} (b - |x_k - x_l|)_+ \le b\,\mathsf{P}$ with $\mathsf{P} = \#\{(k,l) \in \mathcal{I}_T^2 : |x_k - x_l| < b\}$. Let $\mathsf{R}_0 \ge 2$. The pairs with $|\gamma_k - \gamma_l| \le \mathsf{R}_0 b$ number at most $C_1\mathsf{R}_0\mathsf{M}$ by (8.11). For the others, let $\mathsf{R} = 2^i\mathsf{R}_0$, $i \ge 0$, and consider the pairs with $\mathsf{R}b < |\gamma_k - \gamma_l| \le 2\mathsf{R}b$ and $|x_k - x_l| < b$. For them $|e_k - e_l| \ge (\mathsf{R}-1)b \ge \mathsf{R}b/2$, and $b\mathcal{L}(\tau_k) \ge \frac{1}{2}$ for $T \ge T_0$, so $\mathcal{L}(\tau_k)|e_k - e_l| \ge \mathsf{R}/4$ and

$$1 \le \Big(\frac{4\mathcal{L}(\tau_k)|e_k - e_l|}{\mathsf{R}}\Big)^2 \le \frac{32}{\mathsf{R}^2}\Big(\mathcal{L}(\tau_k)^2 e_k^2 + \mathcal{L}(\tau_l)^2 e_l^2\Big)\Big(1 + O(1/\log T)\Big).$$

Summing over the ordered pairs of the block and symmetrizing, their number P_{R} satisfies

$$\mathsf{P}_{\mathsf{R}} \le \frac{65}{\mathsf{R}^2} \sum_{k \in \mathcal{I}_T} \mathcal{L}(\tau_k)^2 e_k^2 \, n_{2\mathsf{R}}(\gamma_k) \le \frac{65}{\mathsf{R}^2}\Big(\sum_k \mathcal{L}(\tau_k)^4 e_k^4\Big)^{1/2}\Big(\sum_k n_{2\mathsf{R}}(\gamma_k)^2\Big)^{1/2} \le \frac{130\sqrt{C_2}\,\mathsf{A}^2\mathsf{H}}{\mathsf{R}}\,\mathsf{M},$$

by Cauchy–Schwarz, the hypothesis and (8.11) (both k and l lie in $\mathcal{I}_T$, so $n_{2\mathsf{R}}$ counts every l of the block). Summing over $i \ge 0$ gives at most $260\sqrt{C_2}\,\mathsf{A}^2\mathsf{H}\mathsf{M}/\mathsf{R}_0$. With $\mathsf{R}_0 = \mathsf{A}\sqrt{\mathsf{H}}\,(260\sqrt{C_2}/C_1)^{1/2}$ (which is ≥ 2 for C_2 large enough) the two parts are equal and $\mathsf{P} \le 2\sqrt{260C_1\sqrt{C_2}}\,\mathsf{A}\sqrt{\mathsf{H}}\,\mathsf{M}$. □

Only the fourth moment of the error of the j-th clock enters. For $j = 2$, with A bounded (shown below), the multiplicity of the second clock is $\ll \sqrt{H_2}\,\mathfrak{b}_T\#\mathcal{I}_T$. The factor $\sqrt{\mathsf{H}}$ is not a loss in the iteration: it is absorbed by the cutoff of the next step.

All further steps

Let $H_0 = \log\log T$, $c_1 = H_0^{-1/2}$, $c_2 = H_1^{-1/2}$ with $H_1 = \log H_0$, and define recursively, with an absolute constant $0 < C_\star \le 1$ fixed in the proof of Theorem 8.11,

$$H_j = 2\log\frac{1}{c_j} \quad (j \ge 1), \qquad c_j = C_\star\, H_{j-1}^{-9/16} \quad (j \ge 3), \tag{8.13}$$

so that $H_2 = \log H_1$ as in Theorem 8.6, $H_j = \frac{9}{8}\log H_{j-1} + 2\log(1/C_\star)$ for $j \ge 3$, and $\log H_{j-1} \le H_j$ for all j. All cutoffs depend only on T. Put

$$A_j(r) = \frac{\|\mathcal{L}(\tau_k)e_{j,k}\|_{2r,\mathcal{I}_T}}{\sqrt{H_j}} \quad (j \ge 1), \qquad A_0(r) = \frac{\|\mathcal{L}(\tau_k)\delta_k\|_{2r,\mathcal{I}_T}}{\sqrt{H_0}},$$

and let $m^* = m^*(T)$ be the first index m with $H_m < H^*$, where H^* is the absolute constant of the next theorem. Since $\log H_{j-1} \le H_j \le \frac{9}{8}\log H_{j-1} + 2\log(1/C_\star)$, the sequence H_j decreases from $\log\log T$ to below H^* in $m^* = \log^* T + O(1)$ steps, where $\log^*$ is the iterated logarithm.

Theorem 8.11 (All Newton-type steps, uniformly)**.** *There are absolute constants $C_\star$, H^*, C and T_0 such that, with the cutoffs* (8.13)*, for every $T \ge T_0$ and every $1 \le m \le m^*$,*

$$\Big\langle \mathcal{L}(\tau_k)^2 e_{m,k}^2 \Big\rangle_{\mathcal{I}_T} = \frac{H_m}{4\pi^2} + \vartheta_m\, C\Big(1 + \sqrt{H_m}\Big), \qquad |\vartheta_m| \le 1, \tag{8.14}$$

and $\|\mathcal{L}(\tau_k)e_{j,k}\|_{4,\mathcal{I}_T} \le C\sqrt{H_j}$ for $1 \le j < m^*$. In particular, at $m = m^* = \log^* T + O(1)$,

$$\frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(x_k^{(m^*)} - \gamma_k\Big)^2 \le \frac{C'}{\log^2 T}\,: \tag{8.15}$$

the clock corrected $\log^* T + O(1)$ times with the Euler phases of the primes up to T^{c_j} locates the ordinates with a mean-square error of $O(1)$ mean spacings, with an absolute constant, using no ordinate in its construction. The cutoffs (8.13) *and the number m^* of steps involve the absolute constants $C_\star$ and H^* fixed in the proof; once they are fixed, every $x_k^{(m)}$ is an explicit expression in k, τ_k and the primes.*

Proof. Multiplicities. Let $\mathfrak{k}_j$ be the least $\mathfrak{k} \ge 1$ with $\int \tilde{n}_j^2 \le \mathfrak{k}\,\mathfrak{b}_T\#\mathcal{I}_T$. Then $\mathfrak{k}_0 = 1$ (the τ_k are separated by at least $\mathfrak{b}_T$, so only the diagonal contributes), $\mathfrak{k}_1 \le C$ (Lemma 8.4) and $\mathfrak{k}_j \le C_3 \max(A_j(2), 1)\sqrt{H_j}$ for $j \ge 2$ (Theorem 8.10 with $\mathsf{A} = \max(A_j(2), 1)$ and $\mathsf{H} = H_j$).

Base. For every $r \ge 1$ with $2r \le \kappa_{\mathrm{F}} \log T$, by Lemma 8.2 with $X = 2$, $|S_2| \le \sigma_2 < 1$ and $0 \le \chi_k \le 1/T$, $\|\mathcal{L}(\tau_k)\delta_k\|_{2r} \le \|u_k(2)\|_{2r} + 2 \le C_0(r^2 + \sqrt{rH_0})$, so $A_0(r) \le C_0(r^2/\sqrt{H_0} + \sqrt{r}) \le 2C_0 r^2$ for $r \ge 1$.

One step. Let $j \ge 1$ and $r \ge 1$ with $X_j \le T^{1/(160r)}$ and $2r \le \kappa_{\mathrm{F}} \log T$. By (8.3), $\mathcal{L}(x_k^{(j-1)}) = \mathcal{L}(\tau_k)(1 + O(T^{-1/2}))$, Lemma 8.2 (with $V \le \log \frac{1}{c_j} + O(1) \le H_j$), Lemma 8.5 at the points $x_k^{(j-1)}$ with multiplicity constant $\mathfrak{k}_{j-1}$, and $\|\chi_j\|_{2r} \ll T^{c_{j-1}-1}$,

$$\sqrt{H_j}\, A_j(r) \le C\Big(r^2 + \sqrt{rH_j}\Big) + 2\sqrt{H_{j-1}}\, A_{j-1}(2r) \cdot \frac{C\mathfrak{k}_{j-1}^{1/(8r)} \sqrt{r}\,(1 + \log X_j)}{\mathcal{L}(\tau_k)} \Big(1 + o(1)\Big) + o(1),$$

and $(1 + \log X_j)/\mathcal{L}(\tau_k) \le 2\pi c_j(1 + o(1))$, $\mathfrak{k}_{j-1}^{1/(8r)} \le \mathfrak{k}_{j-1}^{1/8}$. For $j \le 2$, $\sqrt{H_{j-1}} c_j = 1$ and $\mathfrak{k}_{j-1} \le C$; for $j \ge 3$, $\mathfrak{k}_{j-1}^{1/8}\sqrt{H_{j-1}} c_j \le (C_3 \max(A_{j-1}(2), 1))^{1/8} H_{j-1}^{1/16 + 1/2 - 9/16} C_\star = C_\star (C_3 \max(A_{j-1}(2), 1))^{1/8}$, and $\max(A_{j-1}(2), 1) \le \max(A_{j-1}(2r), 1)$ (monotonicity of the discrete norms), so that $A_{j-1}(2r)\max(A_{j-1}(2r), 1)^{1/8} = \max(A_{j-1}(2r), A_{j-1}(2r)^{9/8})$. Fix $C_\star = \min(1, (4\pi C\, C_3^{1/8})^{-1})$. Then, for all $j \ge 1$ and $r \ge 1$,

$$A_j(r) \le C\Big(\sqrt{r} + \frac{r^2}{\sqrt{H_j}}\Big) + \frac{C_5\sqrt{r}}{\sqrt{H_j}} \max\Big(A_{j-1}(2r), A_{j-1}(2r)^{9/8}\Big), \tag{8.16}$$

with absolute C, C_5, provided $X_j \le T^{1/(160r)}$ and $2r \le \kappa_{\mathrm{F}} \log T$.

Induction. Let $B_\star = \max(2C, 2C_0, 1)$ and $C_6 = \Big(4C/B_\star + 4C_5(4B_\star)^{9/8}/B_\star\Big)^2$. We claim that for every $m \le m^*$,

$$A_j(r) \le B_\star r^2 \qquad (0 \le j \le m - 1,\ 1 \le r \le 2^{m-j}). \tag{8.17}$$

For $j = 0$ this is the base. Suppose it holds at $j - 1$ for $r \le 2^{m-j+1}$, let $1 \le r \le 2^{m-j}$, and assume $H_j \ge C_6 4^{m-j}$, $X_j \le T^{1/(160r)}$ and $2r \le \kappa_{\mathrm{F}} \log T$. Since $A_{j-1}(2r) \le 4B_\star r^2$ and $4B_\star r^2 \ge 1$, (8.16) gives

$$A_j(r) \le C\sqrt{r} + \frac{Cr^2 + C_5(4B_\star)^{9/8} r^{11/4}}{\sqrt{H_j}} \le \frac{B_\star}{2} r^2 + \frac{\Big(C + C_5(4B_\star)^{9/8}\Big) r^3}{\sqrt{H_j}} \le B_\star r^2,$$

because $r^3/\sqrt{H_j} \le r^2 \cdot 2^{m-j}/\sqrt{H_j} \le r^2/\sqrt{C_6}$ and $\sqrt{C_6} \ge 2(C + C_5(4B_\star)^{9/8})/B_\star$.

The three conditions. Let H^* be an absolute constant with $H^* \ge \max(4C_6, 160^2 \cdot 4)$ and so large that $e^{\frac{8}{9}(H - 2\log(1/C_\star))} \ge 4H$ and $e^H \ge 4H$ for $H \ge H^*$. If $m \le m^*$ then $H_{m-1} \ge H^*$ (by definition of m^*), and from $H_{j-1} \ge e^{H_j}$ $(j \le 2)$ or $H_{j-1} = e^{\frac{8}{9}(H_j - 2\log(1/C_\star))}$ $(j \ge 3)$ we get $H_{j-1} \ge 4H_j$ whenever $H_j \ge H^*$; by downward induction, $H_{m-1-i} \ge 4^i H^*$ for $0 \le i \le m - 1$. Hence $H_j \ge 4^{m-1-j} H^* \ge C_6 4^{m-j}$ for $1 \le j \le m - 1$, and $H_{j-1} \ge 4^{m-j} H^* \ge 160^2\, 4^{m-j}$ for $1 \le j \le m$, which gives $c_j \le H_{j-1}^{-1/2} \le 2^{-(m-j)}/160$, i.e. $X_j \le T^{1/(160r)}$ for $r \le 2^{m-j}$, and $X_j \ge 2$ because $c_j \ge C_\star(\log\log T)^{-9/16}$. Finally $2r \le 2^{m-j+1} \le 2^{m^*+1}$, and $m^* = \log^* T + O(1)$ with an absolute $O(1)$, so $2r \le \kappa_{\mathrm{F}} \log T$ for $T \ge T_0$. So (8.17) holds for every $m \le m^*$, which is the statement $\|\mathcal{L}e_j\|_{4,\mathcal{I}_T} \le 4B_\star\sqrt{H_j}$ for $j < m^*$.

Order of the constants. The constants are fixed in this order, and none of them depends on m, j, r or T: first C, C_0, C_3, given by Lemmas 8.2, 8.5 and Theorem 8.10; then $C_\star$, which

depends only on C and C_3; then C_5, the coefficient of (8.16), obtained once $C_\star$ is fixed; then $B_\star$ and C_6; then H^*. With $C_\star$ and H^* fixed, $m^*(T) = \log^* T + O(1)$ with an $O(1)$ depending only on them. Finally T_0 is taken so large that for $T \ge T_0$: the $o(1)$ terms of the one-step bound are at most 1; $2^{m^*(T)+1} \le \kappa_{\mathrm{F}} \log T$; and $c_j \ge C_\star (\log\log T)^{-9/16}$ gives $X_j \ge 2$. The second condition contains the base case, where the orders reach $r \le 2^m \le 2^{m^*}$. With these choices all the conditions used above hold simultaneously for every $m \le m^*(T)$ and every $T \ge T_0$.

The last step. For $m \le m^*$ and $r = 1$, (8.17) at $j = m-1$ gives $A_{m-1}(2) \le 4B_\star$, so the transport term of the m-th step, which is already in units of zeros, satisfies $\|\mathfrak{t}_k^{(m)}\|_{2,\mathcal{I}_T} \le C_5(4B_\star)^{9/8}$ (the transport bound in the proof of (8.16), with $r = 1$), with $\mathfrak{t}_k^{(m)} = S_{X_m}(\gamma_k) - S_{X_m}(x_k^{(m-1)})$. Theorem 5.1 at $X = X_m$ ($2 \le X_m \le T^{1/160}$) gives $\langle u_k(X_m)^2 \rangle = \frac{1}{2\pi^2}\log\frac{1}{c_m} + O(1+\sqrt{H_m}) = \frac{H_m}{4\pi^2} + O(1+\sqrt{H_m})$, the cross term is at most $2\|u(X_m)\|_2\, C_5(4B_\star)^{9/8} \ll 1+\sqrt{H_m}$, and $\chi_{m,k} \ll T^{-1/2}$; (8.2) gives (8.14). At $m = m^*$, $H_{m^*} < H^*$ and $\mathcal{L}(\tau_k) = \frac{\log T}{2\pi}(1 + O(1/\log T))$ give (8.15). □

Corollary 8.12 (Third step)**.** *With $c_3 = C_\star H_2^{-9/16}$ and $H_3 = \frac{9}{8}\log H_2 + 2\log(1/C_\star)$,*

$$\Big\langle \mathcal{L}(\tau_k)^2 e_{3,k}^2 \Big\rangle_{\mathcal{I}_T} = \frac{H_3}{4\pi^2} + O\Big(1+\sqrt{H_3}\Big), \qquad \frac{1}{\#\mathcal{I}_T}\sum_{k\in\mathcal{I}_T}\Big(x_k^{(3)} - \gamma_k\Big)^2 = \frac{H_3 + O(\sqrt{H_3})}{\log^2 T},$$

and the multiplicity of the second clock satisfies $\int \tilde{n}_2^2 \ll \sqrt{H_2}\,\mathfrak{b}_T \#\mathcal{I}_T$.

Proof. Theorem 8.11 with $m = 3$ (for $T \ge T_0$, $H_2 \ge H^*$), and Theorem 8.10 with $\mathsf{A} = A_2(2) \le 4B_\star$ and $\mathsf{H} = H_2$. □

The induction (8.17) bounds the moments of every order $2r \le 2^{m^*-j+1}$ of the j-th error, not only its mean square, and the number of available moments grows with T. This gives tails.

Corollary 8.13 (Tails of the corrected clocks)**.** *With the cutoffs* (8.13)*, for $T \ge T_0$, every $1 \le j < m^*$, every integer $1 \le q \le 2^{m^*-j}$ and every $\lambda > 0$,*

$$\#\Big\{k \in \mathcal{I}_T : \mathcal{L}(\tau_k)|e_{j,k}| > \lambda\sqrt{H_j}\Big\} \le \Big(\frac{B_\star q^2}{\lambda}\Big)^{2q} \#\mathcal{I}_T. \tag{8.18}$$

For fixed j and q the right side is $\ll_q \lambda^{-2q}\#\mathcal{I}_T$ for $T \ge T_0(j,q)$, since $2^{m^*-j} \to \infty$: for every fixed q, and T large in terms of q, the j-th corrected clock misses its ordinate by more than $\lambda\sqrt{H_j}$ positions on a proportion $\ll_q \lambda^{-2q}$ of the indices. The orders available at a given T are $q \le 2^{m^*-j}$; nothing is claimed uniformly in q.

Proof. (8.17) with $m = m^*$ gives $\|\mathcal{L}(\tau_k)e_{j,k}\|_{2q,\mathcal{I}_T} \le B_\star q^2\sqrt{H_j}$ for $j \le m^*-1$ and $q \le 2^{m^*-j}$; apply Chebyshev's inequality to the $2q$-th moment. □

The Newton points of a level need not be in increasing order (Proposition 8.8 bounds the proportion of inversions of the second clock). Sorting the points of the final level does not increase the unweighted error.

Lemma 8.14 (Sorting the final points)**.** *Let $\gamma_1 \le \cdots \le \gamma_{\mathsf{M}}$, let $z_1, \dots, z_{\mathsf{M}}$ be real, and let $z^{\uparrow}$ be the same values in nondecreasing order. Then $\sum_k |z_k^{\uparrow} - \gamma_k|^q \le \sum_k |z_k - \gamma_k|^q$ for every $q \ge 1$, and $\{z_k^{\uparrow}\} = \{z_k\}$ as multisets. In particular, sorting the points $x_k^{(j)}$, $k \in I$, of one level does not increase the unweighted means $\langle |x_k^{(j)} - \gamma_k|^q \rangle_I$ and does not change the multiplicity function $\tilde{n}_j$; it removes the strict inversions $d_{j,k} < 0$ of Proposition 8.8, but not the coincidences $d_{j,k} = 0$: points that coincide remain coincident.*

Proof. For $a \le b$, $x \le y$ and φ convex, $\varphi(a-x) + \varphi(b-y) \le \varphi(a-y) + \varphi(b-x)$ (the points $a - y \le a - x, b - y \le b - x$ have the same sum in both pairings, and the outer pair majorizes the inner one). With $\varphi(t) = |t|^q$, each transposition of an inverted pair of z's does not increase the sum, and finitely many transpositions sort z. Sorting permutes the values, so the multiset, and with it $\tilde{n}_j$, is unchanged, and equal values stay equal. □

The lemma is used only for the final level (Corollary 8.15). It does not show that the Newton-type iteration continued from sorted points satisfies Theorems 8.6, 8.10 and 8.11: their proofs use the weighted norms $\|\mathcal{L}(\tau_k) e_{j,k}\|_{r,I}$, in which the weight is attached to the index, and the identity (8.2), in which the index enters through $k - \frac{1}{2}$; the effect of sorting on these is not controlled here.

Corollary 8.15 (A nondecreasing clock built without ordinates)**.** *Let $\mathcal{I}_T^{\tau} = \{k : T \le \tau_k < 2T\}$, a set of consecutive indices defined by the clock alone. For each T, with the cutoffs and the number of steps chosen at height T, the points $x_k^{(m^*)}$ of Theorem 8.11 are defined by* (8.1) *on the block $\mathcal{J}_T = \{k : T/2 \le \tau_k \le 3T\}$, which contains $\mathcal{I}_T \cup \mathcal{I}_T^{\tau}$ for $T \ge T_0$; let $x_k^{\uparrow}$, $k \in \mathcal{I}_T^{\tau}$, be these values in nondecreasing order. Then $x_k^{\uparrow} \le x_{k+1}^{\uparrow}$, no ordinate enters the construction of the $x_k^{\uparrow}$ (they depend on the τ_k, the primes and the constants $C_\star$, H^* of Theorem 8.11), and for $T \ge T_0$*

$$\frac{1}{\#\mathcal{I}_T^{\tau}} \sum_{k \in \mathcal{I}_T^{\tau}} \bigl(x_k^{\uparrow} - \gamma_k\bigr)^2 \le \frac{C' + 1}{\log^2 T}.$$

Proof. By Lemma 8.14 with $q = 2$ on the index set $\mathcal{I}_T^{\tau}$, the sum is at most $\sum_{k \in \mathcal{I}_T^{\tau}} (x_k^{(m^*)} - \gamma_k)^2$. Theorem 8.11 is a statement over $\mathcal{I}_T$; for the indices of $\mathcal{I}_T^{\tau} \setminus \mathcal{I}_T$ only the localization of the points proved below is used, and it is applied on $\mathcal{J}_T$, while the mean estimates are kept on their original index sets. The output for $k \in \mathcal{I}_T^{\tau}$ does not depend on this choice of block. Since $|\delta_k| \le C_{10}$ for all k (Theorem 3.1), the symmetric difference $\mathcal{I}_T^{\tau} \triangle \mathcal{I}_T$ consists of indices with $|\gamma_k - T| \le C_{10}$ or $|\gamma_k - 2T| \le C_{10}$, and has $O(\log T)$ elements by (2.3). For every $k \in \mathcal{I}_T^{\tau}$ the points $x_k^{(j)}$ lie in $[T - 2T^{1/2}, 2T + 2T^{1/2}]$ and Lemma 8.7 gives $|x_k^{(j)} - \tau_k| = |y_k^{(j)}| \le (\sigma_{X_j} + 1)/\mathcal{L}(\tau_k) \ll T^{1/320}/\log T$, because $X_j \le T^{1/160}$ for every $j \le m^*$ (proof of Theorem 8.11, *the three conditions*, with $r = 1$) and $\sigma_X \ll \sqrt{X}/\log X$. Hence $|x_k^{(m^*)} - \gamma_k| \le C_{10} + O(T^{1/320}/\log T)$ for $k \in \mathcal{I}_T^{\tau} \triangle \mathcal{I}_T$, these indices contribute $O(\log T \cdot T^{1/160}/\log^2 T) = O(T^{1/160})$ to the sum, and (8.15) gives

$$\sum_{k \in \mathcal{I}_T^{\tau}} \bigl(x_k^{\uparrow} - \gamma_k\bigr)^2 \le \frac{C' \,\#\mathcal{I}_T}{\log^2 T} + O\bigl(T^{1/160}\bigr) \le \frac{(C'+1)\,\#\mathcal{I}_T^{\tau}}{\log^2 T}$$

for $T \ge T_0$, since $\#\mathcal{I}_T^{\tau} = \#\mathcal{I}_T + O(\log T)$ and $\#\mathcal{I}_T \asymp T \log T$. □

The cutoff of the third step, T^{c_3} with $c_3 \asymp H_2^{-9/16}$ instead of $H_2^{-1/2}$, is what the multiplicity costs: the exponent $\frac{9}{16} = \frac{1}{2} + \frac{1}{16}$ comes from the factor $\sqrt{H_2}$ of (8.12) through the exponent $\frac{1}{8r} \le \frac{1}{8}$ of Lemma 8.5. The constants of Fujii's and Montgomery–Vaughan's bounds, which grow with the moment order 2^{m-j} used at level j, are absorbed by the normalization $H_j^{-1/2}$, because H_j grows as a tower in $m-j$ while the orders grow geometrically.

For the one-step clock with a fixed cutoff exponent c, bounded mean-square error in units of zeros is equivalent to bounded mean-square transport, $\|r(T^c)\|_{2,\mathcal{I}_T} = O_c(1)$ if and only if $\|\mathfrak{t}(T^c)\|_{2,\mathcal{I}_T} = O_c(1)$ (Remark 7.9). The available transport estimates, Lemma 7.2 and Theorem 7.5, do not establish this boundedness. This limitation concerns the one-step construction; it does not establish a corresponding obstruction for the final cutoff of the iterated construction.

9. Theorem D′: a regularized clock with a fixed cutoff

For a fixed cutoff exponent the one-step clock is controlled in mean square if and only if its transport is (Remark 7.9), and this is not proved here. The construction below avoids the transport from τ_k: it replaces $F_X = N_0 + S_X$ by its best L^4 approximation among increasing piecewise linear functions with slopes comparable to $\mathcal{L}$, and solves the clock equation for that function. A fixed power of the height then suffices, without Newton iteration.

9.1. The construction

For $T \ge T_0$ let $\Lambda_T = \frac{\log T}{2\pi}$, $\mathsf{J}_T = [\frac{T}{2}, 3T]$, $\mathsf{L}_T = \lceil \frac{5}{2} T\Lambda_T \rceil$, $\mathsf{h}_T = \frac{5T}{2\mathsf{L}_T} \le \Lambda_T^{-1}$ and $\mathsf{t}_i = \frac{T}{2} + i\mathsf{h}_T$, $0 \le i \le \mathsf{L}_T$. Let $\mathcal{K}_T$ be the set of continuous functions P on J_T that are linear on each $[\mathsf{t}_{i-1}, \mathsf{t}_i]$ with slope in $[\frac{1}{2}\Lambda_T, 2\Lambda_T]$. For $X \ge 2$ let $F_X = N_0 + S_X$.

Definition 9.1 (Regularized clock)**.** $\widehat{F} = \widehat{F}_{T,X}$ is the element of $\mathcal{K}_T$ that minimizes $\int_{\mathsf{J}_T} |P - F_X|^4\, dt$ over $P \in \mathcal{K}_T$. For every integer k with $\widehat{F}(\frac{T}{2}) < k - \frac{1}{2} < \widehat{F}(3T)$, $\widehat{x}_k = \widehat{x}_k(T, X) \in \mathsf{J}_T$ is the solution of $\widehat{F}(\widehat{x}_k) = k - \frac{1}{2}$, and $\widehat{u}_k = k - \frac{1}{2} - \widehat{F}(\gamma_k)$ when $\gamma_k \in \mathsf{J}_T$.

Lemma 9.2 (The regularized clock is well defined)**.** *For $T \ge T_0$ and $X \ge 2$ the minimizer $\widehat{F}$ exists and is unique; $\widehat{F}$ is strictly increasing with $\frac{1}{2}\Lambda_T \le \widehat{F}' \le 2\Lambda_T$ on each mesh interval; each $\widehat{x}_k$ is unique, obtained by locating the mesh interval where $\widehat{F}$ crosses $k - \frac{1}{2}$ and interpolating linearly; and*

$$\frac{\pi}{\log T} \le \widehat{x}_{k+1} - \widehat{x}_k \le \frac{4\pi}{\log T} \tag{9.1}$$

whenever both points are defined. $\widehat{F}$ and the $\widehat{x}_k$ depend only on T, N_0 and the primes $p \le X$; no value of ζ and no zero enters.

Proof. For $T \ge T_0$ and $t \in [\frac{T}{2} - 1, 3T + 1]$, $N_0'(t) = \mathcal{L}(t) = \frac{1}{2\pi} \log \frac{t}{2\pi} \in [\frac{3}{4}\Lambda_T, \frac{5}{4}\Lambda_T]$. An element of $\mathcal{K}_T$ is determined by its nodal values $v = (v_0, \dots, v_{\mathsf{L}_T})$, and $\mathcal{K}_T$ corresponds to the closed convex polyhedron $\{v : \frac{1}{2}\Lambda_T \mathsf{h}_T \le v_i - v_{i-1} \le 2\Lambda_T \mathsf{h}_T\}$; it contains the interpolation of N_0, whose chord slopes are values of N_0'. The functional $\Phi(v) = \int_{\mathsf{J}_T} |P_v - F_X|^4\, dt$ is continuous; it is strictly convex, because $z \mapsto |z|^4$ is strictly convex and $P_v \ne P_w$ on a set of positive measure when $v \ne w$; and it tends to infinity on the polyhedron as $|v| \to \infty$, because the slopes are bounded and F_X is bounded on J_T, so $|v_0| \to \infty$ and $|P_v - F_X| \to \infty$ uniformly. Hence there

is a unique minimizer. The slope bounds give strict monotonicity and the uniqueness of $\widehat{x}_k$, and $1 = \widehat{F}(\widehat{x}_{k+1}) - \widehat{F}(\widehat{x}_k)$ lies between $\frac{1}{2}\Lambda_T$ and $2\Lambda_T$ times $\widehat{x}_{k+1} - \widehat{x}_k$, which is (9.1). The data of the program are N_0 and S_X. □

9.2. Three lemmas

Lemma 9.3 (Theorem A for Lipschitz models)**.** *Theorems 4.1 and 4.2 hold for every real Lipschitz function F on an open interval containing $[\gamma_{k_1}, \gamma_{k_2}]$, with F' its derivative almost everywhere; in particular for the functions of $\mathcal{K}_T$.*

Proof. The integrands are bounded and measurable. In the proofs, the only use of $F \in C^1$ is the evaluation of $\int_{\gamma_k}^{\gamma_{k+1}} \mathfrak{g}(k - F)F'\,dt$ on a gap. If F is Lipschitz, $t \mapsto \mathfrak{G}(k - F(t))$ is Lipschitz on $[\gamma_k, \gamma_{k+1}]$ ($\mathfrak{G}$ is C^1 and F is bounded there), hence absolutely continuous, with derivative $-\mathfrak{g}(k - F(t))F'(t)$ almost everywhere (chain rule for a C^1 function of a Lipschitz function), and the fundamental theorem of calculus for absolutely continuous functions gives the same value. The rest of both proofs is unchanged. □

Lemma 9.4 (A slope-constrained model of the counting function)**.** *There are absolute constants T_0 and C such that for $T \geq T_0$ there is $P_N \in \mathcal{K}_T$ with $\int_{\mathsf{J}_T} |P_N - N|^4\,dt \leq CT$.*

Proof. Let $\mathsf{a} = \frac{1}{2}\Lambda_T$, $\mathsf{b} = 2\Lambda_T$, $h_j = j/\Lambda_T$, $\phi(x) = \mathsf{a}x$ for $x \geq 0$ and $\phi(x) = \mathsf{b}x$ for $x < 0$, and

$$H(t) = \sup_{s \in \mathsf{J}_T} \Big(N(s) + \phi(t - s) \Big) \qquad (t \in \mathsf{J}_T).$$

Each function $g_s(t) = N(s) + \phi(t - s)$ satisfies $\mathsf{a}(y - x) \leq g_s(y) - g_s(x) \leq \mathsf{b}(y - x)$ for $x < y$, and so does their supremum, which is finite because N is bounded on J_T; and $H \geq N$ (take $s = t$). Thus

$$0 \leq H(t) - N(t) = \max\Big\{ \sup_{s \leq t} \Big(\mathsf{a}(t - s) - (N(t) - N(s)) \Big),\ \sup_{s > t} \Big(N(s) - N(t) - \mathsf{b}(s - t) \Big) \Big\},$$

with $s \in \mathsf{J}_T$. By (2.3) and $|\varpi|/\pi < \frac{1}{100}$, $N(y) - N(x) = N_0(y) - N_0(x) + S(y) - S(x) + \theta$ with $|\theta| < \frac{1}{50}$. Let $D_j^{\pm}(t) = \pm(S(t \pm h_j) - S(t))$.

If $s \leq t$ and $t - s \in [h_j, h_{j+1})$, $j \geq 0$, then $s \leq t - h_j$, $N(s) \leq N(t - h_j)$ and $\mathsf{a}(t - s) < \mathsf{a}h_j + \frac{1}{2}$, so, with $N_0(t) - N_0(t - h_j) \geq \frac{3}{4}j$,

$$\mathsf{a}(t - s) - (N(t) - N(s)) \leq \tfrac{1}{2}j - \tfrac{3}{4}j + |D_j^-(t)| + 1 = |D_j^-(t)| - \tfrac{1}{4}j + 1,$$

with $D_0^- = 0$. If $s > t$ and $s - t \in [h_j, h_{j+1})$, then $N(s) \leq N(t + h_{j+1})$ and $\mathsf{b}(s - t) \geq \mathsf{b}h_{j+1} - 2$, and $N_0(t + h_{j+1}) - N_0(t) \leq \frac{5}{4}(j + 1)$ since $t + h_{j+1} \leq 3T + 1$, so

$$N(s) - N(t) - \mathsf{b}(s - t) \leq |D_{j+1}^+(t)| - \tfrac{3}{4}(j + 1) + 3.$$

Hence, with $Z_j = \max(|D_j^+|, |D_j^-|)$ for $1 \leq j \leq j^* = \lceil \frac{5}{2}T\Lambda_T \rceil + 1$ (the terms with $t - h_j < \frac{T}{2}$ omitted from D_j^-),

$$0 \leq H - N \leq 3 + \sup_{1 \leq j \leq j^*} \Big(Z_j - \tfrac{1}{4}j \Big)_+, \qquad (H - N)^4 \leq 8\Big(81 + \sum_{j=1}^{j^*} \Big(Z_j - \tfrac{1}{4}j \Big)_+^4 \Big).$$

Since $(z-\frac{j}{4})_+^4 \le (4z/j)^4 z^4 = 256\, z^8/j^4$,

$$\int_{\mathsf{J}_T} (H-N)^4\,dt \le 648\cdot\tfrac{5}{2}T + 2048\sum_{j=1}^{j^*} j^{-4}\int_{\mathsf{J}_T} Z_j^8\,dt.$$

By the substitution $u=t-h_j$ in the terms D_j^-, $\int_{\mathsf{J}_T} Z_j^8 \le 2\int_{T/2}^{3T}(S(u+h_j)-S(u))^8 du$, and Theorem 3.8 with $k=4$ on $(\frac{T}{2},T)$, $(T,2T)$ and $(2T,4T)$ gives, with $h_j\le 1$ and $h_j\log(4T)+3\le 8j+3$,

$$\int_{T/2}^{3T}\Big(S(u+h_j)-S(u)\Big)^8 du \ll T\Big(\log((8j+3)e^4)\Big)^4 \ll T\log^4(j+3)\quad (j\le\Lambda_T),$$

For $j>\Lambda_T$, so $h_j>1$, use instead $(S(u+h_j)-S(u))^8 \le 2^7(S(u+h_j)^8+S(u)^8)$ and Theorem 3.3 with $j=4$ on dyadic intervals covering $[\frac{T}{4},6T]$, which contains u and $u+h_j$ since $h_j\le h_{j^*}\le \frac{5}{2}T+1$; this gives $\ll T(\log\log T)^4$. Therefore $\int_{\mathsf{J}_T}(H-N)^4 \ll T + T\sum_{j\ge1} j^{-4}\log^4(j+3) + T(\log\log T)^4\Lambda_T^{-3} \ll T$. Let P_N be the linear interpolation of H at the nodes t_i. Its slopes are chord slopes of H, so $P_N\in\mathcal{K}_T$; on $[\mathsf{t}_{i-1},\mathsf{t}_i]$ both H and P_N lie between $H(\mathsf{t}_{i-1})+\mathsf{a}(t-\mathsf{t}_{i-1})$ and $H(\mathsf{t}_{i-1})+\mathsf{b}(t-\mathsf{t}_{i-1})$, so $|P_N-H|\le(\mathsf{b}-\mathsf{a})\mathsf{h}_T\le\frac{3}{2}$. Hence $|P_N-N|\le H-N+\frac{3}{2}$ and $\int_{\mathsf{J}_T}|P_N-N|^4\ll T$. □

Lemma 9.5 (The prime model on the block)**.** *There are absolute constants T_0 and C such that for $T\ge T_0$ and $2\le X\le\sqrt{T/2}$, with $c=\log X/\log T$,*

$$\int_{\mathsf{J}_T}|F_X-N|^4\,dt \le CT\Big(1+\log(1/c)\Big)^2.$$

Proof. By (2.3), $F_X-N=-(S-S_X)-\varpi/\pi$ with $|\varpi|/\pi<\frac{1}{100}$. Apply (5.11) on $(T',2T']$ for $T'\in\{\frac{T}{2},T,2T\}$, which cover J_T: for $T\ge T_0$, $2\le X\le(T/2)^{1/2}\le T'^{1/2}$, and by Lemma 5.8 the quantity V computed at height T' is $\log(\log T'/\log X)+O(1)=\log(1/c)+O(1)$, with $\log T'/\log T=1+O(1/\log T)$. □

9.3. The theorem

Theorem 9.6 (Theorem D′)**.** *There are absolute constants T_0 and C such that for $T\ge T_0$ and $2\le X\le\sqrt{T/2}$, with $c=\log X/\log T$ and $\widehat{F}$, $\widehat{x}_k$, $\widehat{u}_k$ as in Definition 9.1, and with $\mathcal{I}$ either of the index sets $\mathcal{I}_T$ and $\mathcal{I}_T^\tau=\{k: T\le\tau_k<2T\}$:*

(a) $\widehat{x}_k$ is defined for every $k\in\mathcal{I}_T\cup\mathcal{I}_T^\tau$, and these points are increasing in k with gaps (9.1)*;*

(b) $\langle\widehat{u}_k^4\rangle_{\mathcal{I}}\le C(1+\log(1/c))^2$ and $|\widehat{x}_k-\gamma_k|\le 4\pi|\widehat{u}_k|/\log T$; hence

$$\Big\langle|\widehat{x}_k-\gamma_k|^4\Big\rangle_{\mathcal{I}} \le \frac{C(1+\log(1/c))^2}{\log^4 T},\qquad \Big\langle|\widehat{x}_k-\gamma_k|^2\Big\rangle_{\mathcal{I}} \le \frac{C(1+\log(1/c))}{\log^2 T},\tag{9.2}$$

and for every $\lambda>0$

$$\frac{\#\{k\in\mathcal{I}:|\widehat{x}_k-\gamma_k|>\lambda\cdot 2\pi/\log T\}}{\#\mathcal{I}} \le \frac{C(1+\log(1/c))^2}{(2\pi\lambda)^4};\tag{9.3}$$

(c) $\max_{k\in\mathcal{I}_T\cup\mathcal{I}_T^\tau}|\widehat{x}_k-\gamma_k|\le CT^{1/5}$;

(d) if $X\le T^{1/4}$, *then* $\big\langle (S_X(\gamma_k)-S_X(\widehat{x}_k))^2\big\rangle_{\mathcal{I}}\le C(1+\log(1/c))\big((1+\log X)/\log T\big)^2\ll 1$, *with an absolute implied constant.*

In particular every fixed $0<c<\frac12$ *is admissible for* $T\ge T_0(c)$, *with the constant* C *independent of* c.

Proof. The regularized model is close to N. By the minimizing property, Lemma 9.4 and Lemma 9.5, with $\|\cdot\|_4$ the norm of $L^4(\mathsf{J}_T)$,

$$\|\widehat{F}-N\|_4\le\|\widehat{F}-F_X\|_4+\|F_X-N\|_4\le\|P_N-F_X\|_4+\|F_X-N\|_4\le\|P_N-N\|_4+2\|F_X-N\|_4, \tag{9.4}$$

so $\int_{\mathsf{J}_T}|\widehat{F}-N|^4\le C_{\mathrm{r}}T(1+\log(1/c))^2$ with an absolute C_{r}.

Two good heights. Let C_{g} bound the gaps between consecutive ordinates (Theorem 3.1) and $K=(16C_{\mathrm{r}})^{1/4}(1+\log(1/c))^{1/2}$. By Chebyshev's inequality the set where $|\widehat{F}-N|>K$ has measure at most $T/16$, so there are $t_\flat\in[\frac{T}{2}+C_{\mathrm{g}},\frac{3T}{4}]$ and $t_\sharp\in[\frac{9T}{4},3T-C_{\mathrm{g}}-1]$ with $|\widehat{F}(t_\flat)-N(t_\flat)|\le K$ and $|\widehat{F}(t_\sharp)-N(t_\sharp)|\le K$. Since $c\ge\log 2/\log T$, $K\ll(\log\log T)^{1/2}$.

(a). $\widehat{F}$ is increasing, so $\widehat{F}(\frac{T}{2})\le\widehat{F}(t_\flat)\le N(\frac{3T}{4})+K$ and $\widehat{F}(3T)\ge\widehat{F}(t_\sharp)\ge N(\frac{9T}{4})-K$. For $k\in\mathcal{I}_T$, $N(T)+1\le k\le N(2T)$; for $k\in\mathcal{I}_T^\tau$, $k-\frac12=N_0(\tau_k)\in[N_0(T),N_0(2T))$, and $|N-N_0|\ll\log T$ (Theorem 3.1). Since $N(T)-N(\frac{3T}{4})$ and $N(\frac{9T}{4})-N(2T)$ are $\gg T\log T$, $\widehat{F}(\frac{T}{2})<k-\frac12<\widehat{F}(3T)$ for $T\ge T_0$. Monotonicity and gaps are Lemma 9.2. Both index sets have $\asymp T\log T$ elements, and for $k\in\mathcal{I}_T^\tau$ the ordinate γ_k lies in $[T-C_{10},2T+C_{10}]$ ($|\delta_k|\le C_{10}$, Theorem 3.1).

(b). Let $k_1=N(t_\flat)$ and $k_2=N(t_\sharp)+1$. Then $t_\flat-C_{\mathrm{g}}<\gamma_{k_1}\le t_\flat<\gamma_{k_1+1}$ and $\gamma_{k_2-1}\le t_\sharp<\gamma_{k_2}\le t_\sharp+C_{\mathrm{g}}$, so $[\gamma_{k_1},\gamma_{k_2}]\subset(\frac{T}{2},3T)$ and $\mathcal{I}_T\cup\mathcal{I}_T^\tau\subset\{k:k_1<k<k_2\}$, because $\gamma_{k_1}\le t_\flat\le\frac{3T}{4}<T-C_{10}$ and $\gamma_{k_2}>t_\sharp\ge\frac{9T}{4}>2T+C_{10}$. As $\widehat{F}$ is increasing, $\widehat{u}_{k_1}\ge N(t_\flat)-\frac12-\widehat{F}(t_\flat)\ge-K-\frac12$ and $\widehat{u}_{k_2}\le N(t_\sharp)+\frac12-\widehat{F}(t_\sharp)\le K+\frac12$. Lemma 9.3 with $j=2$, $\Psi_2(u)=u^4+\frac12u^2+\frac{1}{80}\ge u^4$ and $\widehat{F}'\le2\Lambda_T$ gives

$$\begin{aligned}\sum_{k\in\mathcal{I}}\widehat{u}_k^4\le\sum_{k_1<k<k_2}\Psi_2(\widehat{u}_k)&=\int_{\gamma_{k_1}}^{\gamma_{k_2}}(N-\widehat{F})^4\widehat{F}'\,dt-\frac{(\frac12+\widehat{u}_{k_1})^5+(\frac12-\widehat{u}_{k_2})^5}{5}\\&\le2\Lambda_T\int_{\mathsf{J}_T}|N-\widehat{F}|^4\,dt+K^5,\end{aligned}$$

and $\#\mathcal{I}\asymp T\log T$ gives the bound for $\langle\widehat{u}_k^4\rangle_{\mathcal{I}}$, for either index set. Since $\widehat{F}(\gamma_k)-\widehat{F}(\widehat{x}_k)=-\widehat{u}_k$ and $\widehat{F}'\ge\frac12\Lambda_T$, $|\widehat{x}_k-\gamma_k|\le2|\widehat{u}_k|/\Lambda_T=4\pi|\widehat{u}_k|/\log T$, which gives (9.2), the second bound by Cauchy–Schwarz, and (9.3) by Markov's inequality.

(c). Let $k\in\mathcal{I}_T\cup\mathcal{I}_T^\tau$, so that γ_k and $\widehat{x}_k$ lie in J_T, and let $d=|\widehat{x}_k-\gamma_k|>0$. If $\widehat{x}_k>\gamma_k$, then for $t\in(\gamma_k,\widehat{x}_k)$, $N(t)\ge k$ and $\widehat{F}(t)\le k-\frac12-\frac12\Lambda_T(\widehat{x}_k-t)$; if $\widehat{x}_k<\gamma_k$, then for $t\in(\widehat{x}_k,\gamma_k)$, $N(t)\le k-1$ and $\widehat{F}(t)\ge k-\frac12+\frac12\Lambda_T(t-\widehat{x}_k)$. In both cases $\int_{\mathsf{J}_T}|N-\widehat{F}|^4\ge(\frac12\Lambda_T)^4d^5/5$, so $d^5\ll T(1+\log(1/c))^2/\log^4T\ll T$.

(d). By (c), for $k\in\mathcal{I}$ the interval $[\widehat{x}_k-d_k,\widehat{x}_k+d_k]$, $d_k=|\widehat{x}_k-\gamma_k|$, lies in $[T-2T^{1/2},2T+2T^{1/2}]$ for $T\ge T_0$, where the cutoff $\varphi^\sharp$ of Lemma 8.5 equals 1. By (9.1) and $\mathfrak{b}_T=2\pi/\log(3T)<2\pi/\log T$, every interval of length $\mathfrak{b}_T$ contains at most 2 of the points $\widehat{x}_k$,

so $\tilde n \le 2$ and $\int_{\mathbb R}\tilde n^2 \le 2\int_{\mathbb R}\tilde n = 2\mathfrak b_T\#\mathcal I$ in Lemma 8.5, which with $r=1$ and $X \le T^{1/4}$ gives $\langle g(\widehat x_k)^4\rangle_{\mathcal I} \ll (1+\log X)^4$ for $g=\mathcal M(\varphi^\sharp S'_X)$; the proof of that lemma uses of its index set only that it has $\asymp T\log T$ elements and that the points lie in $[T-2T^{1/2},2T+2T^{1/2}]$, so it applies to $\mathcal I_T^\tau$ as well. Since $|S_X(\gamma_k)-S_X(\widehat x_k)| \le 2|\gamma_k-\widehat x_k|\,g(\widehat x_k)$, Cauchy–Schwarz and (9.2) give

$$\Big\langle (S_X(\gamma_k)-S_X(\widehat x_k))^2\Big\rangle_{\mathcal I} \le 4\Big\langle|\gamma_k-\widehat x_k|^4\Big\rangle_{\mathcal I}^{1/2}\Big\langle g(\widehat x_k)^4\Big\rangle_{\mathcal I}^{1/2} \ll (1+\log(1/c))\frac{(1+\log X)^2}{\log^2 T},$$

and $(1+\log X)/\log T \le (1+\frac{1}{\log 2})c$ because $c \ge \log 2/\log T$, while $c^2(1+\log(1/c)) \le 1$. □

9.4. A certified tolerance for the convex program

The minimizer $\widehat F$ is computed by an iterative solver. The following statement shows that an approximate minimizer with a controlled excess of the objective gives points with the same guarantee, and that the excess can be certified from the data of the program alone. Write $P_{\mathbf v}\in\mathcal K_T$ for the function with nodal values $\mathbf v=(v_0,\dots,v_{\mathsf L_T})$,

$$E(\mathbf v)=\int_{\mathsf J_T}|P_{\mathbf v}-F_X|^4\,dt, \qquad E_*=\min_{P_{\mathbf v}\in\mathcal K_T}E(\mathbf v)=E(\widehat F).$$

Proposition 9.7 (Approximate minimizers and a dual certificate)**.** *Let T, X, c be as in Theorem 9.6 and $0\le\eta_{\mathrm{reg}}\le 1$.*

(a) If $P_{\mathbf v}\in\mathcal K_T$ satisfies $E(\mathbf v)\le E_*+\eta_{\mathrm{reg}}T$, and $\widehat x_k^{(\mathbf v)}$ are its level points, $P_{\mathbf v}(\widehat x_k^{(\mathbf v)})=k-\frac12$, then (a)–(d) of Theorem 9.6 hold for these points, with $(1+\log(1/c))^2$ replaced by $(1+\log(1/c))^2+\eta_{\mathrm{reg}}$ in the fourth-moment bounds and in (9.3), and $1+\log(1/c)$ by $1+\log(1/c)+\sqrt{\eta_{\mathrm{reg}}}$ in the second-moment bound; since $\eta_{\mathrm{reg}}\le 1$, all of them hold with C replaced by $2C$.

(b) Let $\mathsf D\mathbf v=(v_i-v_{i-1})_{1\le i\le \mathsf L_T}$, $\mathsf a=\frac12\Lambda_T$, $\mathsf b=2\Lambda_T$, $|\mathsf J_T|=\frac52 T$, and for $\boldsymbol\mu_\pm\in\mathbb R^{\mathsf L_T}$ with nonnegative entries

$$\mathscr L(\mathbf v,\boldsymbol\mu_+,\boldsymbol\mu_-)=E(\mathbf v)+\boldsymbol\mu_+^{\mathsf T}(\mathsf D\mathbf v-\mathsf b\mathsf h_T\mathbf 1)+\boldsymbol\mu_-^{\mathsf T}(\mathsf a\mathsf h_T\mathbf 1-\mathsf D\mathbf v),$$
$$\mathbf r=\nabla E(\mathbf v)+\mathsf D^{\mathsf T}(\boldsymbol\mu_+-\boldsymbol\mu_-), \qquad A_{\mathrm{reg}}=\frac{\mathsf h_T^2}{108\,|\mathsf J_T|},$$
$$B_{\mathrm{reg}}=\mathscr L(\mathbf v,\boldsymbol\mu_+,\boldsymbol\mu_-)-\frac{3}{4^{4/3}}A_{\mathrm{reg}}^{-1/3}\|\mathbf r\|_2^{4/3},$$

where $\partial_i E(\mathbf v)=4\int_{\mathsf J_T}(P_{\mathbf v}-F_X)^3\mathsf B_i\,dt$ and $\mathsf B_i$ is the hat function of the node $\mathsf t_i$. Then $B_{\mathrm{reg}}\le E_*$ for every $\mathbf v\in\mathbb R^{\mathsf L_T+1}$. Consequently, if $P_{\mathbf v}\in\mathcal K_T$, $\boldsymbol\mu_\pm\ge 0$ and $E(\mathbf v)-B_{\mathrm{reg}}\le\eta_{\mathrm{reg}}T$, then $E(\mathbf v)\le E_*+\eta_{\mathrm{reg}}T$.

All quantities in (b) are computed from N_0, the primes $p\le X$ and the mesh; no zero and no value of ζ enters.

Proof. (a) In (9.4) replace $\|\widehat F-F_X\|_4\le\|P_N-F_X\|_4$ by $\|P_{\mathbf v}-F_X\|_4\le(E_*+\eta_{\mathrm{reg}}T)^{1/4}\le\|P_N-F_X\|_4+(\eta_{\mathrm{reg}}T)^{1/4}$, so that $\int_{\mathsf J_T}|P_{\mathbf v}-N|^4\le 8\Big(C_{\mathrm r}T(1+\log(1/c))^2+\eta_{\mathrm{reg}}T\Big)$. The rest of the proof of Theorem 9.6 uses of $\widehat F$ only its membership in $\mathcal K_T$ and this bound; with

$K^4 = 16 \cdot 8(C_{\mathrm{r}}(1+\log(1/c))^2 + \eta_{\mathrm{reg}})$ and $\eta_{\mathrm{reg}} \le 1$ the good heights exist and the boundary terms are bounded as before. In (d), $(1+\log(1/c))^2 + \eta_{\mathrm{reg}} \le 2(1+\log(1/c))^2$.

(b) For real x and d, $(x+d)^4 - x^4 - 4x^3 d - \frac{1}{3}d^4 = 6d^2(x+\frac{d}{3})^2 \ge 0$. Integrating with $x = P_{\mathbf{v}} - F_X$ and $d = P_{\mathbf{d}}$, $E(\mathbf{v}+\mathbf{d}) \ge E(\mathbf{v}) + \nabla E(\mathbf{v})^{\mathsf{T}}\mathbf{d} + \frac{1}{3}\int_{\mathsf{J}_T} P_{\mathbf{d}}^4$. By Cauchy–Schwarz, $\int_{\mathsf{J}_T} P_{\mathbf{d}}^4 \ge |\mathsf{J}_T|^{-1}(\int_{\mathsf{J}_T} P_{\mathbf{d}}^2)^2$, and $\int_{\mathsf{J}_T} P_{\mathbf{d}}^2 = \mathbf{d}^{\mathsf{T}}\mathsf{G}\mathbf{d}$ with the tridiagonal mass matrix G of the hat functions, whose diagonal entries are $\frac{2}{3}\mathsf{h}_T$ ($\frac{1}{3}\mathsf{h}_T$ at the two ends) and whose off-diagonal entries are $\frac{1}{6}\mathsf{h}_T$; Gershgorin's theorem gives $\lambda_{\min}(\mathsf{G}) \ge \frac{1}{6}\mathsf{h}_T$. Hence $\frac{1}{3}\int P_{\mathbf{d}}^4 \ge A_{\mathrm{reg}}\|\mathbf{d}\|_2^4$. The constraint terms of $\mathscr{L}$ are linear in $\mathbf{v}$, so $\mathscr{L}(\mathbf{v}+\mathbf{d},\boldsymbol{\mu}_\pm) \ge \mathscr{L}(\mathbf{v},\boldsymbol{\mu}_\pm) + \mathbf{r}^{\mathsf{T}}\mathbf{d} + A_{\mathrm{reg}}\|\mathbf{d}\|_2^4 \ge \mathscr{L}(\mathbf{v},\boldsymbol{\mu}_\pm) - \frac{3}{4^{4/3}}A_{\mathrm{reg}}^{-1/3}\|\mathbf{r}\|_2^{4/3} = B_{\mathrm{reg}}$ for every $\mathbf{d}$, by minimizing $-\|\mathbf{r}\|_2 s + A_{\mathrm{reg}} s^4$ over $s \ge 0$. For feasible $\mathbf{w}$ (that is $P_{\mathbf{w}} \in \mathcal{K}_T$) and $\boldsymbol{\mu}_\pm \ge 0$ the constraint terms are ≤ 0, so $E(\mathbf{w}) \ge \mathscr{L}(\mathbf{w},\boldsymbol{\mu}_\pm) \ge B_{\mathrm{reg}}$; taking $\mathbf{w}$ the minimizer gives $E_* \ge B_{\mathrm{reg}}$. The last assertion follows. □

A numerical certification would require enclosures of the integrals, the gradient and the inequalities in directed rounding; no solver was run for this paper, and the certificate controls the computation of the minimizer, not the constants T_0 and C.

Remark 9.8. (a) Compared with Theorem 8.11 and Corollary 8.15, Theorem 9.6 reaches the scale of one mean spacing with a fixed cutoff exponent and without Newton iteration, in fourth mean, with increasing and separated points, on the clock window $\mathcal{I}_T^\tau$ as well as on $\mathcal{I}_T$; the dependence on c is explicit and parallel to Theorem 5.1. The separation (9.1) is imposed by the slope constraints; it is not a statement about the spacings of the zeros. (b) The cost is locality: $\widehat{F}$ is the solution of a convex program with $\mathsf{L}_T + 1 \asymp T\log T$ unknowns whose data are S_X on the whole block J_T, and it depends on the block; the points of Theorem 7.3 and of the iteration are explicit expressions index by index. A version on short overlapping windows is not treated here. (c) Part (d) bounds the transport from the regularized points, not from τ_k; for the one-step clock τ_k^X with a fixed cutoff exponent, Theorem 7.5 gives $1 + \log\log\log T$, and $O_c(1)$ is not proved (Remark 7.6). (d) The slope constraints $[\frac{1}{2}\Lambda_T, 2\Lambda_T]$ enclose the main slope $\mathcal{L}$ of the counting function on J_T and are fixed in advance; no zero is used to choose them, and the ordinates enter only in the proofs (through P_N in Lemma 9.4 and the heights $t_\flat$, $t_\sharp$).

10. Theorem E: the pointwise envelope

Let $\mathfrak{a} = \limsup_{t\to\infty} |S(t)|/\log t$. By Theorem 3.2, $\mathfrak{a} \le 0.10076$, and $\mathfrak{a} \le 0.1013$ also follows from [38]; whether $\mathfrak{a} = 0$ is not known.

Proposition 10.1 (The envelope of the clock is the envelope of S)**.** *(a)* $\limsup_{k\to\infty} |\tau_k - \gamma_k| = 2\pi\mathfrak{a}$. *(b) If* $X_k \ge 2$ *and* $\sigma_{X_k} = o(\log\tau_k)$*, in particular if* $X_k \le (\log\tau_k)^{2-\varepsilon}$*, then* $\limsup_{k\to\infty} |\tau_k^{X_k} - \gamma_k| = 2\pi\mathfrak{a}$.

Proof. (a) The values $S(\gamma_k^\pm)$ are limits of values of S at non-ordinates. For $\varepsilon > 0$, (2.8) gives $|s_k| \le (\mathfrak{a}+\varepsilon)\log\gamma_k + O_\varepsilon(1)$, and $|\delta_k| = |s_k|/\mathcal{L}(\xi_k)$ with $\mathcal{L}(\xi_k) = \frac{\log\gamma_k}{2\pi}(1+o(1))$ since $\delta_k = O(1)$; so $\limsup|\delta_k| \le 2\pi\mathfrak{a}$. Conversely, let $t > \gamma_1$ be a non-ordinate, let $\gamma < t < \gamma'$ be the nearest ordinates and $k = N(t)$, so that $\gamma_k = \gamma$ and $\gamma_{k+1} = \gamma'$. On (γ,γ') the function G is continuous and decreasing ($G' = -\mathcal{L}$), so $G(\gamma'^-) \le G(t) \le G(\gamma^+)$, while $s_k = k - \frac{1}{2} - N_0(\gamma) = G(\gamma^+) - \frac{1}{2}$ and $s_{k+1} = k + \frac{1}{2} - N_0(\gamma') = G(\gamma'^-) + \frac{1}{2}$. Hence $s_k \ge G(t) - \frac{1}{2}$ and $s_{k+1} \le G(t) + \frac{1}{2}$, so $\max(|s_k|, |s_{k+1}|) \ge |G(t)| - \frac{1}{2}$, and $|G(t) - S(t)| = |\varpi(t)|/\pi \to 0$. Both

ordinates are within bounded distance of t (Theorem 3.1). Choosing non-ordinates $t \to \infty$ with $|S(t)| \ge (\mathfrak{a}-\varepsilon)\log t$ gives indices $k \to \infty$ with $|\delta_k| = |s_k|/\mathcal{L}(\xi_k) \ge 2\pi(\mathfrak{a}-\varepsilon)+o(1)$.

(b) $|\tau_k^{X_k} - \tau_k| = |S_{X_k}(\tau_k)|/\mathcal{L}(\tau_k) \le \sigma_{X_k}/\mathcal{L}(\tau_k) \to 0$, and $\sigma_X \ll \sqrt{X}/\log X$. □

So the corrected clocks with slowly growing cutoffs have exactly the pointwise envelope of the bare clock, and for every cutoff the envelope of τ_k is that of S. A smaller coefficient for the clock is a smaller coefficient for $S(t)$.

Theorem 10.2 (Theorem E)**.** *Let $\mathfrak{e}$ be continuous on $[\frac12,1]$ with $\mathfrak{e}(1)=0$, such that for every $\varepsilon>0$, $\zeta(\sigma+it) \ll_\varepsilon |t|^{\mathfrak{e}(\sigma)+\varepsilon}$ uniformly for $\frac12 \le \sigma \le 1$ and $|t| \ge 2$, and put $\mathfrak{e}(\sigma) = \frac12 - \sigma + \mathfrak{e}(1-\sigma)$ for $0 \le \sigma < \frac12$. Then*

$$\mathfrak{a} \le \frac{1}{2\pi\log 2}\int_0^1 \frac{\mathfrak{e}(\sigma)}{\sqrt{\sigma(2-\sigma)}}\,d\sigma. \tag{10.1}$$

For $J \ge 0$ let $(\mathsf{k}_0,\mathsf{l}_0) = (\frac{13}{84},\frac{55}{84})$, $(\mathsf{k}_j,\mathsf{l}_j) = \mathscr{A}^j(\mathsf{k}_0,\mathsf{l}_0)$ with $\mathscr{A}(\mathsf{k},\mathsf{l}) = \big(\frac{\mathsf{k}}{2\mathsf{k}+2}, \frac12+\frac{\mathsf{l}}{2\mathsf{k}+2}\big)$, $\sigma_j = \mathsf{l}_j - \mathsf{k}_j$, and let $\mathfrak{e}_J$ be the piecewise linear function through the points (σ_j,k_j), $0 \le j \le J$, and $(1,0)$; the first vertices are $(\frac12,\frac{13}{84})$, $(\frac{139}{194},\frac{13}{194})$, $(\frac{173}{207},\frac{13}{414})$, $(\frac{773}{854},\frac{13}{854})$. Let C_J be the right side of (10.1) for $\mathfrak{e}=\mathfrak{e}_J$. Then $\mathfrak{a} \le C_J$ for every J; the sequence C_J decreases to a limit C_∞ with $0 \le C_J - C_\infty \le 3\mathsf{k}_J\sqrt{1-\sigma_J}/(2\pi\log 2)$, and

$$C_\infty = 0.0968151142790\ldots, \qquad \limsup_{k\to\infty}|\tau_k-\gamma_k| = 2\pi\mathfrak{a} \le 2\pi C_\infty = 0.608307303\ldots; \tag{10.2}$$

for comparison $C_0 = 0.1010228\ldots$ (the exponent $\frac{13}{84}$ alone) and $C_7 = 0.0968160\ldots$. Let $\mathcal{P}$ be the set of pairs consisting of $(\frac{13}{84},\frac{55}{84})$, of $(\mathsf{p}_m,\mathsf{q}_m) = \big(\frac{2}{(m-1)^2(m+2)}, 1-\frac{3m-2}{m(m-1)(m+2)}\big)$ for $3 \le m \le 3000$, and of their images $\mathscr{A}^j$, $1 \le j \le 25$, and let $\mathfrak{e}_\mathcal{P}$ be the lower convex hull on $[\frac12,1]$ of the points $(\mathsf{l}-\mathsf{k},\mathsf{k})$, $(\mathsf{k},\mathsf{l}) \in \mathcal{P}$, and $(1,0)$. Let $C_\mathcal{P}$ be the right side of (10.1) for $\mathfrak{e}=\mathfrak{e}_\mathcal{P}$. Then $\mathfrak{a} \le C_\mathcal{P}$ and

$$C_\mathcal{P} = 0.0966530245541\ldots, \qquad \limsup_{k\to\infty}|\tau_k-\gamma_k| \le 2\pi C_\mathcal{P} = 0.6072888637\ldots < 0.6072889. \tag{10.3}$$

Proof. Uniform bounds. Extend $\mathfrak{e}$ by 0 on $[1,\infty)$ and by $\frac12-\sigma$ on $(-\infty,0)$; the extension $\mathfrak{e}^*$ is continuous ($\mathfrak{e}(0) = \frac12 + \mathfrak{e}(1) = \frac12$). For $\sigma \ge 1$, $\zeta(\sigma+it) \ll \log|t|$ uniformly [37, Thm. 3.5]. For $-\frac{1}{50} \le \sigma < \frac12$, $\zeta(\sigma+it) = \chi(\sigma+it)\zeta(1-\sigma-it)$ with $|\chi(\sigma+it)| = (|t|/2\pi)^{1/2-\sigma}(1+O(1/|t|))$ uniformly [37, (4.12.3)], and $|\zeta(1-\sigma-it)| = |\zeta(1-\sigma+it)|$. Hence $\zeta(\sigma+it) \ll_\varepsilon |t|^{\mathfrak{e}^*(\sigma)+\varepsilon}$ uniformly for $-\frac{1}{50} \le \sigma \le 4$, $|t| \ge 2$.

Backlund's method. Let $0 < \varepsilon_\circ < \frac{1}{100}$, $\sigma_\circ = 1+\varepsilon_\circ$, $\varrho = 1+2\varepsilon_\circ$. Then $1 < \sigma_\circ < \varrho$, $\sigma^\sharp = \frac54 + \frac32\varepsilon_\circ$, $\delta^\sharp = \frac14 + \frac{\varepsilon_\circ}{2} \in (0,\frac12)$, and $\varrho/(\sigma_\circ - \frac12) = 2$. For $s = \sigma_\circ + \varrho e^{i\theta}$, $\Re s \in [-\varepsilon_\circ, 2+3\varepsilon_\circ]$ and $|\Im s| \le 2$, so for T large

$$\begin{aligned}\frac{1}{\varkappa}\log|\mathfrak{f}_\varkappa(s)| &\le \log\max_{\pm}|(s\pm iT-1)\zeta(s\pm iT)| \\ &\le \log(T+4) + (\mathfrak{e}^*(\Re s)+\varepsilon)\log(T+2) + C_\varepsilon =: F_\circ(\theta).\end{aligned}$$

an even continuous function. By Theorem 3.16(a), along $\varkappa_m \to \infty$,

$$\limsup_m \mathcal{Z}_{\varkappa_m}(\sigma_\circ,\varrho) \le \log T \cdot \frac{1}{\pi}\int_0^\pi \Big(\mathfrak{e}^*(\sigma_\circ+\varrho\cos\theta)+\varepsilon\Big)d\theta + O_{\varepsilon,\varepsilon_\circ}(1),$$

the terms $\log(T+4)$ and $-\log\sqrt{\varepsilon_\circ^2+T^2}$ cancelling up to $O(1/T)$. The argument defining $S(T)$ runs along $2\to 2+iT\to\sigma^\sharp+iT\to\frac12+iT$, and on the first two segments $\Re s>1$, where $\log\zeta$ is holomorphic; so $\pi S(T)=\arg\zeta(\sigma^\sharp+iT)+\Delta^{1/2}_{\sigma=\sigma^\sharp}\arg((\sigma+iT-1)\zeta(\sigma+iT))-\Delta^{1/2}_{\sigma=\sigma^\sharp}\arg(\sigma+iT-1)$, with $|\arg\zeta(\sigma^\sharp+iT)|\le\log\zeta(\sigma^\sharp)$ and the last term bounded by π. Theorem 3.16(b) with $\varkappa=\varkappa_m\to\infty$ gives, for T large,

$$|S(T)|\le\frac{\log T}{2\log 2}\cdot\frac1\pi\int_0^\pi\Big(\mathfrak{e}^*(1+\varepsilon_\circ+(1+2\varepsilon_\circ)\cos\theta)+\varepsilon\Big)d\theta+O_{\varepsilon,\varepsilon_\circ}(1).$$

Limit. Divide by $\log T$, let $T\to\infty$, then $\varepsilon\to0$ and $\varepsilon_\circ\to0$; $\mathfrak{e}^*$ is bounded and continuous, so by dominated convergence $\mathfrak{a}\le\frac{1}{2\pi\log2}\int_0^\pi\mathfrak{e}^*(1+\cos\theta)\,d\theta$. The integrand vanishes for $0\le\theta\le\frac\pi2$; on $[\frac\pi2,\pi]$ substitute $\sigma=1+\cos\theta$, $d\sigma=-\sin\theta\,d\theta$, $\sin\theta=\sqrt{\sigma(2-\sigma)}$, which gives (10.1).

The profile. Let $\varepsilon'>0$, $(\mathsf{k}_0,\mathsf{l}_0)=(\frac{13}{84}+\varepsilon',\frac{55}{84}+\varepsilon')$ and $(\mathsf{k}_j,\mathsf{l}_j)=\mathscr{A}^j(\mathsf{k}_0,\mathsf{l}_0)$ for $1\le j\le J$, exponent pairs by Theorem 3.14(a),(c), with $\mathsf{l}_j-\mathsf{k}_j\ge\frac12$; by (b), $|\zeta(\mathsf{l}_j-\mathsf{k}_j+it)|\ll|t|^{\mathsf{k}_j}\log|t|$, and $|\zeta(1+it)|\ll\log|t|$. The points $(\mathsf{l}_j-\mathsf{k}_j,\mathsf{k}_j)$ and $(1,0)$ have decreasing ordinates, and Theorem 3.15 between consecutive points gives the hypothesis of the theorem for the piecewise linear function through them. For $\varepsilon'=0$ these are the vertices of $\mathfrak{e}_J$; the right side of (10.1) depends continuously on ε', so $\mathfrak{a}\le C_J$.

The limit. From $\mathsf{k}_{j+1}=\mathsf{k}_j/(2\mathsf{k}_j+2)$ and $\mathsf{l}_{j+1}-\mathsf{k}_{j+1}=\frac12+(\mathsf{l}_j-\mathsf{k}_j)/(2\mathsf{k}_j+2)$ one gets $\mathsf{k}_j\le2^{-j}\mathsf{k}_0$, $\sigma_j<\sigma_{j+1}<1$ and $1-\sigma_{j+1}=(1-\sigma_j+\mathsf{k}_j)/(2\mathsf{k}_j+2)\le\frac12(1-\sigma_j)+\mathsf{k}_j$, so $\sigma_j\to1$. Each new vertex $(\sigma_{J+1},\mathsf{k}_{J+1})$ lies below the chord from (σ_J,k_J) to $(1,0)$: this is $\mathsf{k}_{J+1}(1-\sigma_J)\le\mathsf{k}_J(1-\sigma_{J+1})$, i.e. $\frac{1-\sigma_J}{2\mathsf{k}_J+2}\le\frac{1-\sigma_J+\mathsf{k}_J}{2\mathsf{k}_J+2}$, which holds. Hence $\mathfrak{e}_{J+1}\le\mathfrak{e}_J$ on $[\frac12,1]$ (they agree on $[\frac12,\sigma_J]$), the reflected parts are ordered in the same way, and $C_{J+1}\le C_J$; the limit C_∞ exists. Since $0\le\mathfrak{e}_J-\mathfrak{e}_{J'}\le\mathsf{k}_J$ for $J'\ge J$, supported on $[\sigma_J,1]$ and, after reflection, on $[0,1-\sigma_J]$, and $\sigma(2-\sigma)\ge\sigma$ on $[0,1]$,

$$\begin{aligned}0\le C_J-C_{J'}&\le\frac{\mathsf{k}_J}{2\pi\log2}\Big(\int_{\sigma_J}^1\frac{d\sigma}{\sqrt{\sigma(2-\sigma)}}+\int_0^{1-\sigma_J}\frac{d\sigma}{\sqrt\sigma}\Big)\\&\le\frac{\mathsf{k}_J\big((1-\sigma_J)\sqrt2+2\sqrt{1-\sigma_J}\big)}{2\pi\log2}\le\frac{3\mathsf{k}_J\sqrt{1-\sigma_J}}{2\pi\log2}.\end{aligned}$$

The values. On each linear piece $\mathfrak{e}(\sigma)=\alpha\sigma+\beta$ of $[0,1]$ a primitive of $\mathfrak{e}(\sigma)/\sqrt{\sigma(2-\sigma)}$ is $-\alpha\sqrt{\sigma(2-\sigma)}+(\alpha+\beta)\arcsin(\sigma-1)$; the vertices are rational, and the program `opc_envelope_iterations_p1.py` (Data availability) evaluates C_J from this primitive at 50 digits for $0\le J\le30$. It gives $C_{30}=0.09681511427905377\ldots$ and $3\mathsf{k}_{30}\sqrt{1-\sigma_{30}}/(2\pi\log2)<5\cdot10^{-15}$, hence $C_\infty\in[C_{30}-5\cdot10^{-15},C_{30}]$ and (10.2); C_7 agrees with the 40-digit quadrature of `opc_envelope_coefficient_v14.py`. Multiply by 2π and use Proposition 10.1(a).

The hull profile. Let $\varepsilon'>0$ and replace each $(\mathsf{k},\mathsf{l})\in\mathcal{P}$ by the exponent pair given by Theorem 3.14(a),(c),(d) with ε' added to k and l for Bourgain's pair and to q_m for Heath-Brown's, before applying $\mathscr{A}^j$. Every point satisfies $\mathsf{l}-\mathsf{k}\ge\frac12$ (for $m\ge3$, $\mathsf{q}_m-\mathsf{p}_m\ge\frac23$, and $\mathscr{A}$ maps $\mathsf{l}-\mathsf{k}\ge\frac12$ to $\frac12+(\mathsf{l}-\mathsf{k})/(2\mathsf{k}+2)\ge\frac12$), so Theorem 3.14(b) gives $|\zeta(\mathsf{l}-\mathsf{k}+it)|\ll|t|^{\mathsf{k}}\log|t|$.

The leftmost point is Bourgain's, at $\sigma = \frac{1}{2}$, and the hull ends at $(1, 0)$; a convex function on $[\frac{1}{2}, 1]$ that is positive before 1 and vanishes at 1 is decreasing, so consecutive vertices have decreasing ordinates and Theorem 3.15 between them gives the hypothesis of the theorem for $\mathfrak{e}_{\mathcal{P}}$. The right side of (10.1) depends continuously on ε', so $\mathfrak{a} \le C_{\mathcal{P}}$. The vertices are rational; the program `envelope_best_profile_v9.py` (Data availability) evaluates $C_{\mathcal{P}}$ from the primitive above in interval arithmetic at 50 digits and returns an interval of width below 10^{-25} around the value (10.3). Every finite set of pairs of Theorem 3.14 that contains Bourgain's pair gives an admissible profile; the truncation $m \le 3000$, $j \le 25$ is only a choice of finite set. □

Remark 10.3. (a) Trudgian [38, §5 of the preprint] obtained $\mathfrak{a} \le 0.1013$ from Huxley's exponent $\frac{32}{205}$ at $\sigma = \frac{1}{2}$ by the same method; Theorem 10.2 replaces the exponents and lets the parameters of the disc tend to their limits. With the explicit exponents used by Bellotti and Wong ($\frac{27}{164}$ at $\sigma = \frac{1}{2}$, Yang's lines $\sigma_j = 1 - j/(2^j - 2)$ with exponents $1/(2^j - 2)$, $4 \le j \le 9$ [3]) and linear interpolation between $\sigma = 0$ and $\sigma = \frac{1}{2}$, as in [3, §3.1], the right side of (10.1) is 0.10074, the limit of their $C_1 = 0.10076$ as their parameters tend to $(1, 1)$; the same right-hand profile reflected by the functional equation gives 0.09783. (b) The hull $\mathfrak{e}_{\mathcal{P}}$ uses only published exponent pairs. Trudgian and Yang [40, Lemma 1.1] give four further pairs, $(\frac{4742}{38463}, \frac{35731}{51284})$, $(\frac{18}{199}, \frac{593}{796})$, $(\frac{2779}{38033}, \frac{58699}{76066})$, $(\frac{715}{10238}, \frac{7955}{10238})$; with them and their $\mathscr{A}$-images added to $\mathcal{P}$ the same program gives $2\pi C < 0.6061342$. Tao, Trudgian and Yang [35] give $(\frac{89}{1282}, \frac{997}{1282})$, $(\frac{652397}{9713986}, \frac{7599781}{9713986})$, $(\frac{10769}{351096}, \frac{609317}{702192})$, $(\frac{89}{3478}, \frac{15327}{17390})$; adding them gives $2\pi C < 0.6060224$. Both are preprints, so these values are not stated in Theorem 10.2. The bound $\zeta(\sigma + it) \ll t^{\kappa(1-\sigma)^{3/2}+\varepsilon}$, $\kappa = \frac{8}{63}\sqrt{15}$, of [16, Thm. 5] lies above $\mathfrak{e}_{\mathcal{P}}$ except for $1 - \sigma < 3 \cdot 10^{-7}$, beyond the last vertex used, where it would change $C_{\mathcal{P}}$ by less than 10^{-15}. The $\mathscr{B}$-process maps the points with $\mathsf{l} - \mathsf{k} \ge \frac{1}{2}$ to $\sigma \le \frac{1}{2}$, where the profile is fixed by the functional equation, and adding the $\mathscr{A}$-images ($j \le 10$) of the $\mathscr{B}$-images of the $\mathscr{A}$-images of the pairs with $m \le 300$ does not lower the value (same program). (c) For the Lindelöf profile ($\mathfrak{e} = 0$ on $[\frac{1}{2}, 1]$) the right side is $C_L = (\frac{\sqrt{3}}{2} - \frac{\pi}{6})/(2\pi \log 2) = 0.07862527\ldots$, that is $2\pi C_L = (\frac{\sqrt{3}}{2} - \frac{\pi}{6})/\log 2 = 0.49401719\ldots$ for the clock. Every admissible profile is at least the Lindelöf profile, so with the disc of the proof no exponent profile gives less. For a general admissible disc, $1 < \sigma_\circ < \varrho$, the Lindelöf profile gives $\big(\sqrt{\varrho^2 - u^2} - u \arccos(u/\varrho)\big)/(2\pi \log(\varrho/u))$ with $u = \sigma_\circ - \frac{1}{2}$; a numerical scan of the admissible discs (program `envelope_disc_scan_v9.py`, floating point, several starting points) finds no value below the corner $\sigma_\circ \to 1$, $\varrho \to 1$ used in the proof, for this and for each of the profiles above; this is a numerical observation, not a proof. This is a limit of the one-disc method, not of $\mathfrak{a}$: on the Lindelöf hypothesis $S(t) = o(\log t)$ [37, Thm. 13.6(A), p. 333], that is $\mathfrak{a} = 0$, by an argument that also uses the zeros near $\frac{1}{2} + it$. (d) Theorem 10.2 is asymptotic: the explicit constants of the bounds for ζ do not enter, which is why non-explicit exponents may be used, and it does not replace the explicit bounds of [3], valid for all $T \ge e$.

11. Theorem F: simple zeros on the critical line from the defect

In this section F is any real function, $u_k = k - \frac{1}{2} - F(\gamma_k)$, and $N_0^s(T, 2T)$ is the number of zeros ρ with $T < \gamma \le 2T$, $\beta = \frac{1}{2}$ and multiplicity 1. For $X \ge 2$ and $j \ge 1$ put

$$Q_{2j}(T, X) = \frac{1}{\#\mathcal{I}_T} \int_T^{2T} \Big(N(t) - F_X(t)\Big)^{2j} F_X'(t)\, dt. \tag{11.1}$$

Lemma 11.1 (Multiplicity of an ordinate)**.** (a) *If an ordinate γ occurs with multiplicity $m \geq 2$ in the list (γ_k), at the indices $k, \dots, k+m-1$, then $u_{k+i} = u_k + i$ for $0 \leq i < m$, and for every $j \geq 1$*

$$\sum_{i=0}^{m-1} \Big(2u_{k+i}\Big)^{2j} \geq m.$$

(b) *If γ occurs with multiplicity 1, the zero with ordinate γ is simple and lies on the critical line.*

Proof. (a) $F(\gamma_{k+i}) = F(\gamma)$ for all i, so $u_{k+i} = u_k + i$. Put $\phi(x) = (2x)^{2j}$: it is even, convex, increasing on $[0, \infty)$ and $\phi(\pm\frac{1}{2}) = 1$. For two reals a, b with $|a| + |b| \geq 1$, convexity and evenness give $\phi(a) + \phi(b) \geq 2\phi(\frac{|a|+|b|}{2}) \geq 2$; this applies to consecutive u_{k+i}, u_{k+i+1}, since $|x| + |x+1| \geq 1$. If m is even, pair the m values consecutively. If $m \geq 3$ is odd, at most one of the m values lies in $(-\frac{1}{2}, \frac{1}{2})$, and the two extreme values u_k and $u_k + m - 1$ differ by $m - 1 \geq 2$, so one of them, say v, has $|v| \geq \frac{1}{2}$; pair the other $m - 1$ values consecutively and add $\phi(v) \geq 1$. In both cases the sum is at least m. (b) If $\rho = \beta + i\gamma$ with $\beta \neq \frac{1}{2}$, then $1 - \bar{\rho} = 1 - \beta + i\gamma$ is a zero with the same ordinate, by the functional equation and the reflection principle; a multiple zero repeats its ordinate. In both cases γ has multiplicity at least 2. □

Theorem 11.2 (Theorem F)**.** *For every real function F, every $j \geq 1$ and every $T > 0$,*

$$N_0^s(T, 2T) \geq \#\mathcal{I}_T - \sum_{k \in \mathcal{I}_T} \Big(2u_k\Big)^{2j}. \tag{11.2}$$

Proof. The set $\mathcal{I}_T = (N(T), N(2T)]$ contains, for every ordinate $\gamma \in (T, 2T]$, all its indices. By Lemma 11.1(b), every index of $\mathcal{I}_T$ whose ordinate has multiplicity one is a simple zero on the critical line, so $N_0^s(T, 2T) \geq \#\mathcal{I}_T - \mathsf{m}_T$, where m_T is the number of indices of $\mathcal{I}_T$ whose ordinate has multiplicity at least 2. By part (a) the indices of each such ordinate contribute at least their number to $\sum_{k \in \mathcal{I}_T} (2u_k)^{2j}$, and the other terms are nonnegative, so $\mathsf{m}_T \leq \sum_{k \in \mathcal{I}_T} (2u_k)^{2j}$. □

Corollary 11.3. *Let $j \geq 1$ and $0 < \theta < \frac{2}{2j+1}$ be fixed. For $F = F_X$ with $2 \leq X \leq T^\theta$, as $T \to \infty$,*

$$\frac{N_0^s(T, 2T)}{\#\mathcal{I}_T} \geq 1 - 4^j \Big\langle u_k(X)^{2j} \Big\rangle_{\mathcal{I}_T}, \qquad \Big\langle u_k(X)^2 \Big\rangle_{\mathcal{I}_T} = Q_2(T, X) - \frac{1}{12} + o(1), \tag{11.3}$$

so that $N_0^s(T, 2T)/\#\mathcal{I}_T \geq \frac{4}{3} - 4Q_2(T, X) - o(1)$ for $X \leq T^{1/2}$, and $N_0^s(T, 2T)/\#\mathcal{I}_T \geq 1 - 16Q_4(T, X) + 8Q_2(T, X) - \frac{7}{15} - o(1)$ for $X \leq T^{1/3}$.

Proof. The first inequality is (11.2) divided by $\#\mathcal{I}_T$. Apply Theorem 4.1 with $F = F_X$, $k_1 = N(T)$ and $k_2 = N(2T)+1$, so that the inner indices are exactly $\mathcal{I}_T$. The interval $[\gamma_{k_1}, \gamma_{k_2}]$ differs from $[T, 2T]$ by the pieces $[\gamma_{k_1}, T]$ and $[2T, \gamma_{k_2}]$, on which N is constant; on such a piece the integrand is $(n - F_X)^{2j} F_X'$ with n fixed, whose integral is $\frac{1}{2j+1}$ times a difference of two values of $(n - F_X)^{2j+1}$. As $|N - F_X| \leq |S| + |\varpi|/\pi + \sigma_X \ll \log T + \sqrt{X}$ near T and $2T$, and the same bound holds for the u_k at k_1, k_2, the end pieces and the boundary terms of

(4.1) are $O((\log T+\sqrt{X})^{2j+1})=O(T^{1-\eta})$ for some $\eta=\eta(j,\theta)>0$, which is $o(\#\mathcal{I}_T)$. Hence $\sum_{k\in\mathcal{I}_T}\Psi_j(u_k)=\#\mathcal{I}_T\,Q_{2j}(T,X)+o(\#\mathcal{I}_T)$. With $\Psi_1(u)=u^2+\frac{1}{12}$ and $\Psi_2(u)=u^4+\frac{1}{2}u^2+\frac{1}{80}$ from (4.2), $\langle u^2\rangle=Q_2-\frac{1}{12}+o(1)$ and $\langle u^4\rangle=Q_4-\frac{1}{2}Q_2+\frac{7}{240}+o(1)$; insert them in the first inequality with $j=1$ and $j=2$. □

Remark 11.4 (What the certificate needs). The cross term $\frac{1}{\#\mathcal{I}_T}\int_T^{2T}(N-F_X)^2S_X'\,dt$, which comes from $dF_X=(\mathcal{L}+S_X')\,dt$, is part of Q_2; no hypothesis on the zeros enters (11.3). The first form gives a positive proportion as soon as $Q_2(T,T^c)\le\frac{1}{3}-\eta$ for one fixed $c\le\frac{1}{2}$, $\eta>0$ and all large T, and the proportion $\frac{2}{3}$ when $Q_2\le\frac{1}{6}$. Theorem 5.1 and (4.3) give $Q_2(T,T^c)=\frac{1}{2\pi^2}\log\frac{1}{c}+O(1+\sqrt{\log(1/c)})$, with an absolute constant that is not computed; so the corollary yields no numerical proportion here. Unconditionally, more than two thirds of the zeros are simple and on the critical line, 0.6725 with the window of Montgomery and Taylor, by Alpöge and Furman [1] and, with another proof, by Lamzouri [25]; Knausgård [20] obtains 0.67353 by a computer-assisted argument. These proofs rest on Montgomery's pair-correlation method in the unconditional form of Baluyot, Goldston, Suriajaya and Turnage-Butterbaugh [2]; Theorem F uses a different integrality step, Lemma 11.1, and a different quantity, Q_2.

12. Theorem G: the finite Euler product in the zeta function of the corrected roots

For a real sequence $\kappa=(\kappa_k)_{k\ge1}$ with $\kappa_k\neq0$ and $\sum_k\kappa_k^{-2}<\infty$ put

$$\zeta_\kappa(s)=\frac{1}{G(s)}\prod_{k\ge1}\Big(1+\frac{s(s-1)}{\frac14+\kappa_k^2}\Big),\qquad G(s)=s(s-1)\pi^{-s/2}\Gamma\Big(\frac{s}{2}\Big).\tag{12.1}$$

The product converges absolutely and locally uniformly, its factors vanish exactly at $s=\frac12\pm i\kappa_k$, and $G(1-s)\zeta_\kappa(1-s)=G(s)\zeta_\kappa(s)$, the functional equation of ζ in the form $\xi(1-s)=\xi(s)$; if all nontrivial zeros of ζ were on the critical line, the Hadamard product of ξ would give $\zeta_\gamma=\zeta$. With $z=s-\frac12$,

$$1+\frac{s(s-1)}{\frac14+t^2}=\frac{t^2+z^2}{t^2+\frac14},\qquad f_z(t)=\log\frac{t^2+z^2}{t^2+\frac14}\quad(\Re z>0,\ t\in\mathbb{R}),\tag{12.2}$$

with the principal logarithm: for $\Re z>0$ and real t, $t^2+z^2\notin(-\infty,0]$, so $f_z(t)$ is holomorphic in z, real for real $z>0$, and $f_z(t)=O_z(t^{-2})$, $\partial_t f_z(t)=2t(\frac14-z^2)/((t^2+z^2)(t^2+\frac14))=O_z(t^{-3})$ as $t\to\infty$, locally uniformly in z. For two such sequences κ,κ' with $\kappa_k-\kappa_k'$ bounded, $\sum_k(f_z(\kappa_k)-f_z(\kappa_k'))$ converges absolutely and is a holomorphic logarithm of $\zeta_\kappa/\zeta_{\kappa'}$ in $\Re s>\frac12$, real on $(\frac12,\infty)$.

Lemma 12.1. *For $\Re z>0$ and $\omega>0$,*

$$\int_0^\infty f_z(t)\cos(\omega t)\,dt=\frac{\pi}{\omega}\Big(e^{-\omega/2}-e^{-\omega z}\Big).$$

Proof. For real $a>0$, $\int_0^\infty\frac{2a\cos\omega t}{t^2+a^2}dt=\pi e^{-a\omega}$ (residue theorem). Differentiation under the integral sign, dominated by $2a_2/(t^2+a_1^2)$ for $a\in[a_1,a_2]\subset(0,\infty)$, gives $\frac{d}{da}\int_0^\infty\log\frac{t^2+a^2}{t^2+1/4}\cos\omega t\,dt=$

$\pi e^{-a\omega}$; both sides of the lemma vanish at $a = \frac12$, so the lemma holds for real $z = a > 0$. Both sides are holomorphic in z on $\Re z > 0$, the left one by dominated convergence ($|f_z(t)| \ll_K 1 + |\log t|$ for $t \le 1$ and $\ll_K t^{-2}$ for $t \ge 1$, uniformly for z in a compact K), and they agree on $(0,\infty)$. □

Lemma 12.2 (The midpoint rule along a monotone model)**.** *Let $\mathsf{F} \in C^1[t_*,\infty)$ with $\mathsf{F}' > 0$ on $[t_*,\infty)$, $\mathsf{F}(t) \to \infty$ and $\mathsf{F}(t_*) = v_* \in \mathbb{Z}$, and let ρ_k be the root of $\mathsf{F}(t) = k - \frac12$, $k > v_*$. Let $\phi \in C^1[t_*,\infty)$ with $\phi \to 0$ at ∞ and $\int_{t_*}^\infty (|\phi|\mathsf{F}' + |\phi'|)\,dt < \infty$. Then, with the 1-periodic sawtooth $\psi(v) = \lfloor v + \frac12 \rfloor - v$,*

$$\sum_{k > v_*} \phi(\rho_k) = \int_{t_*}^\infty \phi(t)\mathsf{F}'(t)\,dt - \int_{t_*}^\infty \phi'(t)\,\psi\Big(\mathsf{F}(t)\Big)\,dt.$$

Proof. Put $g = \phi \circ \mathsf{F}^{-1}$ on $[v_*,\infty)$ and $C(v) = \#\{k > v_* : k - \frac12 \le v\} = \lfloor v + \frac12 \rfloor - v_*$. Then $\sum_{k>v_*} \phi(\rho_k) = \int_{(v_*,\infty)} g\,dC$, and $C(v) - (v - v_*) = \psi(v)$ vanishes at $v = v_*$ and is bounded; integration by parts gives $\int g\,dC = \int_{v_*}^\infty g\,dv - \int_{v_*}^\infty g'(v)\psi(v)\,dv$, the boundary term at ∞ vanishing because $g \to 0$. Substitute $v = \mathsf{F}(t)$, $g'(v)\,dv = \phi'(t)\,dt$. □

Theorem 12.3 (Theorem G)**.** *Let $X \ge 2$ be fixed and M_X the bound for $|S_X'|$ in (2.11). Let $t_1 > 2\pi$ with $\mathcal{L}(t_1) \ge 2M_X$, so that $F_X' \ge \frac12\mathcal{L} > 0$ on $[t_1,\infty)$; let $v_0 = \lceil F_X(t_1) \rceil$, $t_0 = F_X^{-1}(v_0)$, $k_0 = v_0 + 1$, and for $k \ge k_0$ let κ_k be the unique root of $F_X(t) = k - \frac12$ in $[t_0,\infty)$; for $1 \le k < k_0$ let κ_k be any nonzero reals, for instance $\kappa_k = \tau_k$. Let $P_X(s) = \prod_{p \le X}(1 - p^{-s})^{-1}$. Then, for $\Re s > \frac12$,*

$$\sum_{k \ge 1} \Big(f_z(\kappa_k) - f_z(\tau_k)\Big) = \log P_X(s) - \log P_X(1) + H_X(s), \tag{12.3}$$

that is $\zeta_\kappa(s) = \zeta_\tau(s)\,P_X(s)P_X(1)^{-1}e^{H_X(s)}$, where, with $t_0' = N_0^{-1}(0)$ and ψ as in Lemma 12.2,

$$H_X(s) = \sum_{k<k_0} f_z(\kappa_k) - \int_{t_0'}^{t_0} f_z\mathcal{L}\,dt - \int_0^{t_0} f_z S_X'\,dt - \int_{t_0}^\infty \partial_t f_z\,\psi(F_X)\,dt + \int_{t_0'}^\infty \partial_t f_z\,\psi(N_0)\,dt$$

is holomorphic in $\Re s > \frac12$ and is determined by X, N_0, the primes $p \le X$ and the choices of t_1 and of κ_k, $k < k_0$, on which the left side of (12.3) depends in the same way.

Proof. N_0 is increasing on $(2\pi,\infty)$ with $N_0(t_0') = 0$, and τ_k, $k \ge 1$, are the roots of $N_0 = k - \frac12$; F_X is increasing on $[t_0,\infty)$ and the κ_k, $k \ge k_0$, are the roots of $F_X = k - \frac12$ there. Lemma 12.2 applies to both, with $\phi = f_z$ (the integrability conditions hold by the bounds after (12.2), $\mathcal{L} \ll \log t$ and $|S_X'| \le M_X$):

$$\sum_{k \ge k_0} f_z(\kappa_k) = \int_{t_0}^\infty f_z(\mathcal{L} + S_X')\,dt - \int_{t_0}^\infty \partial_t f_z\,\psi(F_X)\,dt,$$
$$\sum_{k \ge 1} f_z(\tau_k) = \int_{t_0'}^\infty f_z\mathcal{L}\,dt - \int_{t_0'}^\infty \partial_t f_z\,\psi(N_0)\,dt.$$

Subtracting, it remains to evaluate $\int_0^\infty f_z S_X'\,dt$. By (1.3) and $\arg(1-w) = -\Im\sum_{m\ge1} w^m/m$ for $|w| < 1$, $S_X(t) = -\frac1\pi \sum_{p\le X}\sum_{m\ge1} \frac{\sin(mt\log p)}{mp^{m/2}}$, so $S_X'(t) = -\frac1\pi \sum_{p\le X}\sum_{m\ge1} \frac{\log p}{p^{m/2}}\cos(mt\log p)$,

uniformly convergent. Since $f_z \in L^1(0,\infty)$, the sum may be integrated term by term, and Lemma 12.1 with $\omega = m\log p$ gives

$$\int_0^\infty f_z S_X'\,dt = -\sum_{p\le X}\sum_{m\ge1}\frac{\log p}{p^{m/2}}\,\frac{p^{-m/2}-p^{-mz}}{m\log p} = \sum_{p\le X}\sum_{m\ge1}\frac{p^{-ms}-p^{-m}}{m} = \log P_X(s) - \log P_X(1),$$

with principal logarithms, $|p^{-s}|<1$. Each term of H_X is holomorphic in $\Re s > \frac12$: the finite sum and the integrals over bounded intervals because $f_z(t)$ is holomorphic in z and locally bounded, the two integrals with ψ by dominated convergence, since $|\psi| \le \frac12$ and $\partial_t f_z(t) = O(t^{-3})$ locally uniformly in z. Exponentiating, and using that both sides of (12.3) are real for real $s > \frac12$, gives the product form. □

Remark 12.4. The zeros of ζ_κ are on the critical line by construction, and ζ_τ, P_X and e^{H_X} are free of ζ: Theorem G separates, in the zeta function of the corrected roots, the part supplied by the primes, which is exactly the finite Euler product P_X, from the part supplied by the clock. Consequently, in $\Re s > 1$, $\zeta(s)/\zeta_\kappa(s) = \big(\zeta(s)/P_X(s)\big)\,P_X(1)\,e^{-H_X(s)}/\zeta_\tau(s)$, and in $\Re s > \frac12$ the only factor on the right whose singularities are not explicit is ζ/P_X, whose zeros and poles there are those of ζ. The roots κ_k, $k \ge k_0$, are the roots t_k^X of Proposition 7.8 on every window where both are defined. For a cutoff growing with the height, as in Theorems B and D, the same computation gives a hybrid form in which the prime p enters from the height where it enters the cutoff; this is not proved here.

13. Conclusions

13.1. What was known

Pointwise, $\delta_k = \tau_k - \gamma_k$ is bounded, with $\limsup|\delta_k| \le 0.6331$ from the explicit bound of Bellotti and Wong [3], and $\limsup|\delta_k| \le 0.6365$ from Trudgian's asymptotic bound [38]. On average, s_k is asymptotically Gaussian with variance $\frac{1}{2\pi^2}\log\log T$, by Selberg's formulae for the Gram displacement proved by Korolev [24, 21], so $\langle\delta_k^2\rangle \sim 2\log\log T/\log^2 T$: the clock is close to the ordinates, while in units of zeros its displacement has unbounded mean square. Selberg's approximation [32, 11] replaces $S(t)$ by short prime sums in t-mean.

13.2. What this paper adds

1. *An exact dictionary between clock-type models and the zeros* (Theorems 4.1 and 4.2). For any C^1 model F and any continuous $\mathfrak{g}$, sums of $\mathfrak{g}$-increments of $k - \frac12 - F(\gamma_k)$ over the ordinates are exactly dF-integrals of $\mathfrak{g}(N-F)$ up to two boundary terms; for powers this gives the even moments with the constant $\frac{1}{12}$ per zero, for exponentials it transfers the signed law of $S(t)$ to s_k (Corollary 4.4).

2. *The prime information needed by the clock* (Theorems 5.1 and 5.12). The corrected clock equation $N_0(t) + S_X(t) = k - \frac12$ holds at the true ordinates with mean-square ordinal defect $\frac{1}{2\pi^2}\log\frac{\log T}{\log X} + O(1+\sqrt{\log(\log T/\log X)})$, also with a cutoff varying at every index; the defect is bounded in mean square if and only if the primes reach a fixed power of the height.

3. *An exact finite-height account* (Theorem 6.1). The gain of the one-step clock is a diagonal prime response minus a cross term, and its residual differs from the defect by a transport term and an explicit nonnegative second-order term.

4. *A computable clock that approaches the zeros* (Theorems 7.3, 8.6 and 8.11). With $X = T^{1/\sqrt{\log\log T}}$ the one-step clock has mean-square error $(\log\log\log T + O(\sqrt{\log\log\log T}))/\log^2 T$ in the ordinates, against $(2+o(1))\log\log T/\log^2 T$ for the bare clock; the m-th Newton-type step has error $\frac{H_m}{4\pi^2} + O(1+\sqrt{H_m})$ in units of zeros with absolute constants, and after $\log^* T + O(1)$ steps the error is $O(1)$ mean spacings, using no ordinate. The proof needs an exact moment recursion, a multiplicity bound for the corrected points of every level (Theorem 8.10) and a cutoff law that absorbs the multiplicity.

5. *A regularized clock with a fixed cutoff* (Theorem 9.6). The best L^4 approximation of $N_0 + S_X$ by piecewise linear functions on a mesh of step at most $2\pi/\log T$ with slopes in $[\frac{\log T}{4\pi}, \frac{\log T}{\pi}]$ gives, without Newton iteration and with $X = T^c$, $c < \frac{1}{2}$, increasing points separated by at least $\pi/\log T$ with $\langle|\widehat{x}_k - \gamma_k|^4\rangle \ll (1+\log\frac{1}{c})^2/\log^4 T$ on the clock window; no ordinate enters the construction, which is global on the block $[\frac{T}{2}, 3T]$, and an approximate minimizer with certified excess has the same guarantee. For the one-step clock from τ_k with a fixed cutoff exponent the error is $\ll (1+\log\log\log T)/\log^2 T$ (Theorem 7.5); whether it is $O_c(1)$ mean spacings is not decided.

6. *Tails and order* (Theorem 8.3, Corollaries 8.13 and 8.15). The defect at the ordinates exceeds λ on at most $2^{2-\sqrt{\lambda/C}}$ of the zeros when the primes reach T^c with c in a range fixed by λ; the errors of the corrected clocks have all moments bounded in units of $\sqrt{H_j}$; and sorting the final clock makes it nondecreasing without increasing its mean-square error.

7. *The pointwise envelope* (Proposition 10.1, Theorem 10.2). $\limsup|\tau_k - \gamma_k| = 2\pi\limsup|S(t)|/\log t$ exactly, no slowly growing prime correction changes it, and Backlund's method with the convex hull of the published exponent pairs of Bourgain and Heath-Brown gives $\limsup|\tau_k - \gamma_k| \le 2\pi C_{\mathcal{P}} < 0.6072889$ (0.6060224 with the pairs of the preprints [40, 35]), with no evaluation of ζ; with the disc of the proof, and numerically among the discs examined, Backlund's method cannot go below 0.4940.

8. *Simple zeros on the critical line from the defect* (Theorem 11.2, Corollary 11.3). An ordinate of multiplicity one carries a simple zero on the line; for any model F the simple zeros on the line in $(T, 2T]$ number at least $\#\mathcal{I}_T - \sum_{k\in\mathcal{I}_T}(2u_k)^{2j}$, and with Theorem A their proportion is at least $\frac{4}{3} - 4Q_2 - o(1)$; a positive proportion would follow from any proved bound $\limsup Q_2(T, T^c) < \frac{1}{3}$ at one fixed c.

9. *The finite Euler product in the zeta function of the corrected roots* (Theorem 12.3). For a fixed cutoff, the zeta function of the roots of the corrected equation in its monotone range is $\zeta_\tau(s)\,P_X(s)/P_X(1)$ times a factor holomorphic and zero-free in $\Re s > \frac{1}{2}$, where ζ_τ is the zeta function of the clock and $P_X(s) = \prod_{p\le X}(1-p^{-s})^{-1}$: the construction reproduces exactly the finite Euler product of the primes it uses.

10. *Comparison with the corrected clock equation solved numerically* (Theorems 5.1, 7.3 and 9.6, Table 2). The clock equation corrected by the first N primes was solved numerically in [26], with an accuracy estimated from known zeros, and with smoothed weights it is the equation of the zeros of Gonek's ζ_X [14]. Here, for $p_N \le T^{1/2}$, its mean-square defect at the true ordinates is $\frac{1}{2\pi^2}\log\frac{\log T}{\log p_N} + O(1+\sqrt{\log(\log T/\log p_N)})$, unbounded for a fixed number of primes; explicit Newton-type steps give points with proved error, and so does, for $X \le \sqrt{T/2}$, a convex regularization with a fixed cutoff exponent.

13.3. Scope

Theorem 4.1 applies to any counting staircase and any C^1 model, in particular to the zeros of a primitive Dirichlet L-function; an analogue of Theorem 5.1 would additionally need Selberg's approximation for $L(s,\chi)$ [33, 11] and uniformity in the conductor.

Declaration of competing interest

The author declares no competing interests.

Data availability

No zero dataset was used to choose a construction or parameter. Two programs, `opc_envelope_coefficient_v14.py` (40-digit quadrature) and `opc_envelope_iterations_p1.py` (closed form at 50 digits, $0 \le J \le 30$), evaluate the integral (10.1) for piecewise linear exponent profiles with rational vertices; they and their outputs are supplied in the accompanying compilation archive, in `code/` and `outputs/`. A third program, `verify_envelope_intervals_v5.py`, encloses C_{30} and the analytic tail by interval arithmetic at 80 digits; its output `envelope_intervals_v5.json` confirms the upper bound $2\pi C_\infty < 0.6083074$. The program `envelope_best_profile_v9.py` builds the lower convex hulls of Theorem 10.2 and Remark 10.3 with exact rational arithmetic and evaluates the integral (10.1) in interval arithmetic at 50 digits (output `envelope_best_profile_v9.txt`); `envelope_disc_scan_v9.py` scans the Backlund disc in floating point (output `envelope_disc_scan_v9.txt`). The numerical manifest identifies each program, its precision and parameters, and the output supporting each displayed decimal. Python 3 and mpmath 1.3.0 were used, with numpy and scipy for the disc scan. The programs evaluate consequences of the analytic formulae; no observed zero ordinate chooses the construction or a parameter.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used AI assistants (Claude, Anthropic, and ChatGPT, OpenAI) to edit the text, check proofs and write programs. The author reviewed and edited the content and takes full responsibility for it.

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